Geofinance: Following Sovereign Capital Home*

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Abstract

I study whether sovereign wealth fund (SWF) private investments change where recipient firms do business, and whether these micro relationships aggregate into broader capital-flow responses. This departs from the standard separation benchmark, in which financier identity should not determine real operating choices or the geography of capital allocation. I develop a geofinance framework in which a strategic financing relationship can induce recipient activity toward the financier's home country and lower bilateral frictions for third-party investors. The model predicts stronger directed activity when recipients have weaker outside options, when the home-country relationship is more valuable, and predicts aggregate crowd-in as bilateral links accumulate. Following SWF private investments, recipients with weaker outside options are more likely to enter SWF home countries than the home countries of non-SWF investors in the same financing round. The response is not explained by firm scale, ownership dilution, or round pricing, and is stronger in sectors where SWF home countries have revealed export disadvantages. Similar business redirection appears during the 2007–2008 financial crisis: banks with larger write-downs increased syndicated lending activity toward sovereign-fund home countries. At the aggregate level, complementary FDI and BIS banking-claims tests show crowd-in toward SWF home countries, with magnitudes of about 5–8 percent across the two designs.

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1 Introduction

In May 2025, Saudi Arabia's Public Investment Fund (PIF) and Google Cloud announced a \$10 billion joint investment to build an artificial intelligence hub in Saudi Arabia.[1] Finance flowed one way while economic activity followed the other. This pattern recurs as sovereign wealth funds (SWFs), managing over \$12 trillion, have become large cross-border private-market investors since the mid-2000s.[2] Yet in the Modigliani–Miller benchmark, the identity of capital providers should not determine a firm's real operating choices or the geography of capital allocation (Modigliani and Miller, 1958). I ask when SWF private investments redirect recipients' cross-border activity toward SWF home countries, and whether the bilateral links created by that redirection crowd in outside capital.

I develop a geofinance framework in which a strategic financier can use an investment relationship to support directed business activity toward its home country. The framework differs from geoeconomic models that emphasize leverage through existing trade and financial dependencies (Clayton, Maggiori and Schreger, 2025, 2026): it studies how capital can help create new bilateral links. The recipient's outside financing options determine how valuable continued access to the relationship is, while the financier's home-country objective determines where the response is directed. SWFs are the empirical focus because they are large cross-border investors with potential home-country objectives, but the mechanism is not unique to sovereign capital.

The mechanism has two margins. At the recipient level, directed activity can move above the commercial benchmark when the value of the relationship covers the private cost of doing business in the financier's home market. This departure is easier to sustain when outside financing options are weaker and when the induced activity is more valuable to the financier. At the aggregate level, directed activity adds to a bilateral relationship stock. As this stock rises, bilateral frictions fall, making subsequent investment in the financier's home country more attractive to outside investors.

The model delivers two empirical implications. First, SWF capital should induce recipients to do more business in SWF home countries than their commercial benchmark would imply, especially on new-entry margins where bilateral links are formed. The effect should be stronger when recipients have weaker outside financing options and when the SWF has greater home-country value in the relevant sector or market. Second, if directed activity builds bilateral links, it should crowd in third-party capital toward SWF home countries. I test these implications in

[1] Google Cloud and PIF Advance AI Hub in Saudi Arabia.

[2] For scale, the \$12 trillion figure is comparable to the \$12.7 trillion in total loans and leases held by U.S. commercial banks (Federal Reserve H.8 release, March 2025). SWFs deployed roughly \$1.28 trillion in cross-border private-market investments over 2002–2022, computed from GSWF and SWFI transaction databases.

four settings: private-company financing rounds, crisis-era bank recapitalizations, bilateral foreign direct investment (FDI), and cross-border banking claims. The first two settings test recipient-level business redirection and its heterogeneity; the last two test whether those directed links are followed by broader capital-flow responses.

I first test the mechanism in private-company financing rounds by measuring new entry into investor home countries after investment. This is a geographic outcome, not the usual SWF outcome of target value, governance, or operating performance (Bernstein, Lerner and Schoar, 2013; Kotter and Lel, 2011; Bortolotti, Fotak and Megginson, 2015): it asks whether sovereign capital changes where recipient firms do business.

That question cannot be answered by comparing raw post-investment entry rates, because SWF-backed private-company deals are highly selected. In the cross-border private-company universe, SWF participation is concentrated in high-valuation firms, larger employers, and sectors that SWFs had previously overweighted, often where the SWF home country has a revealed export disadvantage. SWF investors are also more likely than same-round commercial co-investors to return to the same portfolio company in later private financing rounds. These patterns show both nonrandom selection into SWF capital and the investment relationship structure through which continuation value can operate.

These patterns make across-firm comparisons a poor basis for identifying the mechanism. The identifying test therefore moves within SWF-backed financing rounds. Holding fixed the recipient and financing event, I compare new entry into the SWF investor's home country with new entry into the home countries of same-round non-SWF investors, and ask whether that gap varies with the model's two shifters: recipient outside-option weakness and the strategic value of the home-country link.

Before conditioning on these two shifters, mere SWF presence does not produce a stable average gap in new market entry. In raw comparisons, SWF home countries are less common expansion destinations, partly because same-round non-SWF investors are often based in deep markets such as the United States and the United Kingdom. Once I absorb destination-year conditions and country-pair differences, the average SWF-country gap turns positive but remains statistically indistinguishable from zero. The absence of a stable average gap rules out a simple mechanical story in which SWF-backed firms uniformly enter SWF home countries more often. The evidence must come from the conditional cases where the model predicts directed activity.

The new-market-entry gap rises with recipient outside-option weakness. I proxy for outside-option weakness with the firm's valuation discount, defined as lower post-money valuation relative to same-sector-year peers. In the same within-deal design, a one-standard-deviation increase in this discount raises the SWF–non-SWF new-entry gap by 5.46 percentage points, about 72% of the non-SWF baseline rate. This result should be read within the selected SWF-backed sample. SWFs invest in large, high-valuation private companies in the broad private-company universe; the mechanism test asks whether relatively lower valuations among those selected firms predict a larger expansion tilt toward the SWF home country.

I interpret valuation discount as a reduced-form proxy for weak outside financing options. Private-round valuation is an observable price of external equity, so a low valuation relative to peers is a natural marker of costly outside finance, in the spirit of private-placement discounts (Hertzel and Smith, 1993). The financial-constraints literature often uses firm size and age as indirect proxies, while also warning that common proxies are imperfect (Hadlock and Pierce, 2010; Farre-Mensa and Ljungqvist, 2016). The proxy survives controls and horse races for firm scale, age, employees, ownership-dilution measures, and alternative valuation benchmarks, suggesting that the slope is not driven by a simple size, lifecycle, control-rights, or valuation-benchmark artifact.

The strategic-value margin appears in the sectors where outside-option weakness matters most. Firms with weaker outside options are especially more likely to enter the SWF's home country when that home country has historically low export strength in the firm's sector. These are settings where new links are plausibly more valuable to the SWF because entry does not simply extend an established export specialization. This matches the model's complementarity: weak outside options matter most when the home-country link is most valuable.

The same outside-option margin appears in crisis-era bank recapitalizations. During the 2007–2008 financial crisis, banks with larger pre-crisis write-down exposure had greater demand for outside capital, and SWFs were a major source of recapitalization.[3] In syndicated lending data, high-write-down banks became more likely to enter sovereign-fund home-country markets and increased deal counts toward those destinations after the crisis cutoff. The setting is smaller and crisis-specific, but it tests the same relationship mechanism with sharper variation in recipient need for outside capital. A one-standard-deviation increase in write-down severity is associated with about 19% higher deal counts among positive bank–country–quarter lending cells.

I then test the crowd-in prediction in bilateral FDI. In 2018, the Foreign Investment Risk Review Modernization Act (FIRRMA) tightened U.S. review of foreign government-linked investments. SWF home countries with greater pre-2018 exposure to high-regulatory-risk U.S. deals subsequently attracted more third-party FDI from non-U.S. source countries. The response is concentrated among continuing bilateral relationships rather than new source-country entry. Con-

[3]The International Monetary Fund's *Global Financial Stability Report* (April 2008). Public offerings and other institutional investors were the other major sources of funding.

ditional on positive flows, observed exposure levels imply roughly 5–8% higher post-FIRRMA FDI inflows relative to SWF destinations with zero exposure.

The banking-claims evidence points in the same direction. Using the Bank for International Settlements (BIS) Locational Banking Statistics, I trace lender-country claims on SWF home countries after SWF equity injections into banks. Each additional $1 billion of SWF equity is associated with a 5.8% increase in loans and placed deposits to SWF countries.

The four designs point to the same geofinance mechanism. At the micro level, SWF capital is followed by activity toward SWF home markets, and the response is strongest when recipients have weaker outside options and when the home-country relationship is more valuable. At the aggregate level, FDI and banking-claims outcomes subsequently move toward SWF home countries. Sovereign capital transactions are large, strategic, and infrequent, so no single setting can isolate every margin of the mechanism. I therefore read the designs as complementary evidence: private-company rounds provide a broad mechanism test, bank recapitalizations provide sharper crisis variation in recipient capital need, and the FDI and BIS settings test whether the resulting bilateral links are followed by macro capital-flow responses.

Related literature. The paper first contributes to geoeconomics. Recent theory shows how states project power through existing trade and financial relationships (Clayton, Maggiori and Schreger, 2025, 2026), and network accounts emphasize coercion through control over hubs (Farrell and Newman, 2019). Related empirical work shows that state influence and state-backed finance can redirect economic activity: political influence redirected trade toward American products (Berger et al., 2013), China manages foreign access to its bond market (Clayton et al., 2025), and export-credit agencies raise exports where financing and contracting frictions bind (Kabir et al., 2025). The distinction is formation. These papers study pressure through existing dependencies; I study how state capital can create new bilateral business links and how those links can later support private capital flows.

The paper also contributes to work on SWFs and sovereign investment. Existing studies describe SWF investment strategies, political motives, and differences between government-linked and private acquirers (Chhaochharia and Laeven, 2008; Bernstein, Lerner and Schoar, 2013; Knill, Lee and Mauck, 2012; Karolyi and Liao, 2017). A related strand estimates target-firm announcement, governance, and operating effects (Dewenter, Han and Malatesta, 2010; Kotter and Lel, 2011; Bortolotti, Fotak and Megginson, 2015); see Megginson and Fotak (2015) for a survey. More recent work shows that SWF investment increases home-country exports after a U.S. regulation shock (Zhang and Zhou, 2025). I shift the outcome from target-firm value, governance, or sovereign portfolio allocation to the geography of recipient activity and capital flows. The question is not only whether SWF capital affects recipients, but whether and how it changes where recipient firms, banks, and outside investors do business.

The paper speaks to international finance by studying how bilateral capital-flow frictions can change. Cross-border flows are shaped by gravity, information, institutional, and control frictions (Anderson and van Wincoop, 2003; Portes and Rey, 2005; Wei, 2000; Head and Ries, 2008; Daude and Fratzscher, 2008; Pellegrino, Spolaore and Wacziarg, 2025). But these frictions need not be fixed. Chaney (2014) shows that trade networks lower search costs and make future trade easier. I bring this logic to sovereign-backed capital relationships: SWF investments can create bilateral business links that lower relationship frictions and crowd in subsequent FDI and banking claims.

Finally, the paper speaks to corporate finance by studying when financing relationships affect real decisions beyond the separation benchmark. Existing departures emphasize costly external finance and relationship lending: financing frictions affect investment, and repeated lender relationships shape credit access and future financial business (Myers and Majluf, 1984; Boot, 2000). I study a different departure. The relevant friction is not only that external finance is costly, but that the financier has a strategic objective over the recipient's real activity and the recipient values preserving the relationship when outside options are weak. Capital can therefore support not only investment, but destination-specific business activity.

The remainder of the paper is organized as follows. Section 2 develops the geofinance model and derives the main empirical predictions. Sections 3 and 4 present the micro evidence from the private-company and bank-recapitalization settings. Section 5 studies macro crowd-in using FDI and BIS banking claims. Section 6 concludes.

2 A Framework for Geofinance

I model geofinance as a strategic financing relationship that can move recipient activity toward the financier's home market. Firms differ in their outside financing options: firms facing costlier outside finance value relationship surplus more and are therefore more willing to depart from their commercial benchmark. Financiers differ in the strategic value they attach to home-market activity. Geofinance arises when the relationship supports activity that the financier values but that the firm would not choose at the same level under purely commercial finance.[4]

The mechanism has both a recipient-level and an aggregate margin. At the recipient level, out-

[4]This framework extends geoeconomic models of power through trade and financial networks (Clayton, Maggiori and Schreger, 2025, 2026) by treating financing-relationship surplus as a mechanism for creating new bilateral links.

side financing options govern receptiveness, while the financier's home-country objective governs destination. At the aggregate level, directed activity builds a stock of bilateral links. As this stock rises, bilateral frictions fall, making subsequent investment in the financier's home country more attractive to outside investors.

2.1 Setup

Let $\mathcal{I}$ be a finite set of potential recipient firms. I summarize firm i's outside financing options by an effective marginal outside financing cost $r_i > 0$: the cost of raising marginal capital or preserving comparable relationship support from non-strategic financiers. A high r_i means the firm has weaker outside financing options: alternative financing or relationship terms are costly because of risk, opacity, urgency, limited investor competition, or other outside-option premia. It does not mean that the firm cannot raise funds. For tractability, I summarize firm i's ordinary activity by a pre-optimized value $A_i(r_i)$, which is additively separable from directed business activity and drops out of the marginal conditions below.[5]

Let s index the financier's home market. The firm chooses directed business activity toward that market, $k_{is} \geq 0$, with return $G_i(k_{is})$. This activity carries a proportional cost $\mu(Z)k_{is}$, where Z is the stock of bilateral links. Since $\mu'(Z) < 0$, accumulated links lower the cost of later activities in the financier's home market.[6] I assume $A_i(r_i)$ is well behaved; G_i is twice differentiable, strictly increasing, and strictly concave; $\mu(Z) > 0$ and $\mu'(Z) < 0$; and $U(Z)$ is increasing and weakly concave.

I begin with the firm's allocation under purely commercial finance. Taking Z as given, the firm solves

$$\max_{k_{is} \geq 0} \left\{ A_i(r_i) + G_i(k_{is}) - \mu(Z)k_{is} - r_i k_{is} \right\}. \tag{1}$$

For an interior solution, the first-order condition for directed business activity is

$$G_i'(k_{is}^{\text{out}}) = \mu(Z) + r_i. \tag{2}$$

Activity in the financier's home market is disciplined by two costs: the bilateral operating friction

[5] r_i can include the firm's risk-adjusted cost of capital, outside-option premia from information or agency frictions, regulatory pressure, distress, or market segmentation. In the bank application, this cost reflects balance-sheet distress and regulatory leverage pressure; in the corporate application, it reflects costly outside equity or weak outside financing options. One microfoundation is $A_i(r_i) \equiv \max_{k_{ih} \geq 0}\{F_i(k_{ih}) - r_i k_{ih}\}$, where k_{ih} represents ordinary activity outside the strategic market.

[6] The model treats $\mu(Z)$ as a reduced-form bilateral relationship friction. It can reflect search, information, contracting, certification, regulatory navigation, or coordination costs that decline as bilateral activity accumulates.

$\mu(Z)$ and the firm's outside financing cost r_i. Holding Z fixed, firms with weaker outside financing options choose less directed activity.[7]

2.2 The financing relationship and the firm's response

A strategic financier has cost of funds $\rho \geq 0$ and can deliver relationship surplus through capital, access, certification, guarantees, or future financing opportunities. I represent the relationship as a strategic-finance bundle $\{x_i, r_{is}, k_{is}\}$: capital $x_i \geq 0$, an effective package term r_{is}, and directed activity k_{is}. The term r_{is} is chosen by the financier as part of the bundle. It is a convenient way to express the transferable component of the relationship. It can represent a contracted rate, security terms, guarantees, certification, follow-on access, or other package features.[8]

Ordinary investors can supply capital or general expertise to the same firms, but their capital does not enter with a payoff from activity in their own home countries. The strategic financier differs because the bundle pairs financing with a country-specific relationship channel.

With x_i units of finance, the transferable surplus delivered to the recipient is $(r_i - r_{is})x_i$. In an active surplus-delivering relationship this object is positive, but that is an outcome of the chosen package and the participation constraint, not an imposed primitive. The gross joint surplus from replacing outside finance is $(r_i - \rho)x_i$, divided between recipient surplus and financier margin.[9]

Relationship surplus determines whether the firm accepts the bundle; the country-specific channel determines where the response is directed. This channel can include local customers, procurement, regulatory navigation, certification, local partners, or repeated sovereign financing. I summarize its realized effect on the real allocation by the implementation wedge τ_{is}. Ordinary finance is the limiting case in which capital is supplied on market terms, creates no strategic wedge, and leaves the firm at its commercial benchmark. Strategic finance differs because the relationship changes the relative payoff to directed home-market activity and the surplus makes the bundle acceptable. I interpret k_{is} as sustained through project finance, ring-fencing, covenants, regulatory commitments, relational contracting, certification, or other observable deployment restrictions.

The firm's payoff under the relationship decomposes into operating value and relationship

[7] Low observed activity in the strategic market does not mean that projects there have low gross returns. The home-bias literature emphasizes that foreign activity is limited by information, familiarity, monitoring, contracting, and institutional frictions. These additional frictions are summarized by $\mu(Z)$. Firms may earn positive returns abroad and price risk into those activities, but without a strategic financing relationship they may still choose a smaller foreign footprint than the financier would like.

[8] In a same-round equity syndicate, co-investors may share a headline valuation. The model's r_{is} should be read as an effective package term that delivers relationship surplus.

[9] The same surplus could be delivered through debt, equity, guarantees, certification, direct transfers, or future access. In the transfer-separable benchmark, the instrument does not affect the implemented allocation.

surplus:

$$\Pi_i(x_i, r_{is}, k_{is}, Z) = \underbrace{\text{Fund}_i(k_{is}, Z)}_{\text{operating value given } k_{is}} + \underbrace{(r_i - r_{is})\, x_i}_{\text{relationship surplus}}, \tag{3}$$

where

$$\text{Fund}_i(k_{is}, Z) \equiv A_i(r_i) + G_i(k_{is}) - \mu(Z)k_{is} - r_i k_{is}. \tag{4}$$

The surplus term is the transferable part of the bundle. It compensates the firm for moving away from its commercial outside option. The country-specific relationship is not a generic cash transfer; it enters the real allocation through the implementation wedge below. The firm accepts the relationship whenever

$$\Pi_i(x_i, r_{is}, k_{is}, Z) \geq v_i(Z), \tag{5}$$

where the outside option under commercial finance is

$$v_i(Z) = \max_{k_{is} \geq 0} \left\{ A_i(r_i) + G_i(k_{is}) - \mu(Z)k_{is} - r_i k_{is} \right\}. \tag{6}$$

The implemented activity can be represented by an implicit marginal wedge τ_{is}:

$$G_i'(k_{is}) = \mu(Z) + r_i - \tau_{is}. \tag{7}$$

At a given Z, a positive τ_{is} lowers the effective marginal cost of directed home-market activity and induces $k_{is} > k_{is}^{\text{out}}$. The wedge combines home-market access, strategic requests, relationship value, and enforceability. Financing alone does not imply directed activity: if $\tau_{is} = 0$, the firm may accept capital but remains at k_{is}^{out}.

Finally, let $C \subseteq \mathcal{I}$ denote the contracted set of firms. I take this set as given in the benchmark contracting problem and characterize optimal relationships conditional on C. Firms outside C choose their commercial directed activity given the network stock. The aggregate network stock is

$$Z = \sum_{i \in C} k_{is} + \sum_{j \notin C} k_{js}^*(Z), \tag{8}$$

where the second term captures crowd-in. Under the maintained assumptions, outside firms respond positively to a larger network stock:

$$\frac{\partial k_{js}^*}{\partial Z} = \frac{\mu'(Z)}{G_j''(k_{js}^*)} > 0. \tag{9}$$

For local stability of the network fixed point, I impose

$$0 \leq \sum_{j \notin C} \frac{\partial k^*_{js}}{\partial Z} < 1. \tag{10}$$

2.3 The financier's problem and main results

The financier acts as a Stackelberg leader conditional on a contracted recipient set C. It values the bilateral network stock Z, possibly with different weights across sectors. With strategic-objective weight $\theta > 0$, aggregate funding capacity $\bar{X}$, and convex implementation cost $\frac{c}{2}x_i^2$, it chooses relationship bundles $\{x_i, r_{is}, k_{is}\}_{i \in C}$ and the resulting network stock Z:

$$\begin{aligned}
\max_{\{x_i, r_{is}, k_{is}\}_{i\in C},\, Z} \quad & \sum_{i \in C} \left[(r_{is} - \rho)x_i - \tfrac{c}{2}x_i^2 \right] + \theta U(Z) \\
\text{subject to:} \quad & \Pi_i(x_i, r_{is}, k_{is}, Z) - v_i(Z) \geq 0, \qquad \forall i \in C, \quad [y_i] \\
& \bar{X} - \sum_{i \in C} x_i \geq 0, \qquad [\lambda] \\
& \sum_{i \in C} k_{is} + \sum_{j \notin C} k^*_{js}(Z) - Z = 0, \qquad [\Psi_Z] \\
& x_i \geq 0, \quad k_{is} \geq 0, \qquad \forall i \in C.
\end{aligned} \tag{11}$$

All inequality multipliers are nonnegative. The equality multiplier Ψ_Z is unrestricted. I use $\Psi_Z > 0$ for the case in which the financier has a positive shadow value of expanding the bilateral network. The first-order conditions characterize contracted recipients with $x_i > 0$, interior k_{is}, and an interior contract rate, conditional on C.

In the transfer-separable benchmark, the effective package term divides surplus in the interior pricing regime. Only two terms involve r_{is}: $(r_{is} - \rho)x_i + y_i(r_i - r_{is})x_i$. The first-order condition gives $x_i(1 - y_i) = 0$, so for active funding $x_i > 0$, participation binds with $y_i = 1$. The effective package term r_{is} allocates the joint surplus between the financier and the recipient; it does not determine funding volume or directed activity.[10]

The remaining first-order conditions yield two main results, stated below as propositions. Appendix A provides the full derivations.

[10]The cancellation of r_{is} does not require literal risk neutrality by the recipient. It follows from transfer separability: the effective package term and funding volume enter the recipient's payoff through the relationship surplus $(r_i - r_{is})x_i$. With transfer separability and unrestricted transfers, the package term affects surplus division but not the implemented allocation. The cancellation can fail when the package changes payoff risk, default incentives, limited-liability constraints, or the marginal cost of directed activity.

Proposition 1 (Outside financing options and directed relationships)**.** *For any contracted recipient satisfying the interior benchmark conditions,*

$$\tau_{is}^* = \Psi_Z, \qquad x_i^* = \frac{r_i - \rho - \lambda}{c}. \tag{12}$$

Therefore, holding Z fixed, the interior relationship induces $k_{is} > k_{is}^{out}$ whenever $\Psi_Z > 0$. Active funding for a contracted recipient requires

$$r_i > \rho + \lambda. \tag{13}$$

Weaker outside financing options make the relationship surplus more valuable and make the active-funding condition easier to satisfy. They make the firm more receptive to a strategic financing relationship, but the geography of the response comes from the financier's home-country relationship.

Intuition. The home-market relationship determines which activity the financier wants to induce. The relationship package supplies the participation surplus that makes the requested activity acceptable to the firm. Once participation binds, r_{is} cancels from the funding first-order condition: it allocates surplus, while funding volume depends on the value of replacing outside finance. Thus r_i is a receptiveness margin, not a destination margin. The implementation wedge is pinned down by Ψ_Z, while the level of directed activity depends on Ψ_Z, $\mu(Z)$, and the recipient's outside financing options.

Bilateral efficiency. The allocation is a Coasean outcome between the firm and the financier. The firm alone chooses k_{is}^{out}, satisfying $G_i'(k_{is}^{\text{out}}) = \mu(Z) + r_i$. A financier that values the bilateral link adds Ψ_Z, so the contracted allocation satisfies $G_i'(k_{is}^*) + \Psi_Z = \mu(Z) + r_i$. The relationship moves the firm from its commercial benchmark to the firm-financier contracting optimum. The supporting transfer can be delivered through a lower effective rate, equity terms, debt, guarantees, certification, follow-on access, or cash; in the transfer-separable benchmark, the instrument does not affect the implemented allocation.

The same logic gives the heterogeneity tests. The directed-activity gap, $k_{is} - k_{is}^{\text{out}}$, should be larger when two forces coincide. First, the recipient has weaker outside financing options, so the relationship surplus can support a larger departure from the commercial benchmark. Second, the financier places greater value on activity in that sector or market, so it has more reason to request directed activity.

Proposition 2 (Network multiplier and crowd-in)**.** *Strategic financing relationships affect outside* *firms through the endogenous network stock Z. Under the stability condition in Equation (10), the shadow value of network expansion is*

$$\Psi_Z = \left[1 - \sum_{j \notin C} \frac{\mu'(Z)}{G_j''(k_{js}^*)}\right]^{-1} \left[\theta U'(Z) - \mu'(Z) \sum_{i \in C} (k_{is} - k_{is}^{out})\right]. \tag{14}$$

The multiplier term is positive and finite, and $\Psi_Z > 0$ whenever

$$\theta U'(Z) - \mu'(Z) \sum_{i \in C} (k_{is} - k_{is}^{out}) > 0. \tag{15}$$

Intuition. The first bracketed term is the crowd-in multiplier. It is finite by the stability condition. It equals one when the network does not lower outside frictions ($\mu' = 0$) and exceeds one when a larger network stock lowers outside firms' bilateral frictions ($\mu' < 0$). The second bracket contains the financier's direct value of network expansion and the friction-reducing effect of a larger network. Crowd-in is stronger when $|\mu'(Z)|$ is large and outside firms respond more elastically (small $|G_j''|$).

Extensions. Appendix A studies two extensions. In a repeated relationship, the firm complies because deviating would end future access to the relationship package. In the binding-dynamic incentive compatibility, slack-participation regime, the dynamic extension gives $\tau_{is}^{\text{dyn}} = \beta \tau_{is}^{\text{static}} = \beta \Psi_Z$, where $\beta \in (0, 1)$ is the firm's discount factor. Relational enforcement attenuates the static covenant-enforceable wedge. The appendix also studies a first-best planner, clarifying which externalities the strategic financier does and does not internalize.

2.4 From model to tests

Figure 1 summarizes the mechanism. The model object is the induced gap between directed activity under the relationship and the firm's commercial counterfactual, $k_{is} - k_{is}^{\text{out}}$. This gap is positive only when the financier's strategic value and the feasible relationship surplus support a positive implementation wedge.

The main empirical tests do not observe the commercial counterfactual directly, so they use within-setting benchmarks that absorb generic capital provision. In private-market investments, I compare post-deal new-market entry by the same recipient into the SWF home country with entry into same-round non-SWF co-investor home countries. Commercial co-investors also provide capital and often share the headline round terms, so this comparison separates a SWF home-country relationship from generic financing. In the bank setting, pre-crisis write-downs provide

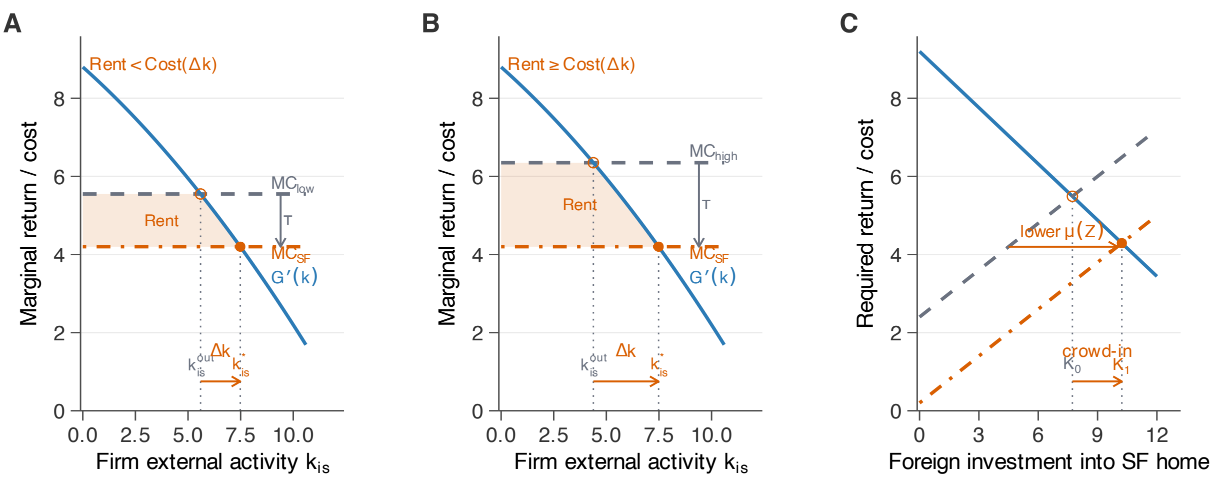

Figure 1: Geofinance: Directed Activity and Aggregate Spillover

Notes: In Panels A and B, the firm's marginal return on directed home-market activity is $G_i'(k_{is})$. The commercial benchmark is $\mathrm{MC} = \mu(Z) + r_i$, and the strategic financing relationship lowers the effective marginal cost to $\mathrm{MC}_{\mathrm{SF}} = \mu(Z) + r_i - \tau_{is}^{\mathrm{imp}}$. The wedge combines the financier's home-market relationship value with the relationship surplus needed to make the bundle acceptable. In the interior benchmark, $\tau_{is}^{*} = \Psi_Z$. Panel A illustrates a case in which the relationship package cannot support the desired directed activity, leaving the recipient at the commercial benchmark or scaling back the action. Panel B illustrates a positive wedge, $\tau_{is}^{\mathrm{imp}} > 0$, which generates $k_{is}^{*} > k_{is}^{\mathrm{out}}$. Panel C shows how the induced activity raises Z, lowers $\mu(Z)$, and expands third-party capital from K_0 to K_1.

sharper variation in recipient demand for outside capital, and I test whether high-need banks redirect lending toward SWF-status countries after the recapitalization window.

Proposition 2 then links these recipient-level responses to aggregate capital flows. If directed activity builds bilateral relationship stock Z, it should lower bilateral frictions and crowd in third-party capital. I study this implication using bilateral FDI inflows and BIS cross-border banking claims.

3 Directed Expansion in Private-Market Investments

The private-company evidence shows a conditional, not uniform, SWF-home-country tilt. In financing rounds with both SWF and non-SWF institutional investors, SWF participation alone does not produce a stable average new-entry gap toward the SWF home country. The gap rises instead among firms raising equity at larger valuation discounts relative to sector-year peers. It is strongest in sectors that SWFs had historically overweighted and where the SWF home country has historically low export strength. The setting therefore tests the model's two margins inside the same financing round: recipient outside-option weakness and the value of the home-country link.

3.1 Private-market investments data and institutional setting

Data and sample. I use PitchBook as the primary data source for private-company financing and later expansion transactions. PitchBook records financing rounds across venture capital, growth equity, buyouts, and M&A, and reports the investor composition of each round, round-level pricing, firm financial characteristics, and subsequent corporate expansion events.[11] These features are necessary for the mechanism test: the same data source identifies SWF participation, non-SWF co-investors in the same round, recipient valuations, and post-deal activity by investor home country.

In terms of data quality, external-source checks suggest that PitchBook is close to Capital IQ and Mergermarket on the margins most relevant for this section. The median disclosed PitchBook SWF-backed deal size and the major target geographies are similar across sources.[12] The sample should nevertheless be interpreted as cross-border SWF participation in private-company financing

[11] PitchBook is a widely used private-market data provider. It is the official data source for National Venture Capital Association, the main U.S. trade association for the venture capital industry.

[12] In the representativeness audit, PitchBook's broad cross-border SWF-backed sample has a median disclosed deal size of \$160 million, compared with \$150 million in Capital IQ and \$225 million in Mergermarket. PitchBook's broad cross-border SWF-backed sample is 29.6% United States targets, 7.0% China/Hong Kong, and 10.2% India; the corresponding shares are 34.7%, 8.7%, and 7.7% in Capital IQ, and 40.0%, 9.4%, and 11.6% in Mergermarket.

transactions, not as a census of all SWF investments. Broad SWF deal sources include domestic investments, real estate, infrastructure, public equity, and fund commitments, which are outside the mechanism tested here.

Directed-expansion outcome. I measure directed activity as post-deal expansion into an investor's home country. Expansion events are subsequent PitchBook-recorded transactions by the financing company in the investor's home country, including acquisitions, joint ventures, corporate transactions, and new private financing rounds. [13] I identify these events by linking financing companies to subsequent PitchBook transactions in each investor's home country.[14] The main outcome is new market entry. It equals one if the target had no recorded activity in the investor's home country before the focal deal but expanded there within five years after the deal.[15] This keeps the main outcome on the entry margin rather than mixing new entry with continuation in countries where the firm was already active.

I use this outcome to test whether post-deal expansion tilts toward SWF home countries relative to other investor home countries, and whether that tilt is strongest when the mechanism predicts it should be. The design therefore requires financing rounds with both an SWF and at least one non-SWF institutional co-investor, so that investor home countries can be compared within the same deal. The unit of observation is an investor–deal pair, and the main new-entry tests keep only investor-country rows where the firm had no recorded prior activity in that investor's home country. This no-prior-history sample contains 3,174 investor–deal pairs across 831 deals from 1998 to 2025.[16]

The sample restriction changes the population, but two facts make the mechanism sample informative. First, the retained mechanism deals are similar to the broader SWF-backed cross-

[13] The included PitchBook transaction types are M&A, buyouts, corporate asset purchases, joint ventures, PE growth/expansion, corporate transactions, seed, early-stage VC, later-stage VC, and private placement transactions.

[14] Appendix B.1 describes the exact normalized-name bridge used to link focal companies to PitchBook investor IDs. I do not use fuzzy matching, alias expansion, or manual predecessor-name stitching. As a result, the measure may miss expansion activity conducted under alternate legal names, subsidiaries, or affiliates that are not linked to the focal firm in PitchBook. This creates potential attenuation from missed true expansions rather than false positives from incorrect matches.

[15] For later financing rounds whose five-year post-deal window extends beyond the latest observed PitchBook expansion data, outcomes are measured through the end of the observed expansion window. I do not condition on firm survival or continued PitchBook coverage; the issue is only calendar observability. Because the regressions include deal-by-year fixed effects, implemented as deal fixed effects, all investor-country rows within a financing round share the same available post-deal window.

[16] In this sample, the typical deal has one SWF and roughly three institutional co-investors: the mean is 4.15 investor rows per deal and the median is 3; 36.6% of deals pair a single SWF with a single non-SWF investor; 89.6% involve only one SWF.

border sample on the main scale and geography margins.[17] Second, the restrictions are imposed for identification and pre-treatment measurement, not on the post-deal expansion outcome. The broad SWF-backed cross-border layer contains 4,667 deals. Requiring a same-round SWF and non-SWF institutional co-investor comparison leaves 2,362 deals, and requiring the valuation-discount measure leaves 833 deals. The final no-prior-history restriction changes the deal count only slightly, from 833 to 831, while removing investor-country rows where the firm had already operated in that country before the round. Thus the restriction narrows the population away from sole-SWF and otherwise non-comparable deals, but it is the restriction that makes the within-deal mechanism test possible. However, I do use the broader cross-border universe when describing SWF participation patterns. Appendix B.1 documents the data construction and cleaning steps.

Table 1 summarizes the sample layers and the role each layer plays in the argument. The broad cross-border universe is used to describe where SWFs participate. The SWF-backed cross-border layer is the external-validity benchmark for the mechanism sample and shows that retained mechanism deals are similar on scale and target geography. The mechanism sample is narrower because it requires the key ingredients for the within-round test: same-round SWF and non-SWF institutional co-investors, non-missing relative valuation, investor-country expansion outcomes, and no prior activity in the investor's home country.

Selected and continuing SWF relationships. SWF-backed private-company deals are selected along firm scale and sector-priority margins. SWF participation is concentrated in large, high-valuation firms that are not necessarily mature: it rises with post-money valuation and employees but falls with firm age. Conditional on valuation, age, employees, target geography, and deal type, deal size is smaller, so the pattern is not simply that SWFs join larger rounds. SWF participation is also higher in sectors that SWFs had previously overweighted in their cross-border portfolios. Appendix Table 17 reports deal-level linear probability models in which the outcome equals one if any cross-border SWF investor participates in the deal.

The relationships also have a continuation component. Appendix Table 18 shows that SWF investors are more likely than same-round non-SWF co-investors to appear again with the same company within five years.

I interpret these selection and continuation patterns as descriptive evidence for the framework's starting point: SWF relationships are selected, continuing relationships whose future value recipients may want to preserve. They do not imply that SWFs generally target distressed or low-valuation

[17] Median deal size is $189 million in the mechanism sample, compared with $160 million in the broader SWF-backed sample; median post-money valuation is $724 million versus $807 million; and the U.S. and advanced-economy target shares are close.

Table 1: SWFs Cross-Border Private Investment Samples

Panel A: Sample Support

	Cross-border universe	SWF-backed cross-border	Main mechanism sample
Unit	Deal	Deal	Investor–deal
Years	1995–2025	1995–2025	1998–2025
Observations	423,850 deals	4,667 deals	3,174 rows; 831 deals
SWF coverage	4,667 SWF-backed deals; 1.10%	61 SWFs; 32 homes	42 SWFs; 20 homes
Role in paper	SWF participation universe	SWF-backed source layer	Within-deal mechanism

Panel B: Deal and Firm Characteristics

	Cross-border universe	SWF-backed cross-border	Main mechanism sample
Median deal size	$7.0M	$160M	$189M
Median post-money valuation	$34M	$807M	$724M
Median revenue	$15M	$115M	$135M
Median firm age	6 yrs	12 yrs	8 yrs
Deal type mix: VC/growth; buyout/M&A; other	46.5%; 45.0%; 8.5%	34.3%; 49.6%; 16.1%	57.3%; 38.1%; 4.6%
Target HQ advanced economy	77.1%	71.8%	68.5%
Target HQ United States	22.8%	29.5%	32.9%

Note: Panel A reports the sample layers used in the SWF selection analysis. The cross-border universe includes completed PitchBook deals with known target home country, sector, and at least one foreign investor. The SWF-backed cross-border layer restricts this universe to deals with at least one cross-border SWF investor. The main mechanism sample further requires same-round SWF and non-SWF institutional co-investors, non-missing relative valuation, investor-country expansion outcomes, and no prior target activity in the investor's home country. Panel B reports deal-level medians and shares for these layers.

private firms in the broad cross-border universe. The mechanism test therefore asks a conditional question: inside selected SWF relationships, does directed activity follow the model's comparative statics in recipient outside options and home-country relationship value?

3.2 Empirical design and results

3.2.1 Within-deal mechanism

The core mechanism test moves within SWF-backed financing rounds. I compare expansion toward the SWF home country with expansion toward the home countries of non-SWF investors in the same financing round. This comparison holds fixed the recipient firm, financing round, timing, and common round-level terms. It asks whether the direction of post-deal activity loads specifically on sovereign investor identity, and whether the SWF-relative entry gap rises when recipient outside options are weaker.

The outside-option proxy is the valuation discount. For each deal d in sector s and year t, I construct it as the negative of the deal's log post-money valuation residual relative to the sector$\times$year mean:[18]

$$\text{ValDiscount}_d = -\left(\log(\text{PostVal}_d) - \overline{\log(\text{PostVal})}_{s(d),t(d)}\right), \tag{16}$$

where the benchmark is computed across all completed deals in sector s and year t from the full PitchBook deal universe.[19] Higher values indicate lower valuations relative to sector×year peers. I interpret this discount as a reduced-form proxy for weaker outside options, not as a direct measure of financing access. Lower relative valuation can reflect costly outside finance, higher perceived risk, less investor competition, or other outside-option premia. The prediction does not require high-discount firms to expand more in absolute terms; it requires the SWF-minus-non-SWF entry gap to rise with valuation discount.

I estimate

$$Y_{jdt} = \beta_1 \,\text{SWF}_{jdt} + \beta_2 \,(\text{SWF}_{jdt} \times \text{ValDiscount}_d) + \alpha_d + \alpha_{j\times t(d)} + \alpha_{j\times hq(d)} + \varepsilon_{jdt}, \tag{17}$$

where j indexes the investor's home country, d indexes the financing round, $t(d)$ is the deal year, and $hq(d)$ is the target firm's headquarters country. Y_{jdt} is new entry into country j for deal d, so the main estimating sample excludes investor-country rows with prior activity in that country. β_1 captures the within-deal SWF–non-SWF expansion differential at the mean valuation discount. β_2 captures whether that differential is larger in higher-discount deals. Deal fixed effects absorb the recipient firm, financing round, deal year, and all deal-level selection margins. Investor-home-country×deal-year fixed effects absorb time-varying differences in destination-country attractiveness, and the investor-home-country×target-HQ-country fixed effects absorb stable country-pair differences.[20] The identifying condition is narrow: conditional on the financing round, destination-year conditions, and stable country-pair differences, the SWF home country is not differentially associated with unobserved pre-planned entry relative to same-round co-investor countries in a way that varies with the recipient's outside option.

Outside-option driver. Table 2 reports the main within-deal estimates. Columns (1)–(3) first show no stable average SWF-country entry premium. Column (1) gives a raw SWF differential of -5.13 percentage points, consistent with destination composition: SWF home countries are often smaller and less common expansion destinations than the United States, the United King-

[18] I use post-money rather than pre-money valuation because PitchBook reports post-money for roughly twice as many completed deals (21.2% versus 11.0% of the universe). The two values are correlated at over 95%.

[19] The measure is winsorized at the 1st and 99th percentiles and standardized to unit standard deviation.

[20] The within-deal investor-level design follows the within-borrower design of Khwaja and Mian (2008), applied here to investor–deal pairs.

dom, and other major non-SWF co-investor homes. With deal fixed effects in Column (2), the differential is -2.50 percentage points. With investor-home-country×deal-year and investor-home-country×target-HQ-country fixed effects in Column (3), the estimate turns positive but remains statistically indistinguishable from zero.

This pattern is useful because non-SWF co-investors also provide capital. If generic post-financing expansion toward investor countries were the mechanism, expansion should tilt similarly toward SWF and non-SWF investor home countries. If sovereign capital generated a uniform home-country pull, the SWF coefficient should be stably positive. The absence of a stable average gap cuts against that simple mechanical story. The next columns therefore test the model's sharper prediction: whether the SWF-relative entry gap rises with recipient outside-option weakness.

The SWF-relative entry gap rises strongly with valuation discount. In the full specification, a one-standard-deviation increase in valuation discount raises the within-deal SWF–non-SWF new-entry gap by 5.46 percentage points, about 72% of the non-SWF baseline rate of 7.59%.[21] This coefficient is a differential slope: it does not imply that high-discount firms expand more in absolute terms, but that their entry tilts more toward the SWF home country relative to same-round non-SWF investor home countries. Columns (4)–(6) show that this interaction is positive without fixed effects, with deal fixed effects, and with the full destination-year and country-pair controls. I interpret valuation discount as the proxy for weaker recipient outside options. The result is the model's outside-option comparative static: the core prediction is not a positive unconditional SWF-country effect, but that the SWF-relative entry gap rises when the recipient's outside option is weaker.

Interpreting valuation discount. The valuation-discount result is not explained by observable scale, dilution, or round-pricing measures. Table 3 shows that several alternative proxies predict the SWF-relative entry gap when entered alone. This is expected: the financial-constraints literature often uses firm size and age as indirect proxies, and dilution can reflect the amount of ownership a firm sells when raising external equity (Hadlock and Pierce, 2010; Farre-Mensa and Ljungqvist, 2016). These variables are therefore related to outside-option weakness, but they do not exhaust it.

When horse-raced against valuation discount, the alternative proxies lose explanatory power. The SWF×valuation-discount slope remains positive when compared with up-round status, deal size scaled by revenue, firm employees, firm age, total dilution, and SWF-bloc dilution. I therefore interpret valuation discount as a reduced-form proxy for weak outside financing options: private-

[21] Out of 100 non-SWF investor–deal pairs in the no-prior-history sample, the target expands into that non-SWF investor's home country within five years in roughly 7.6 cases. The low baseline reflects the sparsity of bilateral cross-border expansion.

Table 2: Outside-Option Weakness and the Within-Deal SWF Entry Gap

	New Entry					
	(1)	(2)	(3)	(4)	(5)	(6)
SWF	-0.0513***	-0.0250***	0.0320	-0.0538***	-0.0295***	0.0086
	(0.0151)	(0.0095)	(0.0263)	(0.0152)	(0.0104)	(0.0277)
SWF × Val. Discount				0.0525***	0.0351***	0.0546***
				(0.0141)	(0.0114)	(0.0200)
Deal-by-year FE (deal FE)	–	✓	✓	–	✓	✓
Investor-home-country × deal-year FE	–	–	✓	–	–	✓
Investor-home-country × target-HQ-country FE	–	–	✓	–	–	✓
R^2	0.010	0.583	0.804	0.061	0.587	0.809
SE clustered by company	✓	✓	✓	✓	✓	✓
Control mean	0.0759	0.0759	0.0759	0.0759	0.0759	0.0759
Observations	3,174	3,174	3,174	3,174	3,174	3,174

Note: Investor–deal level regressions in the no-prior-history mechanism sample. Columns (1)–(3) estimate the average SWF–non-SWF new-entry differential under no fixed effects, deal-by-year fixed effects, and the full fixed-effect specification. Columns (4)–(6) add the SWF×valuation discount interaction under the same sequence of controls. Valuation discount is the standardized proxy for the firm's outside-option weakness, defined in Equation 16. Deal-by-year fixed effects are implemented as deal fixed effects because each deal occurs in one year. The full fixed-effect specification includes deal-by-year fixed effects, investor-home-country×deal-year fixed effects, and investor-home-country×target-HQ-country fixed effects. The new-entry outcome equals one if the target had no prior expansion in the investor's home country before the deal but did expand within five years after the deal, or by the end of the observed PitchBook expansion window for later deals. Control mean is the mean of new entry for non-SWF co-investors. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

round valuation is an observable price of external equity, so a low valuation relative to sector-year peers is a natural marker of costly outside finance, in the spirit of private-placement discounts (Hertzel and Smith, 1993). The table does not make valuation discount a structural measure of outside financing costs, but it shows that the slope contains information beyond observable scale, ownership dilution, and generic round pricing.

3.2.2 SWF-Country Relationship Value and the Driver–Channel Interaction

The outside-option result identifies when the directed-activity wedge bites. The model also predicts where it should bite most strongly: in sectors and destinations where the SWF-country relationship is more valuable. I measure that value first with portfolio tilts, defined as an SWF's cross-border portfolio share in a sector minus the corresponding share for non-government deep-pocket private investors.[22] Appendix Figure 5 shows that these tilts vary sharply across SWF home countries. Appendix Figure 6 then benchmarks the high-tilt cells against early export revealed comparative

[22] For Table 4, the tilt is computed at the individual-SWF level using only portfolio history before the focal deal year. The descriptive heatmap in Appendix Figure 5 reports pooled 2005–2025 home-country tilts. Both measures are count-based rather than value-weighted.

Table 3: Valuation Discount, Observable Pricing, Scale, and Control Proxies

	Pricing / Capital Need		Firm Characteristics		Ownership / Control	
	Up Round	DealSize/ Revenue	log Employees	log Firm Age	Total Dilution	SWF-bloc Dilution
Panel A: Alternative proxies and new entry						
SWF × proxy	-0.0232	0.0414	-0.0555**	0.0034	0.0419**	0.0277**
	(0.0351)	(0.0294)	(0.0223)	(0.0122)	(0.0164)	(0.0139)
Panel B: Horse race with valuation discount						
SWF × proxy	0.0012	0.0179	-0.0244	0.0026	0.0210	0.0086
	(0.0347)	(0.0223)	(0.0209)	(0.0124)	(0.0135)	(0.0113)
SWF × Val. Discount	0.0548***	0.1384***	0.0405**	0.0558***	0.0472**	0.0533***
	(0.0202)	(0.0410)	(0.0198)	(0.0208)	(0.0190)	(0.0194)
Outcome	New Entry	New Entry	New Entry	New Entry	New Entry	New Entry
Deal-by-year FE (deal FE)	✓	✓	✓	✓	✓	✓
Investor-home-country × deal-year FE	✓	✓	✓	✓	✓	✓
Investor-home-country × target-HQ-country FE	✓	✓	✓	✓	✓	✓
Cluster	Company	Company	Company	Company	Company	Company
Observations	3,174	1,407	2,849	2,950	3,123	3,123

Note: Investor–deal level regressions. The dependent variable is new entry into the investor's home country within five years, conditional on no prior activity there before the focal deal. Panel A reports proxy-alone specifications. Panel B horse-races the listed proxy against valuation discount in the same regression. All columns include deal-by-year FE, implemented as deal FE, investor-home-country × deal-year FE, and investor-home-country × target-HQ-country FE. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

advantage (RCA) in the mapped sector.[23] Most high-tilt cells have export RCA below one, so SWFs often overweight sectors in which the home country is not already an export specialist. This pattern points away from a mechanical comparative-advantage story and toward sectors where sovereign investors appear to value new foreign linkages.

Panel A of Table 4 asks whether these sector preferences explain away the valuation-discount result or instead sharpen it. They sharpen it. Across the sector-tilt columns, the SWF×valuation-discount coefficient remains positive after adding individual-SWF sector-tilt controls. The level terms are negative at the mean valuation discount, which means favored sectors are not simply places where SWF-backed firms enter more often unconditionally. The driver–channel interaction is the key object: in the interaction specification, the baseline SWF×valuation-discount slope is 5.81 percentage points; a 10 percentage-point higher prior SWF-investor tilt raises this slope by 1.65 percentage points, implying a 7.46 percentage-point slope at a +10 percentage-point tilt. This is economically meaningful. Because the signed model prediction is positive, the one-sided test is

[23]For country c and PitchBook sector s, RCA is $(X_{c,s}/X_c)/(X_{W,s}/X_W)$, where $X_{c,s}$ is the country's mapped goods-plus-services exports in sector s, X_c is its total goods-plus-services exports, and W denotes the corresponding world totals. RCA above one means the sector accounts for a larger share of the country's exports than of world exports. I use the geometric mean of annual RCA over 2005–2007. The appendix documents the non-overlapping goods-services mapping and source-level checks on world totals.

roughly at the 6% level; the two-sided test is about 12%. I therefore read the result as directional evidence that relationship value and recipient outside options reinforce each other, rather than as a standalone identification result.

Panel B connects the same mechanism to the SWF home country's trade position. The direct export-RCA interaction compares sectors in which the SWF home country has export RCA below one with sectors in which it has RCA above one. The baseline slope in export-specialized sectors is small, 1.78 percentage points, while the export-RCA-below-one differential is 5.69 percentage points. The implied slope in export-disadvantage sectors is therefore 7.47 percentage points, and the corresponding split-sample estimate is 8.10 percentage points. These are trade-position diagnostics and a simple placebo against mechanical comparative advantage: if SWF-country expansion merely followed sectors where the home country was already an export specialist, the slope should be concentrated in RCA-above-one cells. Instead, the slope is larger where the SWF home country lacks export strength in the sector. Together, the sector-tilt and trade-position patterns line up with the model's second margin: outside-option weakness matters most where the SWF-country relationship is plausibly more valuable.

3.2.3 Threats to the mechanism

The remaining checks are organized around the alternative stories they are meant to discipline. First, a Y-side concern is that firms may already have planned expansion into SWF countries before the financing round. Such selection would inflate the within-deal gradient if it were strongest among high-discount firms. The main table narrows this concern by estimating the mechanism on investor-country rows with no recorded prior activity in the investor's home country. This does not rule out hidden plans to enter a new market, but it removes cases in which the firm was already active in that country before the focal round. The destination-fixed matched counterfactual also holds the SWF destination fixed, and the appendix reports same-country restrictions, within-deal permutation inference, and leave-one-out diagnostics.

Second, an X-side concern is that valuation discount may proxy for firm scale, maturity, risk, ownership stakes, or generic round pricing rather than outside options. Table 3 addresses this concern directly. Up-round status, deal size scaled by revenue, firm employees, firm age, total dilution, and SWF-bloc dilution each capture related margins of pricing, scale, or control. Several point in the predicted direction when entered alone, but they lose explanatory power once valuation discount enters the same regression, while the SWF×valuation-discount coefficient remains positive. These checks suggest that the relative valuation measure contains information about outside-option weakness that is not exhausted by simple scale, control-share, or round-pricing

Table 4: SWF New Entry, Portfolio Tilts, and Sector Specialization

	Export Disadvantage			Prior SWF Sector Tilt		
	(1) Level	(2) + VD	(3) Interaction	(4) Level	(5) + VD	(6) Interaction
SWF × Strategic Value	-0.0213	-0.0371	-0.0455	-0.0066	-0.0113	-0.0154
	(0.0256)	(0.0324)	(0.0344)	(0.0119)	(0.0116)	(0.0126)
SWF × Val. Discount		0.0560***	0.0178		0.0563***	0.0581***
		(0.0199)	(0.0244)		(0.0202)	(0.0204)
SWF × Val. Discount × Strategic Value			0.0569*			0.0165
			(0.0328)			(0.0106)
Implied SWF × Val. Discount slope			0.0747***			0.0746***
			(0.0253)			(0.0274)
Outcome	New Entry	New Entry	New Entry	New Entry	New Entry	New Entry
Deal-by-year FE (deal FE)	Yes	Yes	Yes	Yes	Yes	Yes
Investor-home-country × deal-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor-home-country × target-HQ-country FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster	Company	Company	Company	Company	Company	Company
Observations	3,151	3,151	3,151	3,174	3,174	3,174

Notes: The outcome is new entry into the investor's home country within five years. Panel A uses observed investor–deal rows. Columns (1)–(2) add level controls for prior individual-SWF sector tilt and positive tilt; columns (3)–(4) add the corresponding valuation-discount interactions. Sector tilt is the individual SWF's prior cumulative PitchBook cross-border sector share minus the prior non-government deep-pocket benchmark share; one unit equals 10 percentage points. Panel B reports the export-specialization channel. Export RCA is the geometric mean of mapped goods-plus-services RCA over 2005–2007; RCA above one means the sector's export share exceeds the world benchmark. Column (2) reports the differential slope for export RCA below one, column (3) reports the implied slope for export RCA below one, and column (4) reports the split sample for export RCA below one. Fixed effects are shown in the table; deal-by-year fixed effects are implemented as deal fixed effects and absorb sector fixed effects. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

proxies.

Third, the result could come from the arbitrary assignment of SWF labels among investors in the same round. Appendix Table 20 addresses this label threat by randomly reassigning SWF labels among same-deal investors. Panel A applies this exercise to a simple statistic: the SWF–non-SWF difference in new-entry rates after scaling each group by its baseline country deal share. Panel B repeats the exercise for the main regression coefficient on SWF×valuation discount. In both cases, the observed statistic lies outside the placebo distribution from 1,000 within-deal reassignments, so the result is unlikely to come from random same-round investor labels.

Finally, the result could be concentrated in a single SWF country, sector, outcome window, or tail treatment rule. Appendix Figure 7 drops each SWF country in turn and re-estimates the main new-entry specification. The coefficient remains positive across all 20 SWF countries and stays close to the full-sample baseline. Appendix Table 19 shows that the new-entry coefficient remains positive under every sector omission, but it is most sensitive to omitting Consumer Products and Services (B2C).[24] Appendix Tables 21 and 22 show that the result is stable to alternative winsorization and remains positive under 3-, 5-, and 7-year outcome windows. The window estimates rise modestly with the horizon, from 4.79 percentage points at three years to 5.81 percentage points at seven years.

These checks do not turn the PitchBook setting into a natural experiment. In particular, they cannot fully rule out a perfectly tailored pre-planned entry story in which high-discount recipients accept SWF capital exactly when they were already about to enter that SWF's home country, despite having no recorded prior activity there. I next turn to crisis-era bank recapitalizations, where pre-crisis write-down exposure generates quasi-exogenous cross-bank variation in demand for sovereign capital.

4 Directed Lending after Bank Recapitalizations During the Crisis

The bank setting provides corroborating micro evidence on the same two margins: recipient need for sovereign capital and sovereign value from home-country financial links. During the 2007–2008 financial crisis, banks' pre-crisis need for outside capital varied sharply with write-down exposure

[24] PitchBook's primary industry sectors in the valuation-discount co-investment sample are Business Products and Services (B2B), Consumer Products and Services (B2C), Energy, Financial Services, Healthcare, Information Technology, and Materials and Resources. B2C refers to consumer-facing products and services, including retail, restaurants, leisure and hospitality, media, travel, personal services, and consumer goods.

on mortgage-backed and related structured securities, while recapitalizing SWF home countries were generally not export specialists in financial services. In a bank–country–quarter syndicated-loan panel built for 29 globally systemically important banks, I find that high-write-down banks increased syndicated-loan activity toward SWF home countries after 2007Q3, relative to banks with smaller write-downs.

4.1 Data and Empirical Design

Data. I assemble a 2005Q1–2008Q4 bank–country–quarter panel of cross-border syndicated lending from Refinitiv DealScan. Appendix B.2 details the panel construction, parent-bank assignment, and loan-term variables. The bank universe is externally defined by the 2011 Financial Stability Board list of global systemically important banks. The bank-level first-stage sample contains 30 banks, five of which received SWF equity during the 2007–2008 financial crisis. In the realized DealScan lending panel, Wachovia does not appear as a separate parent after its late-2008 acquisition by Wells Fargo, so the lending regressions use 29 bank parents: five treated and 24 controls.

Syndicated lending is sparse at the bank–country–quarter level, so I use two outcomes: an extensive-margin indicator for whether lending begins and an intensive-margin log deal count among cells with positive lending. I do not use allocated loan volume as an outcome because DealScan reports tranche commitments at the syndicate level; mapping dollar amounts to parent banks requires allocation rules, whereas bank participation and deal counts are directly observed. The realized panel contains 163 borrower destination countries, of which 14 are effective SWF-status destinations.

SWF equity injections. Five banks received equity injections from eight SWFs across seven SWF countries, for a total of ten events. I define 2007Q3–2008Q1 as the clean recapitalization window. Banks first reported large structured-credit write-downs in 2007Q3, and SWF equity injections began in 2007Q4. I end the treatment window in 2008Q1 to avoid contamination from later government recapitalization programs and late-crisis private placements.[25] This period also corresponds to the interval in which SWFs were a major source of external recapitalization, providing roughly 39% of all bank capital raised (International Monetary Fund, 2008). Table 5 lists the SWF injection events and the bank write-down ratios used in the exposure variable.

[25] The U.S. Troubled Asset Relief Program and the U.K. bank rescue scheme both began in 2008Q4. Once government equity became available, it became difficult to separate the lending response to SWF capital from the response to public recapitalization. Barclays is the main example: its late-2008 capital raise falls outside the clean window. I therefore keep Barclays in the control group rather than treat it as exposed, and the results are robust to dropping it.

Key regressors. I measure bank capital need with WriteDownRatio_b, defined as cumulative structured-credit write-downs in 2007Q3 and 2007Q4 divided by 2006 common equity.[26] Higher values indicate a larger crisis-era hit to bank capital.

SWFStatus_c indicates whether country c was home to an active cross-border SWF as of 2006. This is an ex ante destination-level treatment: it classifies countries by the pre-crisis presence of sovereign investors, not by whether a particular bank ultimately received capital from that country. The baseline set contains 15 countries: the seven countries that injected equity into treated banks plus eight additional countries with documented pre-2007 cross-border private-market activity.[27] Libya has no realized syndicated-loan observations in the DealScan panel, so the estimating sample contains 14 effective SWF-status destinations.

Because the bank outcome is syndicated lending, SWFStatus_c also has a strategic-value interpretation: it marks destinations where sovereign investors may value new financial-intermediation links. Among the effective SWF-status destinations with early financial-services export-RCA data, seven of eight have RCA below one, meaning that financial services accounted for a smaller share of their exports than of world exports.[28] I therefore interpret SWFStatus_c as a broad ex ante marker for sovereign-capital destinations where financial links may have strategic value, not as a direct measure of realized bank–SWF relationships.

Post_t equals one in quarters at or after 2007Q3. I use 2007Q3 as the cutoff because large structured-credit write-downs first became public in that quarter and recapitalization negotiations began before most SWF injections formally closed in 2007Q4 or 2008Q1. This dating captures the onset of the crisis shock without conditioning on realized bank–SWF match timing. The results are robust to using 2007Q4 or 2008Q1 instead.

Specification and identification. SWF equity receipt is endogenous at the bank level, so I estimate a reduced-form design that interacts bank write-down severity, destination-country SWF status, and the crisis cutoff. Write-downs reflect pre-crisis portfolio exposure and mark-to-market losses rather than country-specific lending choices, so WriteDownRatio_b is plausibly exogenous to the cross-country allocation of lending. SWFStatus_c is defined ex ante using countries with active pre-crisis SWFs. The design is therefore an intent-to-treat specification: it uses broad SWF-

[26] The Bloomberg write-down tracker compiles quarterly mark-to-market losses on subprime RMBS, CDOs, CMBS, leveraged loans, Alt-A securities, credit default swaps, and SIVs reported by financial institutions in their public filings.

[27] The seven injection countries are Singapore, the UAE, Kuwait, Qatar, Saudi Arabia, China, and South Korea. The eight additional countries are Bahrain, Australia, Oman, Malaysia, New Zealand, Brunei, Libya, and Algeria. Norway, Kazakhstan, and the United States are excluded from the baseline definition.

[28] Early financial-services export RCA is measured over 2005–2007 using the same goods-plus-services RCA construction as in the PitchBook section. The seven RCA-below-one countries with available data are Australia, China, Kuwait, Malaysia, New Zealand, Oman, and South Korea; Singapore is the RCA-above-one case.

destination exposure rather than only the ten realized bank–SWF-country injection pairs.[29] The estimating equation on the bank–country–quarter panel is

$$Y_{bct} = \beta\,(\text{WriteDownRatio}_b \times \text{SWFStatus}_c \times \text{Post}_t) + \sum_{\ell} \gamma_\ell \left(\text{WriteDownRatio}_b \times X^{\ell}_{bc} \times \text{Post}_t\right) + \alpha_{bc} + \alpha_{bt} + \alpha_{ct} + \varepsilon_{bct}. \tag{18}$$

where Y_{bct} is new entry or log deal count on cells with positive lending. I do not use allocated loan volume as an outcome because DealScan reports tranche commitments at the syndicate level; mapping dollar amounts to parent banks requires allocation rules, whereas bank participation and deal counts are directly observed. The log-count specification conditions on positive lending, so it describes intensive-margin changes among active bank–country–quarter cells; the new-entry outcome captures the extensive margin.

β asks whether, after 2007Q3, lending activity rises more toward SWF-status destinations for high-write-down banks than for low-write-down banks. $\mathbf{X}_{bc}$ is a vector of pair-level controls at their 2006 values, including the pre-crisis lending share, FTA, common language, and political distance, each interacted with WriteDownRatio_b and Post_t. Bank×country fixed effects absorb time-invariant bilateral relationships, bank×quarter fixed effects absorb bank-specific supply shocks, and country×quarter fixed effects absorb destination-specific demand shocks.[30] Appendix Figure 8 shows that WriteDownRatio is balanced on pre-crisis bank characteristics: none of the five standardized covariates is significant at the 5% level. The identification assumption is that, conditional on these fixed effects and controls, high-write-down banks would not otherwise have increased lending disproportionately toward SWF-status countries after 2007Q3.

4.2 Directed Lending

Validating the capital-need proxy. Table 6 shows that banks with larger crisis write-downs were more likely to receive SWF equity. In the 30-bank G-SIB cross-section, a one-standard-deviation increase in WriteDownRatio (SD ≈ 0.149) is associated with about a 31-percentage-point increase

[29] In the estimation window, the realized G-SIB panel contains 2,664 bank–country pairs, including 346 pairs involving SWF-status destinations. The full-controls specification uses 2,465 bank–country pairs and retains all 346 SWF-status pairs.

[30] Without country×quarter fixed effects, the coefficient on SWFStatus×Post is negative, reflecting the post-crisis contraction in cross-border bank lending (Cetorelli and Goldberg, 2011; De Haas and Van Horen, 2013). The full specification absorbs this aggregate pullback and isolates the relative increase toward SWF-status countries among high-write-down banks.

Table 5: SWF Equity Injections and Bank Write-Down Ratios

Bank	SWF investor	SWF country	Announcement quarter	Injection size ($bn)	Write-down ratio
Citi	ADIA	UAE	2007Q4	7.50	0.199
Merrill Lynch	Temasek	Singapore	2007Q4	4.40	0.703
Morgan Stanley	CIC	China	2007Q4	5.60	0.292
UBS	GIC	Singapore	2007Q4	9.70	0.435
UBS	Saudi	Saudi Arabia	2007Q4	1.75	0.435
Citi	GIC	Singapore	2008Q1	6.88	0.199
Citi	KIA	Kuwait	2008Q1	3.00	0.199
Credit Suisse	QIA	Qatar	2008Q1	3.00	0.255
Merrill Lynch	KIA	Kuwait	2008Q1	2.00	0.703
Merrill Lynch	KIC	South Korea	2008Q1	2.00	0.703

Note: Panel A reports SWF bank recapitalization events in the clean 2007Q3–2008Q1 window. Injection size is in USD billions. Panel B reports the write-down ratio used in the exposure variable, defined as cumulative 2007Q3–2007Q4 structured-credit write-downs divided by 2006 common equity. Full summary statistics are in Appendix D.

in the probability of receiving an SWF injection.[31] This pattern supports interpreting write-down severity as a bank-level proxy for demand for sovereign capital.

Table 6: Bank Write-Downs and SWF Recapitalization

	P(Injection) (1)	asinh(Total Amt.) (2)
WriteDownRatio	2.095**	5.830**
	(0.956)	(2.570)
Banks (total)	30	30
Treated banks	5	5
R^2	0.682	0.661
SE	HC3	HC3

Note: Cross-sectional bank-level OLS on the 30-bank G-SIB universe from the Financial Stability Board's 2011 list. The unit of observation is a bank. The outcome in Column (1) is an indicator for receiving any SWF equity injection. The outcome in Column (2) is asinh(total SWF injection amount, USD billions) across the clean 2007Q4–2008Q1 SWF injection window. WriteDownRatio is measured in raw units on a 0–1 scale. Standard errors are heteroskedasticity-robust HC3, used because clustering is uninformative in this small cross-section. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Main results. Table 7 shows that high-write-down banks increased lending activity toward SWF-status destinations after the crisis cutoff. The effect appears on both margins: banks became more likely to begin lending in SWF-status countries, and among positive-lending cells they increased deal frequency. This is the bank-setting analogue of the model's outside-option prediction. The

[31] As in any linear probability model, fitted probabilities can exceed the unit interval at extreme write-down values in this small sample. The 0.149 standard deviation is computed across the 30-bank bank-level universe; the corresponding realized-panel bank-level standard deviation is 0.152.

relevant object is not an unconditional shift toward SWF-status countries, but whether the SWF-status lending tilt rises with recipient need for sovereign capital, proxied by write-down severity.

The estimates are stable after accounting for pre-crisis lending orientation. Columns (2) and (5) add WriteDownRatio×Pre-share×Post, where Pre-share is the bank–country lending share before the cutoff. This addresses the concern that high-write-down banks were simply expanding in destinations where they were already active before the crisis. Columns (3) and (6) further add regional trade agreement (FTA), common-language, and political-distance controls, fixed at 2006 and interacted with WriteDownRatio and Post. In the full-control specifications, the new-entry coefficient is 0.028 and the log-count coefficient is 1.144. A one-standard-deviation increase in WriteDownRatio, 0.152 in the realized panel, implies about 0.17 log points, or roughly 19%, higher deal counts among positive-lending cells.[32]

Table 7: Bank Write-Down Ratios, SWF-Status Countries, and Syndicated Lending Activity

	New Entry			log(Count)		
	(1)	(2)	(3)	(4)	(5)	(6)
WD × SWFStatus × Post	0.037***	0.036***	0.028***	1.114***	1.010***	1.144***
	(0.007)	(0.007)	(0.008)	(0.308)	(0.330)	(0.381)
WD × Pre-share × Post	–	-0.027*	-0.024	–	-1.635**	-1.508*
		(0.013)	(0.048)		(0.771)	(0.824)
Add. controls (FTA, language, pol. dist.)	–	–	✓	–	–	✓
Bank×Country FE	✓	✓	✓	✓	✓	✓
Bank×Quarter FE	✓	✓	✓	✓	✓	✓
Country×Quarter FE	✓	✓	✓	✓	✓	✓
SE clustered by bank	✓	✓	✓	✓	✓	✓
Observations	37,296	37,296	34,510	10,761	10,761	10,010
R^2	0.260	0.260	0.268	0.809	0.809	0.813

Note: Bank–country–quarter regressions estimated within $[-8, +5]$ quarters of 2007Q3. The treatment variable is $\text{WriteDownRatio}_b \times \text{SWFStatus}_c \times \text{Post}_t$, where Post_t equals one from 2007Q3 onward. Columns progress within each outcome group: Columns (1) and (4) are baseline specifications; Columns (2) and (5) add $\text{WriteDownRatio}_b \times \text{Pre-share}_{bc} \times \text{Post}_t$, where Pre-share is the bank–country average quarterly lending share over the 12 quarters before 2007Q3; Columns (3) and (6) add $\text{WriteDownRatio}_b \times X_{bc} \times \text{Post}_t$ controls, where X_{bc} includes Pre-share, CEPII Gravity's FTA (regional-trade-agreement) indicator, CEPII Gravity's common-official-language indicator, and UN-voting ideal-point distance between the borrower country and the bank parent country, all fixed at pre-crisis values. The new-entry outcome in Columns (1)–(3) is an LPM equal to one in the first post-cutoff quarter with positive lending for bank–country pairs with zero lending in all eight pre-cutoff quarters in the estimation window. The log-count outcome in Columns (4)–(6) is log(deal count), estimated on cells with positive deal counts. All columns include bank×country, bank×quarter, and country×quarter fixed effects. SE clustered by bank. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[32] A placebo-destination exercise that draws 1,000 non-SWF country sets gives a right-tail randomization-inference p-value of 0.074 and a two-sided p-value of 0.138 for the full-controls log-count coefficient. I therefore read the bank evidence as directionally consistent but more finite-sample sensitive than the PitchBook co-investment evidence.

Dynamic evidence. Figure 2 reports the binned event study for log deal count using the baseline specification, with the quarter immediately before the crisis cutoff as the omitted bin.[33] Each coefficient is the event-time slope on $\text{WriteDownRatio}_b \times \text{SWFStatus}_c$ relative to the omitted quarter. The pre-period coefficients are statistically indistinguishable from zero and do not show a systematic upward trend before the cutoff. After the cutoff, the coefficients turn positive: 0.755 in the first post-cutoff bin, 1.655 in quarters 2–3, and 1.203 in quarters 4–5. The pattern points to a sustained post-cutoff increase in deal frequency toward SWF-status destinations among banks with larger write-down exposure, rather than a pre-existing upward trend.

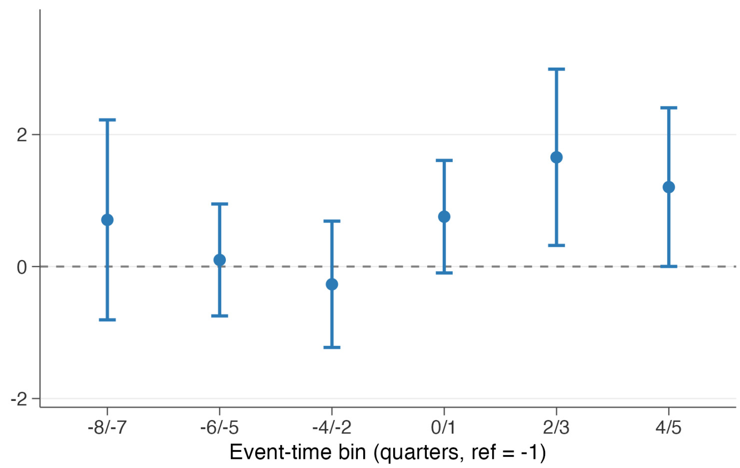

Figure 2: Binned Event Study: Deal Count
Notes: The figure plots coefficients on $\text{WriteDownRatio}_b \times \text{SWFStatus}_c$ by binned event time relative to 2007Q3. The outcome is log(deal count), restricted to bank–country–quarter cells with positive deal counts. The omitted reference bin is $t = -1$, the quarter immediately before 2007Q3. Error bars are 95% confidence intervals, with SE clustered by bank. The baseline specification includes bank×country, bank×quarter, and country×quarter fixed effects.

Threats and residual scope. The bank design leaves three main concerns: bank-side selection into write-down exposure, destination-side definition of SWF-status countries, and finite-sample concentration in a small set of crisis destinations. Appendix Figure 8 addresses the first concern by showing that write-down exposure is not systematically related to observable pre-crisis bank characteristics, including the pre-crisis share of lending to SWF-status countries. The pre-share controls further address the possibility that high-write-down banks were already more exposed to SWF-status destinations, while the bilateral control bundle allows the post-cutoff write-down gradient to vary with standard country-pair frictions. The event study does not show an upward

[33] I focus the main dynamic figure on log deal count because it is the most stable intensive-margin response. I aggregate individual quarters into two-quarter bins because quarterly bank–country lending is sparse, making quarter-by-quarter estimates noisy (Schmidheiny and Siegloch, 2023).

pre-trend in the log-count outcome. Appendix Table 26 addresses cutoff timing by moving the post date from 2007Q3 to 2007Q4 or 2008Q1 while holding the sample window fixed. The log-count response remains positive and significant under both later cutoffs, although the new-entry estimate is more timing-sensitive.

The SWF-status definition is addressed in two ways. First, SWFStatus_c is defined ex ante using pre-crisis sovereign-investor presence, not realized bank–SWF matches. Second, Appendix Table 25 replaces the baseline definition with the 20 PitchBook SWF home countries; the coefficients remain positive, with the strongest response again on the log-count margin. The timing check points in the same direction but is strongest for deal frequency.

The remaining concern is finite-sample concentration. Appendix Figure 9 drops each SWF-status destination in turn and re-estimates the full-controls log-count specification. The coefficient remains positive in every omission and remains statistically significant except when Australia is omitted. Because Australia is part of the ex ante SWF-destination set but is not one of the realized sovereign capital providers in the crisis recapitalization window, this sensitivity is best read as a precision issue rather than evidence that a single capital-provider country drives the result. Appendix Table 27 reinforces this caution: the actual log-count coefficient is positive, but the randomization-inference exercise gives a right-tail p-value of 0.074 and a two-sided p-value of 0.138. I therefore use the bank setting as corroborating evidence for the outside-option margin, not as a standalone estimate of the average effect of sovereign capital in banking.

The two micro settings point to the same two margins: recipient activity tilts toward SWF countries when recipients' outside financing options are weak and when the SWF home-country link has greater strategic value. Section 5 then asks whether these directed links aggregate into lower bilateral frictions by testing broader capital-flow responses in third-party FDI and BIS banking claims.

5 Bilateral Capital-Flow Responses

I now turn from recipient behavior to country-pair capital flows, asking whether directed SWF-backed activity is followed by broader capital-flow responses.

5.1 FDI Responses to FIRRMA

In 2018, the Foreign Investment Risk Review Modernization Act (FIRRMA) expanded CFIUS authority over foreign investments in sensitive U.S. businesses, raising regulatory risk for SWF investments in the United States. I find that SWF countries with greater pre-2018 high regulatory-risk exposure to FIRRMA subsequently attract more FDI from non-U.S. source countries, with the response concentrated among continuing bilateral relationships rather than new source-country entry.

Institutional setting. FIRRMA, enacted in August 2018, expanded the authority of the Committee on Foreign Investment in the United States (CFIUS) to review and restrict foreign acquisitions in sensitive sectors. For SWFs, the relevant margin was large-equity transactions in the United States. Appendix Figure 11 documents the corresponding non-U.S. deal margin: non-U.S. high-dilution SWF deal counts rise after 2018, including among countries with positive pre-FIRRMA U.S. high-dilution exposure. I use this pattern as institutional background rather than as a first-stage regression, because the post-2018 increase is not differential across exposure groups. The reduced-form test below therefore asks whether the countries most exposed to the U.S. shock subsequently attract more third-party FDI.

Exposure construction. The main exposure measure is constructed using the PitchBook data in Section 3 as the 2014–2017 share of a country's SWF co-investment deals that both targeted the United States and involved above-median dilution:

$$\text{Exposed}_i^{\text{HD}} = \frac{\#\{\text{US-targeted, high-dilution deals by SWFs from } i\}_{2014\text{–}2017}}{\#\{\text{deals by SWFs from } i \text{ with non-missing dilution}\}_{2014\text{–}2017}}. \tag{19}$$

High-dilution deals are those with dilution above the median in the SWF co-investment sample. This margin maps to the policy shock because larger equity stakes are more likely to face tighter CFIUS scrutiny. I use the 2014–2017 window because SWF co-investment is thin and volatile before 2014. Under this measure, 9 of the 23 non-U.S. PitchBook SWF home countries have positive exposure;[34] the remaining SWF destinations form the zero-exposure comparison group. Appendix Figure 12 plots the cross-country distribution of the high-dilution FIRRMA exposure measure.

[34]The exposure figure is constructed for the full non-U.S. PitchBook SWF-home-country universe. After merging to the WBG-HBFDI panel and imposing the control coverage used in the main regressions, the estimating sample contains 22 effective SWF-home destinations, as reported in Table 8.

Specification and identification. I estimate whether SWF countries with greater pre-FIRRMA U.S. exposure subsequently attract more third-party FDI from non-U.S. source countries:

$$\text{asinh}(\text{FDI}_{jit}) = \beta\,(\text{Exposed}_i \times \text{Post}_t) + \mathbf{X}'_{jit}\gamma + \alpha_{jt} + \alpha_{ji} + \varepsilon_{jit}, \tag{20}$$

where j indexes source countries, i destination countries, and t years. The outcome is bilateral FDI inflows from the World Bank Harmonized Bilateral FDI Database.[35] The coefficient β compares post-2018 changes in FDI inflows toward more- and less-exposed destinations. Source×year fixed effects absorb shocks to each source country's outward FDI, pair fixed effects absorb time-invariant bilateral affinity, and $\mathbf{X}_{jit}$ adds standard gravity controls like those in the bank setting.[36] Appendix Figure 10 shows that exposure is not strongly related to destination GDP, GDP growth, pre-2018 FDI growth, or the number of source partners. It is higher for SWF destinations with greater pre-2018 U.S. trade exposure and lower pre-2018 FDI inflow levels, so the main table reports specifications that interact the post-FIRRMA period with both U.S. trade share and standardized pre-2018 FDI level. The identifying assumption is that, absent FIRRMA, more-exposed SWF countries would not have experienced systematically different post-2018 changes in FDI from the same source countries.

Main results. Table 8 reports the main FDI estimates. The high-dilution exposure coefficient is positive and stable across specifications in the main sample. With the gravity controls (column 2), the estimate is 0.643. Column (3) adds the two balance controls motivated by Appendix Figure 10: the destination's pre-2018 U.S. trade share interacted with post-FIRRMA and its standardized pre-2018 FDI inflow level interacted with post-FIRRMA. The high-dilution coefficient remains positive at 0.663, while both balance-control interactions are small and statistically insignificant.

Columns (4)–(6) test whether the result depends on the source-country set or on the exposure definition. Column (4) restricts FDI source countries to the 47 non-U.S. countries that appear as targets of high-dilution SWF co-investment deals in PitchBook; the coefficient rises to 1.574. Columns (5)–(6) replace the baseline high-dilution exposure with two broader measures of pre-FIRRMA U.S. exposure: the U.S. share of all deal counts and the U.S. share of all deal values for each SWF country.[37] The estimates remain positive and significant. I use the event study mainly for

[35] Because bilateral FDI flows can be negative or zero, I use the inverse hyperbolic sine transformation.

[36] Controls include free trade agreement (FTA) membership, destination GDP level and growth, asinh(bilateral trade) and diplomatic distance measured by UN-voting ideal-point distance.

[37] The exposure is counted at the SWF-home-country-by-deal level: if two SWFs from the same home country join the same deal, that deal contributes once to the country's count-based and high-dilution exposure measures. The value-weighted exposure measure sums same-country SWF value within a deal before forming the country share.

timing and the decomposition table to separate entry, continuation, and intensive-margin responses behind the asinh outcome.

Table 8: FIRRMA Exposure and Third-Party FDI Inflows to SWF Home Countries

	High-Dilution Exposure				Count	Value
				Target Sources		
	(1)	(2)	(3)	(4)	(5)	(6)
SWF US Exposure × Post	0.560***	0.643***	0.663***	1.574***		
	(0.131)	(0.137)	(0.215)	(0.581)		
SWF US Exposure (count) × Post					0.395**	
					(0.163)	
SWF US Exposure (value) × Post						0.424***
						(0.145)
US Trade Share × Post			-0.109	-0.991	1.241	1.259
			(1.278)	(3.153)	(1.075)	(1.018)
Pre-FIRRMA FDI Level × Post			0.008	0.080	-0.053	-0.031
			(0.075)	(0.152)	(0.063)	(0.065)
asinh(Bilateral Trade)		-0.707**	-0.707**	-1.209**	-0.711**	-0.694**
		(0.297)	(0.302)	(0.606)	(0.302)	(0.302)
Diplomatic Distance		-0.248	-0.249	-0.448	-0.243	-0.231
		(0.210)	(0.211)	(0.505)	(0.211)	(0.210)
Other gravity controls	–	✓	✓	✓	✓	✓
Source × Year FE	✓	✓	✓	✓	✓	✓
Pair FE	✓	✓	✓	✓	✓	✓
SE clustered by pair	✓	✓	✓	✓	✓	✓
Wild-cluster p (destination)	0.007	0.024	0.015	0.056	0.168	0.092
Source countries	224	224	224	47	224	224
Destination countries	22	22	22	22	22	22
Observations	17,591	17,591	17,591	5,529	17,591	17,591
R^2	0.362	0.363	0.363	0.343	0.363	0.363

Note: Dependent variable: asinh(bilateral FDI inflow) from source j to destination i, World Bank Harmonized Bilateral FDI Database, 2014–2020; see Appendix B.3 for data construction. The estimation sample uses the full PitchBook SWF-home-country universe, excluding the United States as an FDI destination; the controlled regressions contain 22 effective destinations. Because bilateral flows include negatives and zeros, the inverse hyperbolic sine is used. Columns (1)–(3): high-dilution exposure (share of above-median-dilution pre-2018 SWF co-investment deals targeting the U.S.) with progressive controls; column (3) adds U.S. trade share × Post and standardized pre-FIRRMA FDI level × Post, where pre-FIRRMA FDI level is the destination's 2014–2017 mean asinh bilateral inflow, standardized across destinations. Columns (4)–(6) include those balance controls: column (4) restricts source countries to the 47 nations where SWFs made high-dilution non-U.S. deals, and columns (5)–(6) use count-based and value-weighted exposure with full controls. All columns include source×year and pair FE. Gravity controls: FTA, log(GDP), GDP growth, asinh(bilateral trade), diplomatic distance. Standard errors are clustered by country pair. The wild-cluster row reports restricted wild-cluster bootstrap p-values clustered by SWF destination country. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dynamic evidence. Figure 3 plots annual event-study coefficients from the balance-control specification in column (3) of Table 8. I use the figure as a timing check rather than as an estimate of the exact adjustment path, because annual FDI flows are volatile and the final post year, 2020, coincides with pandemic-era disruptions to global investment. The pre-FIRRMA leads are statistically indistinguishable from zero.[38] After FIRRMA, the coefficients are positive, with the largest estimate in 2020.

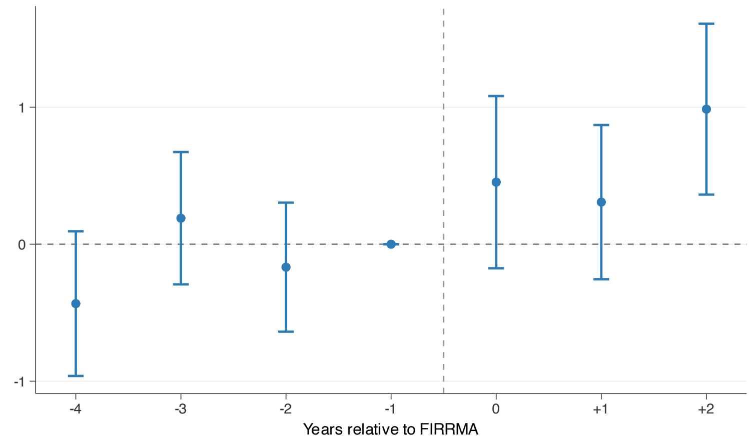

Figure 3: FIRRMA Event Study: FDI Inflows to SWF Home Countries
Notes: Annual event-study coefficients from the balance-control specification corresponding to column (3) of Table 8. The specification includes high-dilution exposure interacted with event time, U.S. trade share × Post, standardized pre-FIRRMA FDI level × Post, gravity controls, source×year FE, and pair FE. Sample: 22 effective non-U.S. SWF home destinations, 2014–2020. The omitted year is 2017. Error bars are 95% confidence intervals with standard errors clustered by country pair.

Margin decomposition. Because bilateral FDI flows can be negative, zero, or positive, the asinh specification mixes entry, continuation, and flow-size changes. Table 9 separates these margins following Haug, Nguyen and Owen (2023). The evidence points to stronger continuation among existing bilateral relationships rather than new relationships from previously inactive source countries. The entry margin does not increase for source–destination pairs without positive pre-FIRRMA inflow (column 1). In contrast, existing relationships strengthen on two margins. Conditional on positive current flows, the intensive-margin coefficient is positive, 0.480, but imprecisely estimated. Among pairs that were already active before FIRRMA, a one-unit increase in high-dilution exposure lowers the annual probability of falling to zero or negative flow by 15.8 percentage points. This pattern is consistent with the model's friction-reduction channel: exposed SWF destinations do not mainly add new source-country relationships, but existing bilateral investment relationships become larger and less likely to disappear.

The decomposition also gives a cleaner economic scale. At observed exposure shares, the conditional log-flow coefficient in column (2) implies increases of about 5.0 percent for Singapore

[38] The joint pre-period test does not reject under pair-clustered standard errors ($p = 0.120$) or destination-clustered standard errors ($p = 0.288$).

(exposure 0.105), 6.9 percent for China (exposure 0.143), and 8.0 percent for Australia (exposure 0.167), among source–destination pairs with positive current FDI flows. The exit-margin coefficient in column (3) implies that high-dilution exposure lowers the annual probability that an already-active pair falls to zero or negative inflows by 1.7 percentage points for Singapore, 2.3 percentage points for China, and 2.6 percentage points for Australia.

Table 9: FIRRMA Spillover: Entry / Intensive / Exit Decomposition

	(1) Entry	(2) Intensive	(3) Exit
SWF US Exposure (high-dilution) × Post	-0.028	0.480*	-0.158***
	(0.022)	(0.252)	(0.050)
SWF US Exposure (count) × Post	-0.043***	0.287*	-0.092**
	(0.014)	(0.166)	(0.036)
SWF US Exposure (value) × Post	-0.041***	0.194	-0.103***
	(0.013)	(0.132)	(0.028)
Outcome	$\mathbf{1}\{\text{flow} > 0\}$	log(flow)	$\mathbf{1}\{\text{flow} \leq 0\}$
Sample	Never-active pre	in_flow > 0	Ever-active pre
Gravity controls	✓	✓	✓
Balance controls	✓	✓	✓
Source × Year FE	✓	✓	✓
Pair FE	✓	✓	✓
SE clustered by pair	✓	✓	✓
Observations	6,683	7,056	10,908

Note: Each cell is a separate regression estimated on the SWF-home destination sample from Table 8. Pre-period is 2014–2017; the FIRRMA shock occurs in 2018. A pair is classified as *never-active pre* if its bilateral inflow was non-positive in every pre-period year, and *ever-active pre* otherwise. Column (1) estimates the *entry* margin: a linear probability model for $\mathbf{1}\{\text{in\_flow} > 0\}$ on the never-active-pre cohort, asking whether HD-exposed pairs are more likely to begin receiving positive bilateral inflows post-FIRRMA. Column (2) estimates the *intensive* margin: an OLS regression of log(flow) on all pair-year observations with positive current flow, regardless of pre-period cohort, asking whether bilateral flow levels rise conditional on remaining positive. Column (3) estimates the *exit* margin: a linear probability model for $\mathbf{1}\{\text{in\_flow} \leq 0\}$ on the ever-active-pre cohort, asking whether already-active pairs are less likely to drop to zero or negative flow. Thus, the entry and exit cohorts are disjoint, but the intensive-margin sample overlaps with both because it conditions on current positive flow. All specifications include the full set of gravity controls (FTA membership, log destination GDP, GDP growth, asinh bilateral trade, diplomatic distance), the two balance controls from Table 8, source×year and pair fixed effects, and standard errors clustered by country pair. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness. Appendix Table 28 reports the same specifications on a broader non-U.S. destination universe. It compares FDI inflows into the 9 positive-exposure SWF destinations with inflows into zero-exposure destinations in the full non-U.S. destination panel. The magnitudes of the coefficients fall by about half but remain significant. I present the SWF-home destination sample as my main specification on conceptual grounds: the geofinance mechanism predicts crowd-in toward SWF home countries specifically, not toward non-SWF destinations.

Appendix Table 29 reports timing and oil-exporter checks using the balance-control specification. Placebo shocks in 2015 and 2016, estimated entirely before FIRRMA, yield small and statistically insignificant coefficients. Dropping oil-exporter SWF countries keeps the estimate positive and statistically significant. Leave-one-positive-exposure-SWF-country-out estimates remain positive throughout (Appendix Figure 13). Dropping China leaves the coefficient essentially unchanged, so the result does not capture the contemporaneous U.S.–China trade war.

FIRRMA identifies crowd-in in bilateral FDI, but exposure varies only across SWF destination countries. I next use pair-level variation to ask whether a similar pattern appears in cross-border banking claims.

5.2 Banking Claims after SWF Equity Injections

Similar crowd-in appears in the banking sector. Using BIS Locational Banking Statistics in a matched stacked-cohort design centered on SWF equity injections into banks, I find that lender-country claims on SWF home countries rise after treatment, build over the first two post-treatment years, and remain elevated through year 5.

Data and treatment. The BIS Locational Banking Statistics report bilateral cross-border claim stocks of banks resident in a reporting country on counterparties in each borrower country. These stocks measure outstanding banking claims rather than new loan originations, so I interpret the estimates as changes in bilateral banking exposure. The data cover claims beyond syndicated lending, including direct loans, debt securities, and deposits, and span 1995–2024 for 31 reporting countries and 218 borrower countries. Appendix B.4 describes the data construction.

I identify SWF bank equity injections from the Global SWF (GSWF) and Sovereign Wealth Fund Institute (SWFI) transaction databases, retaining direct bank equity injections and excluding asset-management mandates, insurance investments, and fund-of-fund commitments. The raw treatment universe contains 107 deals across 55 lender–SWF-country pairs, totaling \$127 billion from 13 SWF countries. After merging to the LBS panel and applying data-availability restrictions, 12 treated pairs remain in the stacked-DiD design.[39] These treated pairs are concentrated among developed-country lenders and SWF-country borrowers.

Specification and design. The treatment variable is *SWF investment intensity*: the total SWF equity injected, in USD billions, into banks in lender country l by SWFs from borrower country b. For example, disclosed Singaporean SWF injections into U.S. banks total \$19.8 billion across three events, generating an intensity of 19.8 for the USA→Singapore pair. This is a pair-level aggregate

[39] Of the 55 treatment pairs, 48 have disclosed deal sizes; of those 48, 12 have lender countries that report to the BIS LBS over the treatment window and enter the matched stacked-DiD design.

treatment: if the same lender–borrower pair receives multiple disclosed injections, I sum them into a single total intensity and assign that pair to a cohort at its first injection year.[40] The estimand is therefore the aggregate change in lender-country banking claims associated with the overall SWF–bank relationship in the pair, not the effect of only the first observed transaction. Within each cohort, I match up to 20 control pairs from the same lender country on pre-treatment log-loan levels and growth rates using Mahalanobis distance.[41] I then stack the cohorts and estimate

$$\log(Y_{lbt}) = \beta \cdot (\text{Intensity}_{lb} \times \text{Post}_t) + \alpha_{g(lb)t} + \alpha_{g(lb),lb} + \varepsilon_{lbt}, \tag{21}$$

where Y_{lbt} is a positive bilateral claim stock, Intensity_{lb} is the pair-specific treatment intensity, and Post_t equals one for periods at or after the cohort's treatment date. Cohort×period fixed effects absorb time-varying shocks common within a cohort. Cohort×pair fixed effects absorb time-invariant bilateral heterogeneity. β estimates whether more intensely treated lender–borrower pairs experience larger post-treatment increases in bilateral claims.

Investment intensity is endogenous: SWFs may target lender countries with stronger pre-existing bilateral ties or better growth prospects. The key assumption is therefore parallel trends within each stacked cohort. Conditional on cohort×period and cohort×pair fixed effects, treated pairs would have followed the same path as their matched controls absent SWF bank injections. Two features make this assumption more plausible. First, Mahalanobis matching aligns controls with treated pairs on pre-treatment log-loan levels and growth rates. Second, treatment is staggered across twelve lender–borrower pairs over 2007–2018, which reduces the scope for any single common shock to explain the full pattern. Although the matched stacks include many control pairs, inference is still based on 12 treated cohorts, so I report wild-cluster-bootstrap p-values in the main table as a small-sample diagnostic.

Main results and magnitude. Table 10 reports the annual main specification using Q4 stocks. Cross-border banking claims from the lender country to the SWF home country rise after SWF bank equity injections. In column (1), total claims rise by 0.058 log points per \$1 billion of SWF equity. Column (2) shows a similar estimate at 0.057 for loans and placed deposits, and column (3) shows a similar estimate at 0.053 for non-bank loans. The similarity across columns suggests that the increase is concentrated in loan-and-deposit claims to SWF-country borrowers rather than

[40] This biases the implied per-dollar effect downward in early years for pairs with later additional events.

[41] Each treated cohort is matched to the $k = 20$ nearest controls within the same lender country by Mahalanobis distance on pre-treatment log-loan level and slope. Post-match absolute standardized differences range from 0.05 to 0.19, below the 0.25 benchmark commonly used to flag large covariate imbalance in matching diagnostics (Stuart, 2010); see Appendix Table 30.

in debt securities or interbank positions. Column (4) shows a positive and significant effect on winsorized standardized loan flows, although that margin is no longer precisely estimated under wild-cluster-bootstrap inference.

The economic magnitude is large. The loans estimate implies roughly 5.8% higher bilateral cross-border loans per additional $1 billion of SWF equity. This corresponds to about $345 million at the pre-treatment treated-pair median and about $600 million at the treated-pair mean. This is a stock interpretation: it describes the average post-treatment outstanding claims, not new loan originations or cumulative annual flows.[42]

Dynamic evidence. Figure 4 reports year-by-year event-study coefficients for total claims, loans and deposits, and non-bank loans. Across the three stock outcomes, the pre-treatment coefficients are centered near zero, which supports the parallel-trends assumption and argues against a simple selection story in which SWF-country claims were already rising before the equity injections. After treatment, the coefficients turn positive, build over the first two post-treatment years, and remain elevated through year 5. The similar timing across total claims and loan-based components suggests that the post-treatment increase reflects a broader rise in lender-country banking exposure to SWF home countries, not only a narrow claim category.

Cross-channel evidence. Table 11 asks whether the banking response is specific to SWF equity injections into banks. I re-estimate the stacked-cohort design around unlisted non-financial SWF investments, excluding pairs with SWF bank equity injections. The estimates are positive but much smaller than the direct bank-channel effects; only non-bank loans are statistically significant. This contrast is informative because a broad country-level information channel would predict similar banking responses after non-financial SWF investments. Instead, banking claims respond much more strongly when SWF capital enters banks directly, consistent with a relationship-friction channel within banking links.

Robustness. These checks address temporal aggregation, placebo outcomes, borrower concentration, and residence-based measurement. Appendix Table 31 re-estimates the design at quarterly frequency with cohort×quarter fixed effects. The coefficient remains close to the annual baseline, so the result is not an artifact of using Q4 stocks. Appendix Table 32 column (5) reports a liability-side test based on BIS liability positions; the estimate is positive but statistically indistinguishable from

[42]The scale calculation uses, for each treated lender–borrower pair, the pair's mean pre-treatment loans and placed deposits in the annual stacked sample. Across the 12 treated pairs, the median is $5.9 billion and the mean is $10.3 billion.

Table 10: Cross-Border Banking Claims Spillover

	(1) Total Claims	(2) Loans & Placed Dep.	(3) Non-Bank Loans	(4) Wins. Std. Flow Loans
SWF Invest ($bn) × Post	0.0581**	0.0566**	0.0530***	0.0252**
	(0.0195)	(0.0193)	(0.0151)	(0.0113)
Outcome transform	log	log	log	wins. flow/pre-stock
Observations	3,259	3,258	3,250	3,259
Cohort × Year FE	✓	✓	✓	✓
Cohort × Pair FE	✓	✓	✓	✓
Cluster	Cohort	Cohort	Cohort	Cohort

Note: Matched stacked-cohort DID using BIS Locational Banking Statistics (residence-based). Treatment intensity: total SWF equity injection in $bn into banks domiciled in the reporting country. When a lender-country × SWF-country pair receives multiple disclosed injections, those amounts are summed into a single pair-level intensity and the cohort is anchored at the pair's first injection date. Columns (1)–(3): log annual cross-border claim stocks (including loans, deposits and debt securities), using positive Q4 end-of-year stocks; column (2) restricts to loans & deposits; column (3) further restricts to non-bank counterparties. Column (4): reported annual LBS loan flow, summed from quarterly flows, divided by pre-treatment mean loan stock and winsorized at the 1st and 99th percentiles. Controls are Mahalanobis-matched on pre-treatment log-loan level and slope. Annual main specification uses a ±6-year window around treatment; quarterly dynamics are reported in the appendix. SE clustered by cohort, defined by the treated lender-country × SWF-country pair and its first injection year. Wild-cluster-bootstrap-t p-values clustered by cohort are 0.007, 0.009, 0.014, and 0.216 for columns (1)–(4), respectively. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

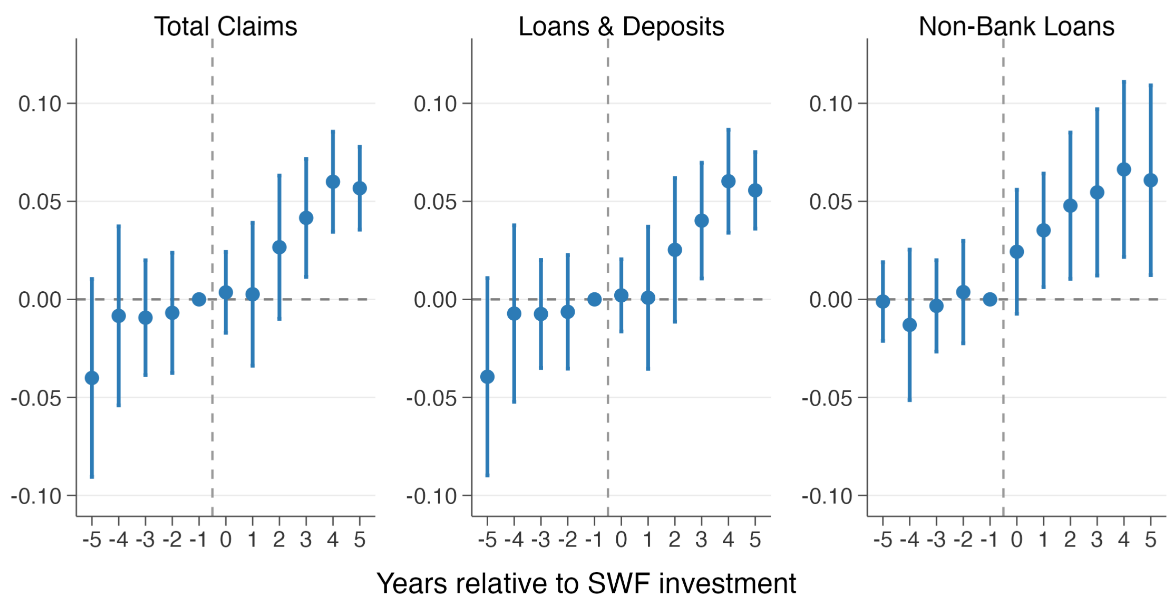

Figure 4: Cross-Border Claims: Annual Intensity-Weighted Event Study

Notes: Year-by-year stacked-cohort event study for BIS LBS cross-border claims. Panels report log total claims, log loans and deposits, and log non-bank loans. Treatment intensity: SWF equity injection ($bn). Reference year: $t-1$. Controls are Mahalanobis-matched using pre-treatment log-loan level and slope. Main annual design uses Q4 end-of-year stocks in a ±6-year window around treatment with cohort×year and cohort×pair FEs; SE clustered by cohort. Quarterly event-study dynamics are reported in the appendix figure.

Table 11: Cross-Channel Banking Claims Response from SWF Firm Investments

	(1) Total Claims	(2) Loans & Deposits	(3) Non-Bank Loans
Treatment intensity × Post	0.0128	0.0133	0.0135**
	(0.0091)	(0.0093)	(0.0064)
Observations	23,138	22,983	21,526
Cohort × Year FE	✓	✓	✓
Cohort × Pair FE	✓	✓	✓
Cluster	Cohort	Cohort	Cohort

Note: Stacked-cohort DiD built around firm-channel treatment events, defined as the first SWF investment in an unlisted non-financial firm in each lender–borrower pair over 2002–2022 (GSWF/SWFI source; finance and listed/public-market deals excluded). The sample is restricted to pairs with no SWF bank equity injection in the same country pair, so the firm-channel effect is estimated independently of the bank equity injections mechanism. Treatment intensity is total SWF firm investment in the pair (USD bn); it enters interacted with Post. Outcomes are log annual BIS LBS claims measured using positive Q4 end-of-year stocks: total claims, loans and deposits, and loans to non-bank counterparties. Treated pairs are matched within lender country to up to 20 controls on pre-treatment log loans and log-loan growth rates. Cohort×year and cohort×pair fixed effects; SE clustered by cohort. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

zero. The asymmetry between claim-side and liability-side responses indicates that the bilateral expansion is concentrated in cross-border lending toward the SWF country rather than having more deposits from the SWF country.

Figure 14 drops each treated lender–borrower cohort in turn and re-estimates the intensity-weighted specification for loans. The coefficient remains positive throughout, so no single cohort drives the result.

Appendix Table 32, columns (6)–(7), compares the LBS estimates with BIS Consolidated Banking Statistics (CBS), which assign claims by parent-bank nationality rather than office residence. The CBS estimates on total claims (0.043) and on international claims (0.060) are close to the LBS total-claims baseline (0.058), which suggests that the result is not solely explained by the residence-based reporting convention.

Taken together, the regulatory-shock and bank-injection exercises provide complementary evidence on the model's crowd-in implication. Both show increases of roughly 5–8 percent toward SWF home countries. The responses appear mainly on continuation or intensive margins, where lower bilateral frictions should matter most.

6 Conclusion

Sovereign capital does more than finance firms. In the framework and evidence in this paper, SWF investments create strategic financing relationships that can redirect recipient activity toward investor home countries. The redirection is conditional: it appears when recipients have weaker outside financing options and when the home-country link has greater strategic value. Those same links can matter beyond the direct recipient. By adding to a stock of bilateral relationships, directed activity can lower frictions for later investors, connecting a micro contracting margin to the geography of cross-border capital flows.

The evidence follows this logic across four settings. In private-market financing rounds, recipient firms tilt new entry toward SWF home countries relative to same-round non-SWF investor countries when outside options are weaker. In crisis-era bank recapitalizations, banks with greater capital need redirect syndicated lending toward SWF-status destinations. In aggregate data, SWF-linked relationships are followed by higher third-party FDI inflows and larger BIS banking claims toward SWF home countries. These estimates should not be read as a single treatment effect of sovereign capital. They are complementary tests of the same mechanism.

Three limitations define the scope of the evidence. First, the reduced-form estimates identify responses to observed SWF investments, but they do not quantify counterfactual changes in sovereign capital policy. Second, SWF transactions are large and infrequent, which limits treatment variation in each empirical setting. Third, the evidence focuses on SWFs' private investments, while their public-market portfolios remain outside the analysis. Whether passive public equity holdings can create similar relationship channels is therefore an open question.

The geofinance framework extends beyond sovereign wealth funds. It applies whenever a financier is willing to trade purely financial return, relationship services, or future access for directed activity. Development finance institutions, state-owned enterprises, and strategic corporate investors are natural settings in which the same logic may apply.

Appendix Overview

A Model Derivations and Extensions

A.1 Benchmark regime and proofs of the main results

The propositions in Section 2 use an interior benchmark for contracted recipients: funding, directed activity, outside-firm responses, and the contract rate are interior. The aggregate-capacity constraint may bind, with multiplier λ. The home-market value $A_i(r_i)$ is additively separable and independent of k_{is}, r_{is}, and Z, so it drops out of the derivative calculations below. This subsection derives the formulas; Appendix A.2 gives the KKT system and corner cases.

The Lagrangian is

$$\begin{aligned}\mathcal{L} = &\sum_{i\in C}\left[(r_{is}-\rho)x_i - \tfrac{c}{2}x_i^2\right] + \theta U(Z)\\ &+ \sum_{i\in C} y_i\left[\Pi_i(x_i, r_{is}, k_{is}, Z) - v_i(Z)\right]\\ &+ \lambda\left[\bar{X} - \sum_{i\in C} x_i\right]\\ &+ \Psi_Z\left[\sum_{i\in C} k_{is} + \sum_{j\notin C} k_{js}^*(Z) - Z\right].\end{aligned} \tag{22}$$

The inequality multipliers y_i and λ are nonnegative. The equality multiplier Ψ_Z is unrestricted in sign.

Contract rate r_{is}. From Equation (3), $\partial\Pi_i/\partial r_{is} = -x_i$. Stationarity gives

$$\frac{\partial\mathcal{L}}{\partial r_{is}} = x_i - y_i x_i = 0. \tag{23}$$

For any contracted recipient with $x_i > 0$ and an interior contract rate,

$$y_i = 1. \tag{24}$$

The participation constraint therefore binds, and the contract rate is pinned down by the reservation value.

Financing volume x_i. Using $\partial\Pi_i/\partial x_i = r_i - r_{is}$,

$$\frac{\partial\mathcal{L}}{\partial x_i} = (r_{is}-\rho) - cx_i + y_i(r_i - r_{is}) - \lambda = 0. \tag{25}$$

Substituting $y_i = 1$ yields

$$x_i^* = \frac{r_i - \rho - \lambda}{c}, \tag{26}$$

so the contract rate r_{is} drops out of the marginal funding condition.

External action k_{is} (Proposition 1). From the operating payoff in Equation (3),

$$\frac{\partial \Pi_i}{\partial k_{is}} = G_i'(k_{is}) - \mu(Z) - r_i. \tag{27}$$

The financier's first-order condition is

$$\frac{\partial \mathcal{L}}{\partial k_{is}} = y_i\left[G_i'(k_{is}) - \mu(Z) - r_i\right] + \Psi_Z = 0. \tag{28}$$

Using $y_i = 1$ and rearranging,

$$G_i'(k_{is}) = \mu(Z) + r_i - \Psi_Z. \tag{29}$$

Comparing this expression with Equation (7) gives

$$\tau_{is}^* = \Psi_Z. \tag{30}$$

Aggregate network stock Z (Proposition 2). Differentiating $\mathcal{L}$ with respect to Z gives

$$\frac{\partial \mathcal{L}}{\partial Z} = \theta U'(Z) + \sum_{i \in C} y_i \left(\frac{\partial \Pi_i}{\partial Z} - \frac{\partial v_i}{\partial Z}\right) + \Psi_Z \left[\sum_{j \notin C} \frac{\partial k_{js}^*(Z)}{\partial Z} - 1\right] = 0. \tag{31}$$

Maintaining that Z enters firm value only through the cost term $-\mu(Z)k$, and applying the envelope theorem to the outside option,

$$\frac{\partial \Pi_i}{\partial Z} - \frac{\partial v_i}{\partial Z} = -\mu'(Z)\left(k_{is} - k_{is}^{\text{out}}\right). \tag{32}$$

Total differentiation of the outside-firm condition

$$G_j'(k_{js}^*) = \mu(Z) + r_j$$

yields

$$G_j''(k_{js}^*)\frac{\partial k_{js}^*}{\partial Z} = \mu'(Z) \qquad \Longrightarrow \qquad \frac{\partial k_{js}^*}{\partial Z} = \frac{\mu'(Z)}{G_j''(k_{js}^*)}. \tag{33}$$

Substituting these expressions into the Z first-order condition and using $y_i = 1$ gives

$$\Psi_Z = \left[1 - \sum_{j \notin C} \frac{\mu'(Z)}{G''_j(k^*_{js})}\right]^{-1} \left[\theta U'(Z) - \mu'(Z) \sum_{i \in C} \left(k_{is} - k^{\text{out}}_{is}\right)\right]. \tag{34}$$

This is the expression used in Proposition 2. $\square$

Benchmark validity conditions. The closed-form expressions above characterize the benchmark candidate when the following conditions hold: (i) contracted recipients have $x^*_i > 0$, equivalently $r_i > \rho + \lambda$; (ii) directed activity is interior, with $G''_i(k_{is}) < 0$, so $\chi_i = 0$ and the solution to the k_{is} first-order condition satisfies $k_{is} > 0$; (iii) outside firms are on an interior differentiable response margin with $G''_j(k^*_{js}) < 0$; (iv) the aggregate feedback satisfies $0 \leq \sum_{j \notin C} \partial k^*_{js}(Z)/\partial Z < 1$; and (v) the contract rate is interior.

A.2 KKT conditions and corner regimes

To describe corner regimes, add the non-negativity constraints $x_i \geq 0$ and $k_{is} \geq 0$, with multipliers $\nu_i \geq 0$ and $\chi_i \geq 0$. Relative to Equation (22), define

$$\widetilde{\mathcal{L}} = \mathcal{L} + \sum_{i \in C} \nu_i x_i + \sum_{i \in C} \chi_i k_{is}.$$

KKT system. Primal feasibility is given by the constraints in Equation (11), together with $x_i \geq 0$ and $k_{is} \geq 0$. The inequality multipliers satisfy

$$y_i \geq 0, \qquad \lambda \geq 0, \qquad \nu_i \geq 0, \qquad \chi_i \geq 0, \qquad \Psi_Z \in \mathbb{R}.$$

Complementary slackness is

$$\begin{aligned} y_i \left[\Pi_i(x_i, r_{is}, k_{is}, Z) - v_i(Z)\right] &= 0, \\ \lambda \left(\bar{X} - \sum_{i \in C} x_i\right) &= 0, \\ \nu_i x_i &= 0, \qquad \chi_i k_{is} = 0. \end{aligned}$$

The stationarity conditions for the firm-specific controls are

$$\begin{aligned}
\frac{\partial \widetilde{\mathcal{L}}}{\partial r_{is}} &= x_i - y_i x_i = 0,\\
\frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} &= (r_{is} - \rho) - c x_i + y_i (r_i - r_{is}) - \lambda + \nu_i = 0,\\
\frac{\partial \widetilde{\mathcal{L}}}{\partial k_{is}} &= y_i \left[G_i'(k_{is}) - \mu(Z) - r_i\right] + \Psi_Z + \chi_i = 0.
\end{aligned}$$

The Z stationarity condition is unchanged except that y_i need not equal one:

$$\theta U'(Z) + \sum_{i \in C} y_i \left(\frac{\partial \Pi_i}{\partial Z} - \frac{\partial v_i}{\partial Z} \right) + \Psi_Z \left[\sum_{j \notin C} \frac{\partial k^*_{js}(Z)}{\partial Z} - 1 \right] = 0. \tag{35}$$

If outside firms are at an interior solution, their response is characterized by

$$\frac{\partial k^*_{js}}{\partial Z} = \frac{\mu'(Z)}{G_j''(k^*_{js})}.$$

[43] Under the maintained concavity and regularity assumptions, this system provides the local characterization used in the benchmark regime and in the corner cases below.

Contracted-recipient benchmark regime. If $x_i > 0$ and $k_{is} > 0$, complementary slackness implies $\nu_i = \chi_i = 0$. Provided the contract rate is interior, the contract-rate condition then pins down $y_i = 1$, and the benchmark formulas above follow immediately. The aggregate-capacity constraint may still bind. This is the regime used for the propositions in the main text.

Firms outside the contracted set. If a candidate contract is not activated, the firm is excluded from C and is treated as an outside firm in the main text. No strategic relationship is formed, no directed action is contracted, and the contract rate is economically irrelevant. The propositions focus on contracted recipients because the empirical predictions concern firms and banks that receive strategic financing relationships.

No induced action or corner deployment. Positive funding need not imply distorted real activity. If $\Psi_Z = 0$ and both the contract allocation and commercial benchmark are interior, the financier

[43] If an outside firm is at a boundary, the derivative is replaced by the relevant local one-sided derivative, or by zero when the boundary locally fixes k^*_{js}, as implied by that firm's own KKT conditions. The main benchmark assumes interior outside responses.

leaves the firm at $k_{is} = k_{is}^{\text{out}}$. If $\Psi_Z < 0$, the financier would prefer less external deployment than the commercial benchmark.

With no tied-financing constraint, $x_i > 0$ is compatible with the corner $k_{is} = 0$. Such a contract is funding without positive directed home-market activity, so it lies outside the directed-action benchmark.

Binding aggregate capacity. If $\lambda > 0$, aggregate funding capacity binds. A positive λ enters each contracted recipient's funding condition negatively, compressing funding all else equal:

$$x_i^* = \frac{r_i - \rho - \lambda}{c}.$$

The contract-rate cancellation survives because r_{is} still drops out of the funding first-order condition when $y_i = 1$.

A.3 First-best benchmark

The baseline model characterizes the strategic financier's contracting problem, not a welfare optimum. This subsection defines a first-best planner that directly chooses deployment and makes clear which externalities the strategic financier does and does not internalize.

Setup. A first-best planner chooses external deployment $k_{\ell s}$ for every firm $\ell \in \mathcal{I}$ and values the aggregate network stock at $W(Z)$, with $W' > 0$ and $W'' \leq 0$, separately from the friction-reduction channel $\mu(Z)$. As in the main model, ordinary home-market activity is pre-optimized into $A_\ell(r_\ell)$, which is constant in the external-deployment problem. The planner inherits the same relationship-surplus mechanism as the baseline model, so the comparison isolates the internalization of network externalities. Unlike the strategic financier, the planner problem does not separate contracted recipients from outside firms. The planner's external-deployment problem is

$$\max_{\{k_{\ell s}\}_{\ell \in \mathcal{I}}} \sum_{\ell \in \mathcal{I}} \Big[A_\ell(r_\ell) + G_\ell(k_{\ell s}) - \mu(Z) k_{\ell s} - r_\ell k_{\ell s} \Big] + W(Z), \tag{36}$$

where $Z = \sum_{\ell \in \mathcal{I}} k_{\ell s}$, subject to $k_{\ell s} \geq 0$ for all $\ell \in \mathcal{I}$.

First-best optimality. Let

$$Z^{FB} = \sum_{\ell \in \mathcal{I}} k_{\ell s}^{FB}.$$

Consider an interior planner allocation satisfying the relevant second-order conditions, with $k_{\ell s}^{FB} > 0$ and $G_\ell''(k_{\ell s}^{FB}) < 0$ for each firm ℓ. Because the planner directly controls every firm's directed activity, changing $k_{\ell s}$ changes Z one-for-one. The first-order condition internalizes how a higher Z lowers the bilateral-friction cost on aggregate directed home-market activity:

$$G_\ell'(k_{\ell s}^{FB}) - \mu(Z^{FB}) - r_\ell - \mu'(Z^{FB})Z^{FB} + W'(Z^{FB}) = 0. \tag{37}$$

Using the policy-wedge notation

$$G_\ell'(k_{\ell s}) = \mu(Z) + r_\ell - \tau_{\ell s},$$

the first-best wedge is

$$\tau_{\ell s}^{FB} = -\mu'(Z^{FB})\,Z^{FB} + W'(Z^{FB}). \tag{38}$$

At a given aggregate stock, the social wedge is common across firms. The planner applies it to all firms, while the strategic financier contracts only with firms in C. Since $\mu'(Z) < 0$, the friction-reduction component is positive when $Z^{FB} > 0$. A positive wedge lowers the effective marginal cost of directed home-market activity relative to the private benchmark.

Comparison with the strategic financier. The planner and the strategic financier differ in which payoffs enter the marginal choice. The strategic financier values the whole network stock through $\theta U(Z)$. It does not, however, maximize total firm surplus, so the friction-reduction term enters only through the participation constraints of contracted recipients. Ignoring outside-firm feedback for exposition, the strategic wedge is common across contracted recipients and, for any $i \in C$, is

$$\tau_{is}^* = \theta U'(Z) - \mu'(Z)\sum_{n\in C}\left(k_{ns} - k_{ns}^{\text{out}}\right), \qquad \tau_{is}^{FB} = W'(Z) - \mu'(Z)Z.$$

The sum over $n \in C$ appears because a marginal increase in Z changes the participation value of all contracted recipients, not only recipient i. These objects need not be ordered. The planner internalizes the social value of lower bilateral frictions for all firms; the strategic financier internalizes its own network payoff and the compensation needed to contract with C. Which wedge is larger depends on $\theta U'(Z)$ versus $W'(Z)$, the contracted set, outside-firm feedback, and the allocation being compared.[44] The first-best benchmark is therefore a diagnostic of which externalities the

[44] If a planner allocation is at a corner, the comparison is read through the corresponding KKT inequality rather than the interior first-order condition. A weak ranking can be obtained under restrictive conditions. If rate interiority holds, outside-firm feedback is shut down $\left(\sum_{j\notin C}\partial k_{js}^*(Z)/\partial Z = 0\right)$, $k_{is}^{\text{out}} \geq 0$, and $\theta U'(Z) \leq W'(Z)$, then $\sum_{i\in C}(k_{is} - k_{is}^{\text{out}}) \leq$

financier does and does not internalize, not a monotone bound.

A.4 Dynamic incentive compatibility and relational contracts

The baseline model assumes that the strategic financier can enforce the required external action k_{is} through legally binding covenants. Let

$$\tau_{is}^{\text{static}} \equiv \tau_{is}^{*} = \Psi_Z$$

denote the wedge under perfect covenant enforcement. This subsection keeps the contracted-recipient benchmark regime, but replaces legal enforcement with relational enforcement through the threat of termination. In the binding dynamic incentive compatibility (DIC) regime, the financier may leave rents to sustain compliance, so the static rent-extraction result $y_i = 1$ need not apply.

Environment. The relationship repeats indefinitely with common discount factor $\beta \in (0, 1)$. I study a stationary extension of the static contract: the same $\{x_i, r_{is}, k_{is}\}$ is offered each period, and Z is not separately accumulated over time. A deviating firm accepts the relationship package but chooses k_{is}^{out}. The deviation is detected after one period, and future access to the contract ends.

The dynamic incentive compatibility constraint. Let

$$B_i(k, Z) = G_i(k) - \mu(Z)k - r_i k$$

denote the part of operating payoff that depends on directed home-market activity after ordinary activity has been pre-optimized. The one-period deviation gain from ignoring the directed-action requirement is

$$D_i(k_{is}, Z) \equiv B_i(k_{is}^{\text{out}}, Z) - B_i(k_{is}, Z).$$

Since k_{is}^{out} is the private optimum at fixed Z, $D_i(k_{is}, Z) \geq 0$, with equality only when the mandated action coincides with the private optimum. This is the operating payoff the firm recovers by ignoring the directed-action requirement. Financing terms cancel in D_i because the firm receives the same financing rate whether it complies or deviates. Let

$$R_i = (r_i - r_{is})x_i$$

$\sum_{i \in C} k_{is} \leq Z$, so $\tau_{is}^{FB} \geq \tau_{is}^{*}$ at the common Z. Once these restrictions are relaxed, no global ranking follows.

be the per-period relationship rent. Stationary participation requires

$$R_i - D_i(k_{is}, Z) \geq 0, \qquad [y_i] \tag{39}$$

with multiplier $y_i \geq 0$. Compliance gives present value $(R_i - D_i(k_{is}, Z))/(1-\beta)$ relative to the outside option. A one-shot deviation gives R_i today and ends future surplus. The dynamic incentive compatibility constraint is therefore

$$\frac{R_i - D_i(k_{is}, Z)}{1-\beta} \geq R_i,$$

or equivalently

$$\beta R_i = \beta(r_i - r_{is})x_i \geq D_i(k_{is}, Z). \qquad [\xi_i] \tag{40}$$

Let $\xi_i \geq 0$ denote the DIC multiplier. When the DIC binds with $D_i(k_{is}, Z) > 0$, the financier leaves rents to preserve compliance: $R_i = D_i/\beta > D_i$. Participation is then slack, so complementary slackness gives $y_i = 0$. Static rent extraction would set $R_i = D_i$ and leave no continuation surplus, so costly directed action requires extra rent under relational enforcement. The relevant nontrivial relational regime is therefore the binding-DIC, slack-participation regime analyzed below.

If no admissible contract satisfies the DIC, induced action cannot be sustained and the financier must either relax the required action toward k_{is}^{out} or exit the relationship. The binding-DIC analysis assumes the required rent can be delivered through an interior rate choice; otherwise, the financier must relax the action or exit.

Interior wedge under binding DIC. Adding the stationary participation term $y_i[R_i - D_i(k_{is}, Z)]$ and the DIC term $\xi_i[\beta R_i - D_i(k_{is}, Z)]$ to the Lagrangian gives the rate condition

$$x_i - y_i x_i - \beta\xi_i x_i = 0, \tag{41}$$

so for a contracted recipient with $x_i > 0$,

$$y_i + \beta\xi_i = 1. \tag{42}$$

The x_i first-order condition is

$$(r_{is} - \rho) - cx_i + y_i(r_i - r_{is}) - \lambda + \beta\xi_i(r_i - r_{is}) = 0. \tag{43}$$

Using $y_i + \beta\xi_i = 1$, the contract rate again cancels and the funding formula is preserved:

$$x_i^* = \frac{r_i - \rho - \lambda}{c}. \tag{44}$$

Under the maintained interior-deployment assumption, so $\chi_i = 0$, the k_{is} condition is

$$y_i\left[G_i'(k_{is}) - \mu(Z) - r_i\right] + \Psi_Z - \xi_i \frac{\partial D_i(k_{is}, Z)}{\partial k_{is}} = 0. \tag{45}$$

Because

$$\frac{\partial D_i(k_{is}, Z)}{\partial k_{is}} = -\left[G_i'(k_{is}) - \mu(Z) - r_i\right] = \tau_{is},$$

this condition implies

$$\tau_{is}^{\text{dyn}} = \frac{\Psi_Z}{y_i + \xi_i}.$$

In the binding-DIC regime with positive deviation gain, participation is slack, so $y_i = 0$ and $\xi_i = 1/\beta$. Therefore the dynamic wedge is

$$\tau_{is}^{\text{dyn}} = \beta\Psi_Z = \beta\,\tau_{is}^{\text{static}}. \tag{46}$$

Thus relational enforcement scales the static wedge by β, holding the contracted set and static network multiplier Ψ_Z fixed. The extension does not re-solve the network fixed point under dynamic enforcement.

Proposition 3 (Local attenuation under relational contracts)**.** *In this reduced-form relational-contract extension with common discount factor β, binding DIC, and slack participation, compliance sustained by the threat of relationship termination rather than by legal covenants attenuates the implementability wedge according to*

$$\tau_{is}^{dyn} = \beta\,\tau_{is}^{static} = \beta\Psi_Z.$$

The same funding formula and contract-rate cancellation are preserved.

Empirical support for the relational-contract assumption. The relational-contract interpretation requires persistent ties. I treat the available evidence as supportive but not identifying. Supplemental persistence tables show repeated SWF participation in PitchBook funding histories and add-on SWF injections in the bank sample. These patterns indicate durable investor–recipient relationships, while leaving the contractual mechanism itself unobserved.

Connecting the extensions. When the benchmark wedge is non-negative, the dynamic extension attenuates it:

$$\tau_{is}^{\text{dyn}} \leq \tau_{is}^{\text{static}}$$

because $\beta \in (0, 1)$. The first-best planner provides a contrasting benchmark because it directly controls all firms and internalizes the full network effect. Table 12 summarizes these regimes. The table is not an additional empirical test; it clarifies which assumptions strengthen or weaken the model's directed-action prediction.

Table 12: Implementability Wedge Across Regimes

Regime	Wedge	Key determinant
First-best planner	$\tau_{\ell s}^{FB} = -\mu'(Z^{FB})Z^{FB} + W'(Z^{FB})$	Direct command over all firms
Static strategic financier	$\tau_{is}^{\text{static}} = \Psi_Z$	Covenant enforcement
Dynamic relational contract (binding DIC)	$\tau_{is}^{\text{dyn}} = \beta\Psi_Z$	Firm patience β
Purely commercial finance	$\tau_{\ell s} = 0$	No strategic objective or directed-action requirement

Note: The first-best wedge is evaluated at the first-best allocation. The first-best planner directly controls all firms; the strategic financier operates through contracts with a subset. The dynamic row reports the binding-DIC, slack-participation regime in an infinitely repeated relationship with discount factor β. The static and dynamic strategic-financier rows use the same static contracted set and static-equilibrium network multiplier Ψ_Z; the dynamic row applies the reduced-form β attenuation rather than re-solving a separate dynamic network fixed point.

B Data Construction

B.1 PitchBook Co-Investment Panel

This subsection documents the PitchBook panel behind Section 3. The construction choices are part of the empirical design: the sample keeps deals with both SWF and non-SWF investors, drops same-country comparisons that would mechanically favor the SWF, and uses an exact normalized-name bridge to PitchBook investor IDs to avoid fuzzy-name false positives.

Source tables. I link five PitchBook tables: *Deal* (2.5M transactions with deal date, size, post-money valuation, revenue, type, and status), *Investor* (400K profiles with type, headquarters country, and AUM), *DealInvestorRelation* (3.0M investor–deal participation records), *CompanyInvestorRelation* (2.4M company–investor relationship records), and *Company* (1.0M firms with headquarters country, sector, founding year, and financials). All monetary values are in USD millions.

SWF classification. I start from PitchBook's "Sovereign Wealth Fund" investor type, which includes 98 investors across 51 countries. I then exclude 28 entities that are not SWFs, such as state pension funds, regional development agencies, and charitable foundations, and add 7 government entities that function as de facto SWFs, including the Future Fund of Australia, the Hong Kong Monetary Authority, and the Monetary Authority of Singapore. The resulting classification is then filtered by the co-investment and valuation-discount regression sample; 20 SWF home countries contribute SWF investor–deal observations to the main regression sample. The replication files report the full investor list.

Co-investment identification. A deal enters the co-investment sample if it includes at least one SWF and at least one deep-pocket non-SWF investor, defined as PE/buyout, VC, asset manager, hedge fund, investment bank, university, family office, growth/expansion, government investor or mutual fund. Of 2.5M completed deals with valid dates, 6,789 satisfy this filter.

Panel filters.

1. *Investor join*: DealInvestorRelation and CompanyInvestorRelation are linked to Investor by the PitchBook investor identifier. Rows whose investor IDs do not appear in Investor are dropped rather than imputed, which is conservative but avoids assigning SWF status or investor home country using uncertain identifiers.

2. *Target-country assignment*: I use the company-level headquarters country when available. If it is missing, I fall back only to validated site-location tokens: exact country names are accepted, and exact US state names or postal abbreviations are mapped to the United States. Unvalidated strings remain missing.

3. *Same-country drop*: Remove investor–deal pairs where the target headquarters country equals the investor's home country, avoiding the mechanical confound that firms do more business at home.

4. *Same-as-SWF drop*: Remove non-SWF investors from the same country as the deal's SWF, since these cannot contribute clean cross-country within-deal variation.

5. *Both-arm requirement*: After each filter, verify that every deal still contains at least one SWF and one non-SWF row; drop deals that lose an arm.

6. *Contamination filter*: For multi-round companies, drop later-round rows where the *same investor* (by ID) appeared in an earlier round, to avoid double-counting repeat relationships. Later-round rows with new investors are retained.

7. *Deduplication*: Remove exact duplicate investor–deal rows inherited from duplicates in the raw DealInvestorRelation table.

The final panel contains 9,646 investor–deal pairs across 2,362 deals and 2,308 companies, with 2,677 SWF rows and 6,969 non-SWF rows. Rows with unresolved target country are dropped before the cross-border filters because they cannot establish whether the investor and target are foreign to one another. Most of the attrition from the 6,789 candidate co-investment deals to the final 2,362 deals comes from enforcing clean cross-country comparisons, requiring both investor arms after those exclusions, and dropping contaminated later rounds with repeat investors. The regression samples are smaller because the financing proxies impose additional non-missingness restrictions: the parent valuation-discount sample contains 3,456 investor–deal pairs across 833 deals, the common valuation-discount-plus-dilution sample contains 3,403 pairs across 821 deals, and revenue-scaled proxy diagnostics use the smaller revenue-available subsample. The main new-entry mechanism sample then excludes investor-country rows with recorded prior activity in that country, leaving 3,174 investor–deal pairs across 831 deals.

Outcome construction. I identify expansion events through two relationship routes after linking focal companies to PitchBook investor IDs by exact normalized name: lower-cased, trimmed company name must equal lower-cased, trimmed investor name. In route 1, the panel company appears as a new investor, acquirer, follow-on investor, or add-on sponsor in PitchBook's DealInvestorRelation table for expansion-type deals, including M&A, buyout/LBO, corporate asset purchase, joint venture, PE growth/expansion, corporate, seed, early-stage VC, later-stage VC, and PIPE. In route 2, the panel company appears in CompanyInvestorRelation with an active, former, acquirer, or add-on sponsor relationship. I combine the two routes and deduplicate them on company, target country, and date, which yields 12,221 unique expansion events. Focal IPO and PIPE rounds are excluded from the co-investment panel; PIPE can enter only as a later expansion event. I do not use fuzzy matching, alias expansion, or manual predecessor-name stitching, so any resulting measurement error should attenuate the estimates by missing true expansions rather than inflate them through false matches.

For each investor–deal pair, I construct two outcomes:

- *New entry* (Y_{ext}): equals 1 if the target expanded into the investor's home country within 5 years after the deal, or through the end of the observed expansion window for later deals, and had no prior expansion activity in that country before the deal. The main mechanism regressions are estimated on investor-country rows with no recorded prior activity in that country; rows with pre-deal activity are retained in the broader valuation sample for intensive-margin and prior-orientation diagnostics rather than coded as new-entry controls.

- *Count* (asinh(Count)): inverse hyperbolic sine of the number of expansion events in the investor's home country over the same observed post-deal window.

Valuation discount. For each deal d in industry sector s at year t:

$$\text{ValDiscount}_d = -\Big(\log(\text{PostVal}_d) - \overline{\log(\text{PostVal})}_{s(d),t(d)} \Big),$$

where the benchmark $\overline{\log(\text{PostVal})}_{s,t}$ is computed from the full PitchBook deal universe, not just the co-investment panel. The sign flip ensures that higher values correspond to lower valuations relative to peers, which I interpret as evidence of weaker outside options, that is, higher ω_i. I winsorize the variable at the 1st and 99th percentiles and standardize it to unit SD. I construct dilution (= dealsize/postvaluation) analogously, winsorize it at the 99th percentile, and standardize it.

Revenue-scaled round measures. Some proxy diagnostics use revenue-scaled round measures. Revenue in PitchBook is deal-specific rather than taken from the static Company.dta snapshot. Some pre-revenue firms, mainly in biotech, mining exploration, and early-stage technology deals, report zero or negative revenue. I exclude those deals from log revenue-scaled calculations and propagate the resulting missing values through the relevant regression samples. This does not affect the main valuation-discount specification.

Pre-deal geographic orientation. For each investor–deal pair, I compute the target company's pre-deal expansion share toward the investor's home country: the number of expansion events in that country during the three years before the deal divided by the total number of expansion events across all countries in the same window. This variable is used in diagnostics on the broader valuation sample. In the main no-prior-history mechanism sample, investor-country rows with recorded pre-deal activity in the investor's home country are excluded.

Table 13 summarizes the final investor–deal panel used in the main PitchBook regressions. It connects the construction steps above to the paper's within-deal design by showing the outcome rates, SWF and non-SWF row counts, and the valuation-discount and dilution measures used in the mechanism checks.

Table 13: PitchBook Co-Investment Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (investor–deal level)						
New entry (5yr post-deal)	3,174	0.06	0.24	0.00	0.00	0.00
Expansion count (5yr)	3,174	0.12	0.72	0.00	0.00	0.00
asinh(expansion count)	3,174	0.08	0.34	0.00	0.00	0.00
SWF investor	3,174	0.29	0.46	0.00	0.00	1.00
Valuation discount (std.)	3,174	0.08	0.97	-0.62	0.02	0.71
Dilution (std.)	3,123	0.03	0.99	-0.86	-0.41	1.39
Deal size ($M)	3,123	906.31	2581.83	60.00	180.00	574.90
Pre-acq. geographic share	3,174	0.00	0.00	0.00	0.00	0.00
Baseline (country deal share)	3,174	0.13	0.18	0.01	0.02	0.44
Panel B: Panel Dimensions						
Investor–deal pairs (regression obs.)	3,174					
Deals	831					
Target companies	800					
SWF investor–deal pairs	935 (29.5%)					
Unique SWF investors	42					
SWF home countries	20					
Non-SWF investor–deal pairs	2,239					
SWF Investment Breadth (per SWF investor):						
Deals per SWF	22 (median 5, range 1–243)					
Target countries per SWF	5.9 (median 3)					
Sectors per SWF	3.4 (median 3)					
Panel C: Broader Clean Co-Investment Panel						
Investor–deal pairs	9,646					
Deals	2,362					
Target companies	2,308					
SWF investor–deal pairs	2,677					
Non-SWF investor–deal pairs	6,969					
SWF investors	48					
SWF home countries with any row	24					

Notes: Panel A reports investor–deal level variables used in the main no-prior-history mechanism regressions: new entry and expansion-count outcomes within 5 years of the deal, or through the end of the observed PitchBook expansion window for later deals, the SWF-investor indicator, the standardized valuation discount and dilution measures, deal size in USD millions, and baseline destination activity measures. Panel B reports dimensions for the main no-prior-history mechanism sample with non-missing valuation discount. Panel C reports the broader cleaned co-investment panel before imposing the valuation-discount restriction, with 9,646 investor–deal pairs, 2,362 deals, and 2,308 target companies. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 DealScan Syndicated Lending Panel

This subsection documents the construction of the bank–country–quarter syndicated lending panel from Refinitiv DealScan.

Raw data. The source file is the DealScan general-syndicate extract, which contains lender–tranche-level records for syndicated loan facilities worldwide. Each row represents one lender's participation in one tranche. I retain the following fields: lender parent name and operating country, borrower country, deal and tranche identifiers, deal active date, tranche type, tranche amount (USD converted), lender share (USD), number of lenders, primary role, loan purpose fields, tranche maturity dates, tenor, average life, all-in spread drawn (bps), margin (bps), and fee variables (upfront, commitment, and annual fees in bps).

Lender parent assignment. DealScan assigns each lender to a parent entity. I apply one override: any row whose lender name contains "Merrill Lynch" (case-insensitive) is reassigned to the parent "Merrill Lynch," regardless of DealScan's current parent mapping. This preserves Merrill Lynch as a distinct treated unit in the pre-2008 period rather than merging Merrill-branded subsidiaries into post-acquisition parents such as BofA Securities.

Bank–country lending shares. The main outcomes do not use allocated loan volume. DealScan records lender participation directly, so new entry and deal count can be constructed from observed bank–tranche links. I use DealScan commitment amounts only to construct the pre-crisis bank–country lending-share control. For that control, I use reported lender shares when available and otherwise follow the standard lead-arranger allocation convention in DealScan (Sufi, 2007; Ivashina and Scharfstein, 2010). Because this allocation rule is needed only for the pre-share control, not for the outcomes, the main bank results are not driven by allocated loan-volume measurement.

Aggregation to bank–country–quarter. Deal counts are summed within each (lender parent, borrower country, year-quarter) cell to produce the *book panel*. A *balanced panel* is then constructed by cross-joining all observed (bank, country) pairs with all observed quarters, filling missing cells with zero deal count.

Summary statistics. Table 14 documents the bank–country–quarter panel used in the crisis design. It reports the outcome variables, the ex ante SWF-status indicator, the bank-level write-down ratio, and the pair-level controls used in the fully controlled specification. The table connects the appendix construction to Table 7: the balanced panel contains many zero cells, so the paper reports new entry separately and estimates log count on the positive-lending subsample.

Table 14: DealScan Bank–Country–Quarter Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (bank–country–quarter level)						
Deal count	37,296	1.90	8.44	0.00	0.00	1.00
log(deal count), positive cells	10,761	1.148	1.058	0.000	1.099	1.792
New entry (LPM)	37,296	0.005	0.073	0.000	0.000	0.000
Lending share (within-bank quarter)	37,296	0.011	0.061	0.000	0.000	0.001
Write-down ratio (bank)	37,296	0.096	0.133	0.022	0.065	0.092
SWF status (country)	37,296	0.130	0.336	0.000	0.000	0.000
Post ($\mathbf{1}[t \geq$ 2007Q3$]$)	37,296	0.429	0.495	0.000	0.000	1.000
WD × SWFStatus	37,296	0.013	0.062	0.000	0.000	0.000
WD × Pre-share	37,296	0.0012	0.0102	0.0000	0.0000	0.0001
WD × FTA	37,254	0.022	0.068	0.000	0.000	0.000
WD × Common language	36,960	0.018	0.073	0.000	0.000	0.000
WD × Political distance	34,804	0.164	0.324	0.012	0.060	0.165
Panel B: Panel Dimensions						
Bank–country–quarter observations	37,296					
Banks	29 (5 treated, 24 control)					
Borrower countries	163 total (14 SWF-status)					
Quarters in window	14 ($[-8, +5]$ around 2007Q3)					

Notes: Panel A reports bank–country–quarter level variables used in the main continuous DiD (Table 7). Outcomes are deal count, log deal count on positive cells, and the new-entry LPM. The bank-level write-down ratio and country-level SWF-status indicator define the treatment interaction WD × SWFStatus, and the additional pair-level controls (Pre-share, FTA, common language, political distance) enter the fully saturated specification interacted with Post. Panel B reports panel dimensions: 29 realized G-SIB bank parents (5 treated, 24 control), 163 borrower countries in total (14 SWF-status), and the $[-8, +5]$ quarter window around 2007Q3, that is, eight pre-treatment quarters and six quarters from 2007Q3 through 2008Q4. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Bilateral FDI and Gravity Controls

The bilateral FDI data used in Sections 5.1 and related spillover tests are drawn from the World Bank Harmonized Bilateral FDI Database (Steenbergen et al., 2022). This subsection documents the cleaning steps applied to the raw database before estimation.

Source. The WBG-HBFDI database harmonizes four underlying sources: OECD Bilateral FDI Statistics (BMD4 and historical BMD3 series, 2005–2019), the IMF Coordinated Direct Investment Survey (CDIS, 2009–2019), UNCTAD Bilateral FDI Statistics (2001–2012), and China's official FDI bulletins (2003–2019). The database covers 247 reporting economies and 247 partner economies over 2001–2021, with four variables per pair-year: inward/outward flows and inward/outward stocks, all in USD millions. I use inward flows as the outcome variable: for a reporting economy i (destination) and partner economy j (source), the inward-flow series $\text{FDI}^{\text{in}}_{ji,t}$ represents FDI flowing from j into i in year t.

Source deduplication. The harmonized database is constructed through a sequential augmentation process, where each source fills gaps left by higher-priority sources (Table 6 of Steenbergen et al. 2022). The priority hierarchy for inward flows is: (1) OECD-BMD4 from the host (reporting) economy; (2) OECD-BMD4 mirror from the partner economy; (3) China source; (4) UNCTAD; (5) IMF-CDIS. Despite this sequential design, the 2022 vintage of the database contains approximately 39,500 duplicate pair-year observations, concentrated entirely in 2018–2019 where OECD and CDIS coverage overlaps. Of these, roughly 7,500 pairs have conflicting values from different sources (e.g., OECD-BMD4 reports +532 USD million while CDIS reports −108,582 USD million for the same pair-year). I deduplicate by retaining the highest-priority source for each pair-year, breaking ties by preferring the reporter economy ("r") over the partner economy ("p") as the data origin. This reduces the raw database from 588,058 to 548,487 observations.

Estimation window and gravity controls. The cleaned WBG-HBFDI database covers 2001–2021, but the FIRRMA regressions use the 2014–2020 panel after imposing the event window around the 2018 shock and merging the required controls. Bilateral gravity variables are drawn from CEPII Gravity V2022 (Head and Mayer, 2014): FTA membership, destination GDP (used in log current-USD form), bilateral trade flows (Comtrade), and GDP growth. Diplomatic distance is measured as ideal-point distance from the UN General Assembly voting database (Bailey, Strezhnev and Voeten, 2017). CEPII trade coverage is the binding control and is available through 2020, so the main panel ends in 2020. The 2014 GDP growth observation uses the 2013–2014 lag. Pairs missing gravity controls are dropped from the estimation sample.

Winsorization. Bilateral FDI flows contain extreme values driven by financial-center conduit transactions, such as Luxembourg→Ireland flows above \$100 billion in a single year. To limit the influence of those outliers without dropping observations, I winsorize inward flows at the 1st and 99th percentiles before applying the asinh transformation. I do this within each estimation window, including the main 2014–2020 panel and each placebo window.

Negative and zero flows. Bilateral FDI flows can be negative when disinvestment, profit repatriation, or intercompany loan repayment exceeds new investment in a given year. In the control-complete FIRRMA estimation sample (2014–2020, excluding US pairs), approximately 17% of observations have negative inward flows and 58% are exact zeros. The inverse hyperbolic sine transformation accommodates both: $\operatorname{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$ is defined for all real x, approximately linear near zero, and approximately $\ln(2|x|)$ for large $|x|$.

Summary statistics. Table 15 documents the source×destination×year panel used in the FIRRMA design. The table links the raw FDI data to Table 8 by reporting the transformed FDI outcome, the exposure measures, the gravity controls, and the two balance controls after the 2014–2020 sample restriction. It also makes the estimation support explicit: the crowd-in estimates are not estimated on the full WBG-HBFDI universe, but on the subset with the controls needed for the source×year and pair-fixed-effect specification.

B.4 BIS Locational and Consolidated Banking Statistics

The main spillover test in Table 10 uses BIS Locational Banking Statistics (LBS), which report claims by office location or residence. The robustness check in Table 32 uses BIS Consolidated Banking Statistics (CBS), which report claims by the nationality of the parent bank.

SWF investment treatment universe. SWF bank equity injections are identified from the Global SWF (GSWF) and Sovereign Wealth Fund Institute (SWFI) transaction databases, supplemented and cross-checked against the IFSWF member-fund deal disclosures, S&P Capital IQ, Mergermarket, and PrivCo. Across all sectors, the cleaned and deduplicated unlisted private-market SWF deal universe covers about \$1.28 trillion of cross-border investment over 2002–2022. Within that broader universe, I retain direct bank equity injections, dropping asset-management mandates,

Table 15: FIRRMA Bilateral FDI Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (source $\times$ destination $\times$ year, 2014–2020)						
Bilateral FDI inflow ($M, winsorized)	17,591	104.3	649.9	0.0	0.0	2.2
asinh(bilateral FDI inflow)	17,591	0.639	3.248	0.000	0.000	1.520
Exposure: high-dilution (destination)	17,591	0.237	0.350	0.000	0.091	0.400
Exposure: count-based (destination)	17,591	0.332	0.375	0.000	0.244	0.667
Exposure: value-weighted (destination)	17,591	0.358	0.429	0.000	0.080	0.927
Post ($\mathbf{1}$[year $\geq$ 2018])	17,591	0.433	0.495	0.000	0.000	1.000
FTA membership	17,591	0.204	0.403	0.000	0.000	0.000
log(destination GDP)	17,591	20.279	1.487	19.509	19.849	21.228
Destination GDP growth	17,591	0.007	0.104	-0.046	0.015	0.067
asinh(bilateral trade)	17,591	0.510	0.967	0.001	0.034	0.501
Diplomatic distance (UN voting)	17,591	0.854	0.754	0.135	0.667	1.512
US trade share (destination)	17,591	0.090	0.065	0.042	0.079	0.118
Pre-FIRRMA FDI level (std., destination)	17,591	-0.082	0.891	-0.654	-0.204	0.724
Panel B: Panel Dimensions						
Source–destination–year observations	17,591					
Source–destination pairs	3,093					
Source countries	224					
Destination countries	22					
Destinations with positive HD exposure	9					
Years (FIRRMA shock 2018)	2014–2020 (4 pre + 3 post)					

Notes: Panel A reports source$\times$destination$\times$year variables used in the FIRRMA crowd-in regression (Table 8). The outcome is asinh of the winsorized bilateral FDI inflow. Three exposure measures are constructed at the destination level from pre-2018 SWF deal characteristics: high-dilution (above-median dilution), count-based, and value-weighted. Gravity controls follow CEPII Gravity V2022 plus UN-voting diplomatic distance. The balance controls are U.S. trade share and standardized pre-FIRRMA FDI level; the latter is the destination's 2014–2017 mean asinh bilateral inflow, standardized across destinations. Panel B reports panel dimensions: source–destination–year observations, source/destination country counts, and the count of destinations with positive high-dilution exposure. The 2014–2020 window provides four pre-FIRRMA years and three post years. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

insurance investments, and fund-of-fund commitments. The resulting bank-treatment master contains 107 bank-injection deals totaling \$127 billion across 55 lender–SWF-country pairs and 13 SWF source countries. The BIS spillover design treats these as pair-level aggregate relationships: if a lender-country × SWF-country pair receives multiple disclosed injections, I sum them into a single total intensity and anchor the cohort at the pair's first injection date. After BIS reporting, disclosure, timing, and matching restrictions, 12 treated lender–borrower pairs enter the stacked DiD used in Table 10. The firm-channel treatment in Table 11 further restricts this universe to unlisted non-financial deals over 2002–2022, totaling \$1.13 trillion in Appendix Table 16.

Data cleaning. The BIS introduced the instrument breakdown (loans vs. debt securities) in 1995, so I restrict the sample to 1995–2024. Negative stock positions (fewer than 0.1% of observations), which arise from exchange-rate valuation adjustments and short positions in debt securities (Bank for International Settlements, 2019), are floored at zero because they are reporting and valuation artifacts rather than economically meaningful negative lending stocks. The main stock specifications use log claims on strictly positive Q4 stock positions; in the matched annual regression sample this restriction drops one loans-and-deposits observation and nine non-bank-loan observations. Annual aggregation uses the Q4 end-of-year stock and drops pair-years without a Q4 observation, following Avdjiev et al. (2020). For the flow column, I use reported LBS loan flows, sum quarterly observations to the pair-year level, and scale the annual flow by the pair's pre-treatment mean loan stock; I do not construct flows from changes in stocks. Matching uses the pre-treatment mean and linear trend of log(loans), standardized within the treated-control pool.

Summary statistics. Table 16 documents the matched stacked-cohort panel used in the BIS spillover regression. It connects the BIS data construction to Table 10 by reporting both the LBS outcomes and the SWF equity-injection treatment intensity after coverage, timing, and matching restrictions. The table also shows the cost of the design: the matched panel is cleaner than the raw treatment universe, but it covers only the country pairs for which BIS reporting and pre-treatment matching variables are available.

C Additional PitchBook Results

This appendix collects descriptive selection evidence, robustness checks, and supporting results for the PitchBook co-investment analysis in Section 3. The active checks cover SWF participation patterns, SWF country-sector tilts, follow-on financing, sector heterogeneity, leave-one-SWF-country

Table 16: BIS LBS Stacked-Cohort Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (cohort × pair × year, ±6-year window)						
Total cross-border claims ($M)	3,259	22976.2	48149.6	973.0	5896.0	20846.0
Loans & placed deposits ($M)	3,258	18392.3	37497.6	684.0	5094.4	17571.5
Non-bank loans ($M)	3,250	5442.6	11114.3	285.5	1922.5	5764.8
log(total claims)	3,259	8.325	2.275	6.880	8.682	9.945
log(loans & deposits)	3,258	8.093	2.332	6.528	8.536	9.774
log(non-bank loans)	3,250	7.141	2.072	5.654	7.561	8.660
SWF intensity ($bn)	3,259	0.267	1.730	0.000	0.000	0.000
SWF intensity × Post ($bn)	3,259	0.144	1.276	0.000	0.000	0.000
Post ($\mathbf{1}[\text{year} \geq \text{cohort treat}]$)	3,259	0.541	0.498	0.000	1.000	1.000
Treated pair indicator	3,259	0.048	0.213	0.000	0.000	0.000
Panel B: Panel Dimensions						
Cohort–pair–year observations	3,259					
Cohorts (lender–borrower × first-treatment year)	12					
Reporting countries (full annual LBS panel)	31					
Borrower countries (full annual LBS panel)	218					
Treated lender–borrower pairs	12					
Control pairs (Mahalanobis-matched)	172					
Event-time window	$[-6, 6]$ around cohort treatment					
Panel C: SWF Investment Treatment (treated pairs only)						
Bank-channel deals (raw)	107					
Bank-channel lender–SWF-country pairs (raw)	55					
SWF home countries with bank injections	13					
Total bank-channel SWF equity ($bn)	127					
SWF equity per pair, post-disclosure filter ($bn)	48	2.65	4.76	0.11	0.43	2.73
Pairs in stacked DiD (LBS-reporting + disclosed)	12					
SWF equity per stacked-DiD pair ($bn)	12	5.57	5.99	0.73	4.15	8.32
Non-financial unlisted SWF deals (2002–2022, cross-channel)	2,636					
Non-financial unlisted firm-channel pairs	378					
Total non-financial unlisted SWF investment ($bn)	1134					
Firm-investment intensity per pair ($bn)	378	3.00	9.02	0.10	0.46	1.87

Notes: Panel A reports cohort×pair×year variables used in the BIS spillover regression (Table 10). Outcomes are total cross-border claims, loans and placed deposits, and non-bank loans, both in raw USD millions and as log-transformed positive stocks. The treatment intensity is total SWF equity injected (USD billions) into banks domiciled in the lender country, by SWFs from the borrower country, interacted with the post-treatment indicator. When a pair receives multiple disclosed injections, those amounts are summed into a single pair-level total and switched on from the pair's first injection date. Panel A pools treated and matched-control rows; conditional-on-treatment summaries are reported in Panel C. Panel B reports panel dimensions. Panel C summarizes the SWF investment treatment universe: bank-channel deals and lender–SWF-country pairs (raw, before LBS coverage and timing restrictions) with the per-pair SWF equity-investment distribution, and the analogous non-financial firm-investment universe used as treatment in the cross-channel test (Table 11). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

sensitivity, permutation inference, valuation-discount winsorization, alternative outcome windows, same-country co-investor handling, and destination-fixed matched comparisons.

C.1 SWF Participation Patterns

Table 17 reports the descriptive deal-level selection exercise discussed in Section 3. The outcome equals one if any cross-border SWF investor participates in the completed PitchBook deal. This table characterizes the selected private-market relationships that enter the later mechanism tests; it is not an identification design for directed expansion.

Table 17: Characteristics of Cross-Border Deals Receiving SWF Capital

	Any cross-border SWF participation		
	(1) Firm chars.	(2) + SWF tilt	(3) + RCA < 1
log post-money valuation, std.	0.0231***	0.0222***	0.0221***
	(0.0022)	(0.0019)	(0.0019)
log deal size, std.	-0.0054***	-0.0049***	-0.0048***
	(0.0021)	(0.0019)	(0.0019)
log firm age, std.	-0.0050***	-0.0054***	-0.0053***
	(0.0008)	(0.0007)	(0.0007)
log number of employees, std.	0.0049***	0.0048***	0.0048***
	(0.0008)	(0.0007)	(0.0007)
Lagged aggregate SWF sector tilt, std.		0.0016***	0.0017***
		(0.0005)	(0.0005)
Share of active SWF countries with export RCA < 1, std.			-0.0008
			(0.0006)
Target-HQ country × sector × year FE	Yes	–	–
Year FE	–	Yes	Yes
Coarse deal-type FE	Yes	Yes	Yes
Target geography controls	–	Yes	Yes
SE clustered by company	Yes	Yes	Yes
Mean dep. var.	0.0158	0.0158	0.0158
Observations	94,519	94,462	94,462

Note: Deal-level linear probability models in the completed cross-border PitchBook deal universe. The outcome equals one if any cross-border SWF investor participates in the deal. Coefficients are in probability units, so 0.01 equals one percentage point, and continuous regressors are standardized. Column (1) includes target-HQ-country×sector×year and coarse deal-type fixed effects. Columns (2)–(3) include year and coarse deal-type fixed effects because the sector-level tilt and RCA measures vary at the sector level. Columns (2)–(3) also include controls for target headquarters in an IMF advanced economy and target headquarters in an offshore financial center or unknown jurisdiction. The lower observation count relative to the full cross-border universe reflects complete-covariate availability, especially post-money valuation and employee counts. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 SWF country-sector portfolio tilts

Figure 5 reports sector tilts for the 20 SWF home countries in the PitchBook valuation-discount sample, using a common 2005–2025 base period. For each home country and primary sector, I compute the sector's share of cross-border SWF investor–deal observations for SWFs headquartered in that country and subtract the corresponding share among cross-border investor–deal observations backed by non-government deep-pocket private investors. The measure is count-based, not value-weighted, and is not residualized by year.

The tilts vary sharply across SWF home countries, but they do not map mechanically to home-country revealed comparative advantage. Figure 6 focuses on country–sector cells with large positive tilts and benchmarks each cell against the SWF home country's early revealed comparative advantage (RCA) in the mapped sector. For country c and PitchBook sector s, I compute

$$\text{RCA}_{c,s} = \frac{X_{c,s}/X_c}{X_{W,s}/X_W},$$

where $X_{c,s}$ is the country's mapped goods-plus-services exports in sector s, X_c is total goods-plus-services exports, and W denotes the corresponding world totals. RCA above one means that the sector accounts for a larger share of the country's exports than of world exports. The benchmark uses the geometric mean of annual RCA over 2005–2007, the earliest service-export window with broad country coverage.[45]

The sector mapping is deliberately non-overlapping. For mixed sectors such as IT, mapped goods and services components are summed before forming $X_{c,s}$ and $X_{W,s}$. Goods exports use four-digit SIC codes: Energy uses coal, oil and gas, and petroleum products; Materials uses natural resources, basic materials, and non-pharmaceutical chemicals; B2C uses consumer goods, cosmetics, household appliances, and audio/video goods; B2B uses industrial products, machinery, transport equipment, and non-medical instruments; IT uses computer hardware, communications equipment, and electronic components; and Healthcare uses pharmaceuticals and medical or ophthalmic instruments. Services exports enter through balance-of-payments categories: Financial Services uses financial services only; IT uses telecommunications, computer, and information services; B2B uses other business services; Healthcare uses health services plus health-related travel; and B2C uses travel and personal-cultural services net of health components. This construction is a revealed-comparative-advantage proxy, not a complete production taxonomy.

[45] World totals are constructed from the same source files used for country totals. Sanity checks confirm that no mapped sector total exceeds the corresponding goods, services, or combined export total. The seven mapped PitchBook sectors cover about 88% of world goods-plus-services exports in 2005–2009, so the RCA denominator remains the full export basket rather than only mapped sectors.

The benchmark separates high-tilt cells into aligned and non-aligned cases. Most high-tilt cells with available export data have RCA below one. A smaller set lines up with revealed comparative advantage: China overweights Materials, while Oman, Qatar, and the United Arab Emirates overweight Energy. The more common low-RCA cells include Kuwait's finance overtilt, Saudi Arabia's B2C overtilt, and China's energy and B2B overtilts. The narrow reading is that sector tilts proxy SWF-specific sector weights, not a simple rule of investing only where the home country already has an export advantage. In strong sectors, directed business links can complement existing domestic clusters. In weak sectors, they can be read as consistent with diversification or capability-building motives, subject to the coarse PitchBook-sector aggregation.

C.3 Follow-on Financing

Table 18 reports two pieces of evidence on follow-on financing and investor continuation. Panel A asks whether an SWF investor is more likely to return to the same company. Let c index target companies, j investors, and d financing rounds. Columns (1)–(2) estimate

$$\text{Return}_{cjd}^{5y} = \alpha_d + \beta\,\text{SWF}_{jd} + X'_{cjd}\gamma + \varepsilon_{cjd},$$

where Return_{cjd}^{5y} equals one if the exact investor appears in any later completed PitchBook deal record for the same company within five years. This broad follow-on measure includes later private financings, IPOs, PIPEs, M&A, joint ventures, asset purchases, and corporate transactions when PitchBook records the same investor-company relationship again. It is therefore a continuation-relationship measure, not a fresh-capital-only reinvestment measure. The original-round fixed effect compares SWF and non-SWF investors in the same original financing round. Column (1) includes no controls. Column (2) adds investor j's own prior follow-on rate, computed from earlier portfolio-company relationships whose five-year return window had closed before round d. This is an investor-history control, not a measure of the target firm's prior fundraising. Column (2) also controls for same-home-country co-investors in the original round. The SWF coefficient is 3.37 percentage points in column (1) and 2.29 percentage points in column (2), so SWF investors are more likely than same-deal non-SWF co-investors to return.

Panel B then classifies the same valuation-discount focal-deal universe by realized later deal activity. Among the 819 classifiable focal SWF-backed deals, 405 have no later completed private PitchBook deal with usable investor links, 251 have later private deals but no later SWF participation, and 163 have at least one later private deal with SWF participation. The re-up firms are selected: their median post-money valuation is \$1.0 billion, their mean valuation discount is

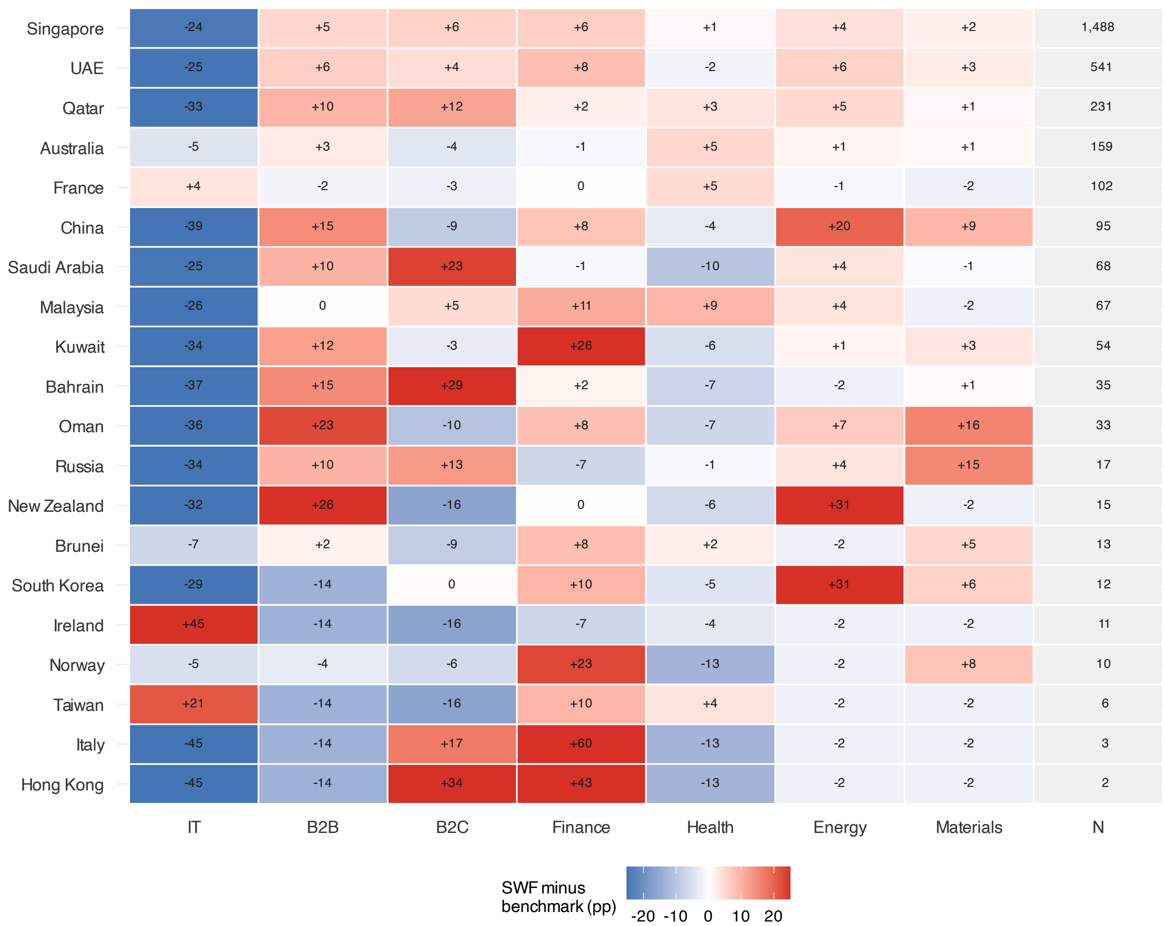

Figure 5: SWF Country-Sector Portfolio Tilts

Notes: Rows are the 20 SWF home countries that contribute SWF investor–deal observations to the PitchBook valuation-discount sample. Colored cells report each country's pooled 2005–2025 cross-border sector share minus the corresponding 2005–2025 cross-border benchmark share for non-government deep-pocket private investors, in percentage points. SWF and benchmark portfolios are counted at the investor × deal level; when multiple SWFs from the same home country join the same deal, each contributes one portfolio row before pooling to home-country shares. Deal-investor rows with Add-on Sponsor status are excluded from this descriptive portfolio-count exercise. The final column reports the number of cross-border SWF investor–deal observations used to compute each SWF home country's sector shares. The measure is count-based.

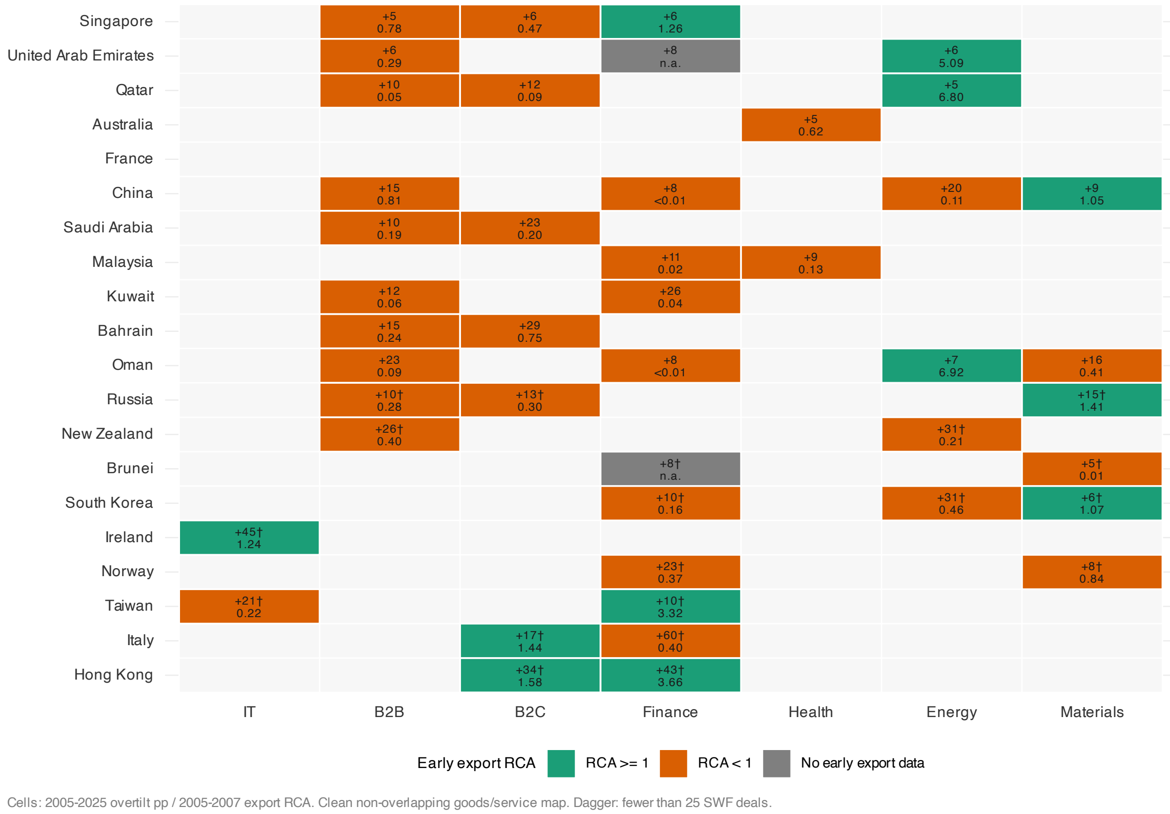

Figure 6: High-tilt sectors and early revealed comparative advantage, all SWF home countries

Notes: Rows are the 20 SWF home countries that contribute SWF investor–deal observations to the PitchBook valuation-discount sample. Colored cells report 2005–2025 country–sector overtilts of at least 5 percentage points and indicate whether the SWF home country's 2005–2007 export RCA in the mapped PitchBook sector is above or below one. France has no highlighted cell because all of its exact sector tilts are below 5 percentage points in absolute value. Grey cells indicate that no early export-RCA benchmark is available for the mapped sector. Cell labels report the portfolio overtilt in percentage points and the export-RCA value; a dagger (†) after the overtilt number marks cells from home countries with fewer than 25 total cross-border SWF investor–deal observations. Deal-investor rows with Add-on Sponsor status are excluded from the descriptive portfolio-count measure. Goods sectors use a clean non-overlapping four-digit SIC mapping; services use balance-of-payments services categories. Constructing the same export-RCA diagnostic with cumulative SWF portfolios through 2015 or 2018 gives qualitatively similar patterns, although very early sector-tilt support is limited. In the regression design, sector tilts are constructed from prior cumulative individual-SWF portfolios before the focal deal year, and the valuation-discount regressions retain the original sample for comparability.

lower, and they raise more later rounds. Their cross-border SWF-row expansion rate is 3.9 percent, compared with 3.7 percent among firms with later private deals but no SWF re-up and 1.2 percent among firms with no later private deal.

Table 18: Follow-on Financing and Re-up Anatomy

Panel A: Exact investor continuation

	Same-Investor Return	
	Base	+ Controls
SWF investor	0.0337***	0.0229**
	(0.0114)	(0.0111)
Investor baseline return controls	–	✓
Other same-country investor controls	–	✓
Observations	3,174	3,174
Deals / firms	831 / 800	831 / 800
Original-round FE	✓	✓
SE clustered by company	✓	✓

Panel B: Focal SWF-backed deal groups

	(1)	(2)	(3)
Group	A1	A2	B
Definition	No later financing	Later, no SWF	Later, SWF re-up
Focal SWF deals	405	251	163
Share of focal deals	49.5%	30.6%	19.9%
Median post-money ($M)	533	721	1,000
Median employees	640	350	420
Mean valuation discount	0.31	0.01	-0.21
Mean later rounds	0.00	1.78	2.88
Later-stage VC share	20.5%	61.0%	66.9%
SWF-row expansion rate	1.2%	3.7%	3.9%

Note: Both panels use the valuation-discount sample. Panel A reports investor–deal regressions where the outcome equals one if the exact investor appears in any later completed PitchBook deal record for the same company within five years. This broad continuation measure includes later private financings, IPOs, PIPEs, M&A, joint ventures, asset purchases, and corporate transactions when PitchBook records the same investor-company relationship again. Panel B collapses the sample to focal SWF-backed deals and classifies them using later completed private PitchBook deal records within five years, excluding IPO and PIPE later rounds and requiring usable investor links. A1 has no such later private deal; A2 has a later private deal but no SWF participation; B has at least one later private deal with SWF participation. SWF-row expansion rate is measured on cross-border SWF investor rows. The Panel B groups are selected after the focal deal and should be read as descriptive continuation anatomy, not as a causal re-up design. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4 Sector Heterogeneity

Table 19 asks whether the PitchBook result is a sector artifact. It drops each primary industry sector in turn and re-estimates the main specification. The new-entry estimate remains positive under every omission, but it is most sensitive to Consumer Products and Services (B2C): omitting that sector lowers the coefficient from 0.055 to 0.024 and removes conventional statistical significance. The count-based appendix outcome is also most sensitive to omitting B2C, falling from 0.065 to 0.044. This pattern suggests that consumer-facing targets contribute meaningfully to the directed-expansion result.

Table 19: Sector Heterogeneity: Leave-One-Sector-Out

	New Entry	asinh(Count)
Baseline (all sectors)	+0.055***	+0.065***
Business (B2B)	+0.062***	+0.072***
Consumer (B2C)	+0.024	+0.044
Energy	+0.055***	+0.064***
Fin. Services	+0.059***	+0.065***
Healthcare	+0.058**	+0.068***
Info. Technology	+0.065**	+0.067**
Materials	+0.054***	+0.065***

Notes: Each row drops one PitchBook primary industry sector (Business Products and Services (B2B), Consumer Products and Services (B2C), Energy, Financial Services, Healthcare, Information Technology, or Materials and Resources) and re-estimates the baseline specification (SWF × Valuation Discount with deal-by-year FE, implemented as deal FE, plus investor-home-country × deal-year FE and investor-home-country × target-HQ-country FE; SE clustered by company). New Entry = **1**[no pre-deal activity in investor country AND post-deal activity]. Asinh(Count) uses all post-deal activity (not conditioned on pre=0). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Leave-One-SWF-Country-Out

Figure 7 drops each SWF investor home country in turn. This check addresses the main external-validity concern in PitchBook: the identifying variation is concentrated in a small number of active SWF countries. The coefficient remains positive under every omission, preserving the sign of the result, but the sensitivity to Singapore and the UAE qualifies how broadly the PitchBook estimate should be generalized.

C.6 Permutation Inference

Table 20 asks whether the PitchBook pattern could be generated by randomly labeling same-round investors as SWFs. The exercise leaves the realized expansion outcomes unchanged. It also leaves each deal's investor set, investor home countries, valuation discount, and number of SWF investors fixed. Within each deal, I randomly reassign the SWF labels and recompute two statistics. Panel A uses a model-free statistic: the difference between SWF and non-SWF new-entry rates after scaling each group by its baseline country deal share. Panel B re-estimates the main regression and stores the placebo coefficient on SWF×valuation discount.

C.7 Winsorization Sensitivity

Table 21 tests whether the valuation-discount result is driven by extreme valuation residuals. It re-estimates the main specification with no winsorization and with a tighter two-sided winsorization cutoff. The coefficient is stable across the non-winsorized and winsorized specifications, so the valuation-discount result is not an artifact of a small number of extreme observations.

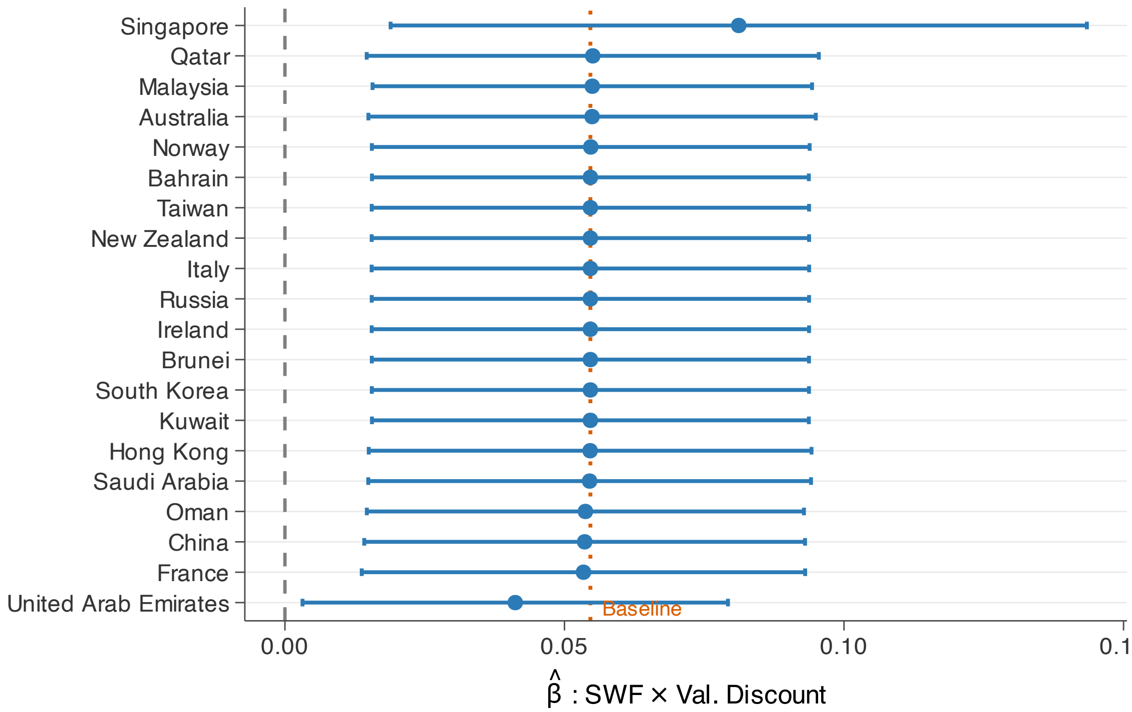

Figure 7: Leave-One-SWF-Country-Out: SWF × Val. Discount Coefficient

Notes: Each row drops one SWF investor home country and re-estimates the full-FE valuation-discount specification (deal-by-year FE, implemented as deal FE, plus investor-home-country × deal-year FE and investor-home-country × target-HQ-country FE). Orange dashed line: full-sample baseline. Error bars: 95% CIs; SE clustered by company. Singapore and the UAE together account for 73% of SWF investor–deal rows and 88% of SWF new-entry events in the valuation-discount sample. Dropping the UAE attenuates the estimate, while dropping Singapore amplifies it; both omissions leave the coefficient positive and statistically significant.

Table 20: Permutation Inference

	Observed	Null Mean [SD]	RI $p_{\text{two-sided}}$
Panel A: Model-free ratio statistic			
SWF vs. non-SWF ratio	0.877	-0.033 [0.017]	0.000
Panel B: Regression coefficient			
New Entry	0.0546***	-0.0003 [0.0088]	0.000
asinh(Count)	0.0645***	-0.0002 [0.0122]	0.000
Permutations		1,000	

Note: Within-deal random reassignment of SWF investor labels (1,000 permutations). Panel A tests a model-free ratio statistic: $\mathrm{E}[Y|\mathrm{SWF}=1]/\mathrm{E}[\mathrm{baseline}|\mathrm{SWF}=1] - \mathrm{E}[Y|\mathrm{SWF}=0]/\mathrm{E}[\mathrm{baseline}|\mathrm{SWF}=0]$; RI p = share of permuted ratios with $|\mathrm{ratio}^{\mathrm{perm}}| \geq |\mathrm{ratio}^{\mathrm{obs}}|$. Panel B permutes the SWF indicator and re-estimates the main specification (deal-by-year FE, implemented as deal FE, plus investor-home-country × deal-year FE and investor-home-country × target-HQ-country FE; SE clustered by company); RI p = share of permuted SWF × Val. Discount coefficients with $|\hat{\beta}^{\mathrm{perm}}| \geq |\hat{\beta}^{\mathrm{obs}}|$. Stars on observed coefficients reflect asymptotic p-values from the main regression. Two-sided p-values reported. Significance levels: $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

Table 21: Winsorization Sensitivity

	Outcome: New Entry (5yr)		
	Main: winsor p1/p99	No winsorization	Winsor p5/p95
	(1)	(2)	(3)
SWF × Val. Discount	0.0520***	0.0511***	0.0482***
	(0.0192)	(0.0192)	(0.0187)
Deal-by-year FE (deal FE)	Yes	Yes	Yes
Investor-home-country × deal-year FE	Yes	Yes	Yes
Investor-home-country × target-HQ-country FE	Yes	Yes	Yes
Observations	3,174	3,174	3,174

Note: Each column re-standardizes valuation discount under a different tail treatment and re-estimates the main specification (deal-by-year FE, implemented as deal FE, plus investor-home-country × deal-year FE and investor-home-country × target-HQ-country FE; SE clustered by company). Column (1) is the main-text specification, which winsorizes valuation discount at the 1st and 99th percentiles. Column (2) uses the raw valuation discount without winsorization. Column (3) winsorizes valuation discount at the 5th and 95th percentiles. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.8 Alternative Outcome Windows

Table 22 reports two checks on the timing of the PitchBook outcome. Panel A asks whether the new-entry result depends on the post-deal window used to construct the dependent variable, while keeping the main regression sample fixed. The 3-year, 5-year, and 7-year estimates are all positive, with coefficients rising from 0.0479 to 0.0546 to 0.0581.

Panel B addresses right-censoring by holding the baseline five-year new-entry outcome fixed and restricting the sample to deals with at least 3, 5, or 7 years of observable post-deal expansion history. The SWF × valuation-discount coefficient remains positive: 0.0674 in the at-least-3-year sample, 0.0492 in the at-least-5-year sample, and 0.0807 in the at-least-7-year sample. Precision declines as the restrictions reduce the regression sample to 2,303, 1,186, and 734 investor–deal rows, respectively.

C.9 Same-Country Non-SWF Co-Investors

The baseline sample drops non-SWF investors headquartered in the same country as the deal's SWF because their outcome, expansion into the investor's home country, is mechanically identical to the SWF outcome. If an SWF-backed firm expands into the SWF home market, a same-country non-SWF co-investor inherits the same outcome without adding independent within-deal variation. Table 23 reports two diagnostics: an expanded sample that keeps those rows, and a stricter sample that drops the entire financing round whenever it contains a same-country non-SWF co-investor. The estimates remain positive across entry and expansion count, so the result is not driven by how

Table 22: Alternative Outcome Windows and Follow-up Restrictions

Panel A: Alternative post-deal outcome windows

	New Entry		
	3-year	5-year	7-year
	(1)	(2)	(3)
SWF × Val. Discount	0.0479**	0.0546***	0.0581***
	(0.0193)	(0.0200)	(0.0200)
Deal-by-year FE (deal FE)	Yes	Yes	Yes
Investor-home-country × deal-year FE	Yes	Yes	Yes
Investor-home-country × target-HQ-country FE	Yes	Yes	Yes
SE clustered by company	Yes	Yes	Yes
Observations	3,174	3,174	3,174

Panel B: Minimum observable post-deal follow-up

	Baseline 5-year New Entry		
	≥3-year	≥5-year	≥7-year
	(1)	(2)	(3)
SWF × Val. Discount	0.0674**	0.0492	0.0807
	(0.0275)	(0.0417)	(0.0769)
Deal-by-year FE (deal FE)	Yes	Yes	Yes
Investor-home-country × deal-year FE	Yes	Yes	Yes
Investor-home-country × target-HQ-country FE	Yes	Yes	Yes
SE clustered by company	Yes	Yes	Yes
Non-SWF mean	0.0889	0.0793	0.0892
Observations	2,303	1,186	734

Note: Panel A constructs the new-entry outcome using different post-deal windows (3, 5, or 7 years) and re-estimates the main specification on the same regression sample as the main table. Panel B uses the baseline five-year new-entry outcome and restricts the sample by minimum observable follow-up, measured relative to the last observed PitchBook expansion date (February 9, 2025). New entry equals one if the target expands into the investor's home country within the relevant window and there was no prior activity in that country before the focal deal. All columns include deal-by-year FE (implemented as deal FE), investor-home-country × deal-year FE, and investor-home-country × target-HQ-country FE; SE are clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

these mechanically overlapping co-investors are handled.

Table 23: Same-Country Non-SWF Co-Investors: Stability Check

	Expanded Sample		Drop Same-Country Deals	
	New Entry (1)	asinh(Count) (2)	New Entry (3)	asinh(Count) (4)
SWF × Val. Discount	0.0408***	0.0923***	0.0491***	0.1131***
	(0.0134)	(0.0205)	(0.0170)	(0.0280)
SWF	-0.0081	-0.0198	0.0002	0.0341
	(0.0149)	(0.0205)	(0.0330)	(0.0463)
Deal-by-year FE (deal FE)	Yes	Yes	Yes	Yes
Investor-home-country × deal-year FE	Yes	Yes	Yes	Yes
Investor-home-country × target-HQ-country FE	Yes	Yes	Yes	Yes
SE clustered by company	Yes	Yes	Yes	Yes
Observations	3,737	3,737	2,920	2,920

Notes: Investor–deal level panel. Dependent variables are new market entry and asinh(expansion count), measured within 5 years post-deal or through the end of the observed PitchBook expansion window for later deals. Columns (1)–(2) use the expanded sample that keeps non-SWF investors whose home country matches the deal's SWF home country. Columns (3)–(4) drop the entire financing round if it contains any such same-country non-SWF co-investor. Outcomes are constructed from the same PitchBook expansion-event matching procedure used in the main specification. All columns use the sector-year valuation-discount benchmark, include deal-by-year FE, implemented as deal FE, investor-home-country × deal-year FE, and investor-home-country × target-HQ-country FE, and cluster SE by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.10 Destination-Fixed Matched Counterfactual

The within-deal specification in Section 3 compares SWF and non-SWF co-investors in the same financing round, which absorbs deal-level heterogeneity. The matched counterfactual in Table 24 asks a different, across-firm question: whether main-sample SWF-backed firms expand more into the SWF's home country than observationally similar non-SWF-backed firms expand into the same country.

The treatment unit is a deal–candidate-country pair. Treated observations are the SWF deal–country pairs from the 831-deal no-prior-history mechanism sample. Control observations are non-SWF-backed cross-border PitchBook deals paired to the same candidate SWF country and retained only when they share an exact cell with a treated pair. The baseline exact cell is candidate country × three-year deal-year bin × PitchBook sector × target-HQ region. Same-country pairs are excluded, and both treated and control rows are required to have no recorded pre-deal activity in the candidate country.

The estimating equation is

$$Y_{dc} = \beta\,\mathrm{SWFPair}_{dc} + X_d'\Gamma + \alpha_{m(dc)} + \delta_{t(d)} + \eta_{c\times h(d)} + \varepsilon_{dc}, \tag{47}$$

where Y_{dc} equals one if the recipient firm in deal d records an expansion into candidate country c within five years of the deal, or before the PitchBook observation endpoint for later deals. SWFPair_{dc} identifies main-sample SWF-backed deal–country pairs. The fixed effects are match-cell, deal-year, and candidate-country×target-HQ-country effects. X_d contains the deal and firm controls added in columns (2)–(4): standardized log post-money valuation, asinh deal size, asinh employees, founding year, and the firm's prior investor-country footprint. Columns (3)–(4) further tighten common support by augmenting the exact cell with tercile bins for size, employees, age, valuation, and prior footprint as shown in the table.

The level comparison is weak once it is tied to the main mechanism sample and tightened on observable support. The exact-cell estimate is 0.71 percentage points (SE 0.47), and adding firm and deal controls gives 0.49 percentage points (SE 0.63). Requiring common support on size, employees, and age moves the estimate to −0.35 percentage points (SE 0.25), and further matching on valuation and prior footprint gives −0.53 percentage points (SE 0.30). The table is therefore a diagnostic counterfactual rather than the paper's main identifying comparison: the core evidence comes from the within-deal SWF–non-SWF gradient with recipient valuation discount.

Table 24: Destination-Fixed Matched Counterfactual

	Exact FE (1)	Exact + Controls (2)	CEM: Size, Emp., Age (3)	CEM: All Controls (4)
SWF-backed deal–country pair	0.71	0.49	-0.35	-0.53^{*}
	(0.47)	(0.63)	(0.25)	(0.30)
SWF-backed mean (%)	2.44	2.44	2.43	2.70
Matched control mean (%)	1.20	1.20	1.76	2.32
Match-cell FE	Yes	Yes	Yes	Yes
Deal-year FE	Yes	Yes	Yes	Yes
SWF country × target-HQ FE	Yes	Yes	Yes	Yes
Regression controls	No	Yes	Yes	Yes
CEM on size, employees, age	No	No	Yes	Yes
CEM on valuation and prior footprint	No	No	No	Yes
Outcome window	Observed	Observed	Observed	Observed
Observations	72,482	72,321	10,775	4,393
SWF-backed pairs	900	739	699	556
Match cells	373	373	563	492

Notes: Deal–candidate-country pair regressions. Treated observations are SWF-backed deal–country pairs from the main no-prior-history PitchBook mechanism sample. Controls are non-SWF-backed cross-border PitchBook deals paired to the same candidate SWF country and retained within the exact or coarsened-exact match cells described in the text. The dependent variable equals one if the recipient firm records a post-deal expansion into the candidate country within five years, or before the PitchBook observation endpoint for later deals. Columns (1)–(2) use the exact cell candidate country × three-year deal-year bin × sector × target-HQ region; columns (3)–(4) augment the match cell with tercile bins for the listed CEM variables. All columns include match-cell, deal-year, and candidate-country×target-HQ-country fixed effects. Regression controls are standardized log post-money valuation, asinh deal size, asinh employees, founding year, and prior investor-country footprint. Standard errors are clustered by company and candidate country. Coefficients and standard errors are reported in percentage points. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Bank Results

This appendix reports robustness checks for the bank syndicated lending results in Section 4. The checks address the main design risks: whether write-down exposure proxies for pre-crisis bank characteristics, whether any single SWF destination drives the estimate, whether alternative SWF-treatment definitions change the estimand, whether the result depends on the post-cutoff date, and whether placebo destination sets reproduce the result.

D.1 Pre-Crisis Balance: Write-Down Ratio

Figure 8 asks whether the bank write-down ratio is simply proxying for pre-crisis bank characteristics. It regresses WriteDownRatio on five standardized covariates computed from the bank–country–quarter panel over 2004Q3–2007Q2. Four covariates are pre-window levels, and the fifth is a within-window growth measure that compares 2004Q3–2005Q4 with 2006Q1–2007Q2. The balance check is reassuring overall: all five coefficients are statistically insignificant, including pre-crisis SWF-country lending share. The main specification additionally absorbs bank–country level differences through bank×country fixed effects and the WD × Pre-share × Post control. Taken together, the figure argues against the most direct continuation story that distressed banks were already systematically more tilted toward SWF-country lending before the crisis.

D.2 Leave-One-SWF-Country-Out

Figure 9 drops each SWF destination country in turn from the full-controls log-count DID specification. This addresses the bank-sample concentration concern: with a finite set of SWF destinations, the estimated reallocation could in principle be carried by one destination market. The coefficient remains positive in every omission and remains statistically significant at the 5% level except when Australia is omitted. The exercise therefore rules out a sign reversal or a result driven mechanically by one country.

D.3 Alternative SWF Treatment Definitions

The baseline bank specification uses an ex ante destination-level treatment. SWFStatus_c equals one for countries that had active cross-border sovereign investors before the crisis: the 7 countries whose funds later injected equity into treated banks plus 8 additional pre-2007 SWF countries in the

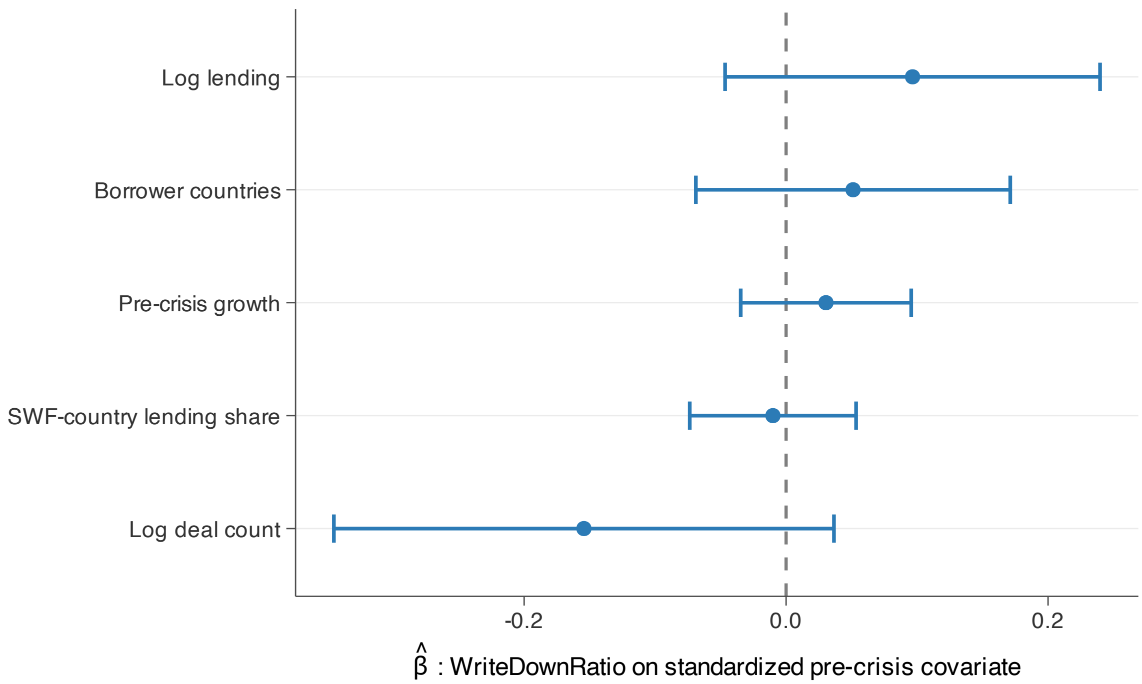

Figure 8: Bank Pre-Crisis Balance: WriteDownRatio on Standardized Covariates

Notes: Cross-sectional OLS of the bank-level WriteDownRatio on five standardized pre-crisis covariates (N=29 realized G-SIB bank parents). Pre-crisis covariates aggregate the bank–country–quarter panel over 2004Q3–2007Q2: total cross-border lending (USD), deal count, number of borrower countries, share of total lending to SWF home countries, and a pre-crisis lending growth measure defined as the log of total lending in 2006Q1–2007Q2 relative to 2004Q3–2005Q4. Each covariate is standardized to mean zero and unit standard deviation. Error bars: 95% confidence intervals from OLS standard errors. Coefficients near zero indicate balance; any residual level differences are absorbed by the Bank×Country and WD × Pre-share × Post controls in the main specification.

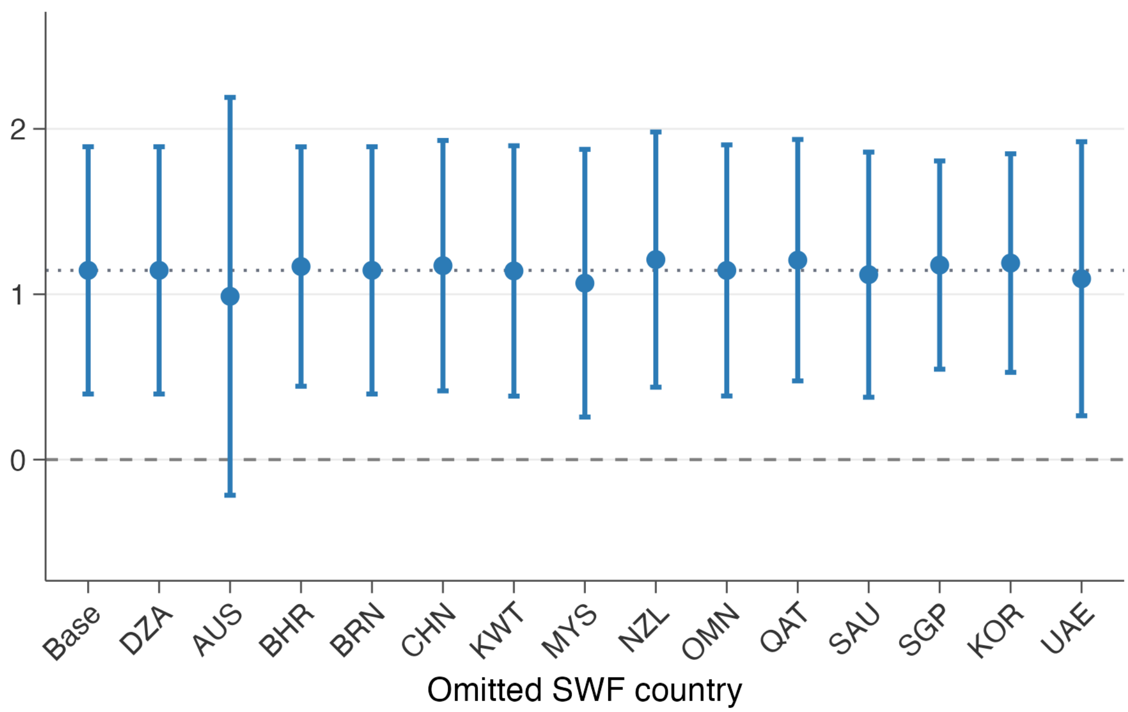

Figure 9: Leave-One-SWF-Country-Out: Deal Count

Notes: Each point is the WD coefficient from the log(deal count) specification, restricted to cells with positive deal count, omitting one SWF country from the destination set. Dashed line: full-sample baseline. Error bars: 95% CIs; SE clustered by bank. Full-controls specification with three-way FEs.

defined set.[46] This definition is chosen before observing the realized crisis-era bank–SWF matches. It asks whether banks with greater demand for sovereign capital increased lending activity toward sovereign-capital destinations, not only toward the specific countries that ultimately provided their equity.

Table 25 replaces the baseline SWF-status set with the 20 PitchBook SWF home countries that contribute to the valuation-discount regression sample. This check uses the same full-controls specification as the main bank table. The coefficients remain positive across both reported outcomes. New entry is positive but imprecise, 0.0092; the log-count coefficient is 0.7863 and significant at the 1% level. The PitchBook-based definition therefore preserves the sign of the bank result, although the strongest response remains on the deal-frequency margin. I treat this as a destination-set diagnostic rather than a preferred specification, because the baseline bank definition is chosen ex ante for the pre-crisis syndicated-lending setting.

Table 25: Alternative SWF Treatment Definition: PitchBook SWF Home Countries

	New Entry (1)	log(Count) (2)
Panel A: PitchBook SWF home countries (20 countries)		
WD × SWF (PitchBook) × Post	0.0092 (0.0072)	0.7863*** (0.2794)
Bank × Country FE	✓	✓
Bank × Quarter FE	✓	✓
Country × Quarter FE	✓	✓
SE clustered by bank	✓	✓
Observations	34,510	10,010

Note: Bank–country–quarter panel around 2007Q3, $[-8, +5]$ quarters. The table replaces the baseline ex ante SWF-status destination set with the 20 PitchBook SWF home countries that contribute to the valuation-discount regression sample. All columns use the full-controls specification with bank×country, bank×quarter, and country×quarter fixed effects; SE clustered by bank. Columns include WD × Post interactions with pre-lending share, FTA, common language, and UN-voting political distance. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.4 Alternative Post Cutoffs

The baseline bank specification defines Post_t from 2007Q3, when large structured-credit write-downs first became public and recapitalization negotiations began. Table 26 repeats the full-controls specification while holding the original bank–country–quarter sample window fixed and moving the post cutoff to 2007Q4 or 2008Q1. This isolates the dating choice rather than changing the destination definition or the set of banks.

[46] The defined set contains 15 countries in total. Libya has no lending observations in the realized panel, leaving 14 effective SWF-status destinations in the estimation sample.

The intensive-margin result is stable under the later cutoffs. The log-count coefficient is 0.994 when post begins in 2007Q4 and 1.177 when post begins in 2008Q1, compared with 1.144 in the baseline. The extensive-margin estimate is more timing-sensitive: the new-entry coefficient falls to 0.001 under the 2007Q4 cutoff and is 0.018 under the 2008Q1 cutoff. I therefore read the timing check as supporting the deal-frequency margin while showing that the entry margin depends more on how the crisis onset is dated.

Table 26: Alternative Post Cutoffs

	New Entry			log(Count)		
Post cutoff	≥2007Q3 (1)	≥2007Q4 (2)	≥2008Q1 (3)	≥2007Q3 (4)	≥2007Q4 (5)	≥2008Q1 (6)
WD × SWFStatus × Post	0.028^{***} (0.008)	0.001 (0.008)	0.018^{**} (0.008)	1.144^{***} (0.381)	0.994^{**} (0.377)	1.177^{***} (0.419)
Observations	34,510	34,510	34,510	10,010	10,010	10,010
Full controls	✓	✓	✓	✓	✓	✓
Bank×Country FE	✓	✓	✓	✓	✓	✓
Bank×Quarter FE	✓	✓	✓	✓	✓	✓
Country×Quarter FE	✓	✓	✓	✓	✓	✓
SE clustered by bank	✓	✓	✓	✓	✓	✓

Note: Bank–country–quarter panel held fixed to the main $[-8, +5]$ window around 2007Q3. Columns are grouped by outcome and repeat the post-cutoff definition within each outcome group. The reported coefficient is $\text{WriteDownRatio}_b \times \text{SWFStatus}_c \times \text{Post}_t$. New Entry is recomputed for each cutoff as the first post-cutoff positive-deal quarter for bank–country pairs with zero deals before that cutoff within the fixed window. Full controls are WriteDownRatio_b interacted with Pre-share, FTA, common language, political distance, and Post_t. Three-way fixed effects are bank×country, bank×quarter, and country×quarter. SE clustered by bank. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.5 Randomization-Inference Placebo Tests

Table 27 draws placebo destination sets from non-SWF countries and re-estimates the continuous DID. It asks whether high-write-down banks expanded toward arbitrary non-SWF foreign markets after the crisis or specifically toward the actual SWF-country geography. Under the log-count specification, the actual coefficient is positive but does not exceed the placebo 95th percentile; the right-tail RI p-value is 0.074 and the two-sided RI p-value is 0.138. The RI evidence is therefore conservative: it does not reproduce the conventional log-count significance, but it confirms that the actual estimate is positive while highlighting finite-destination uncertainty.

Table 27: Randomization-Inference Placebo Tests

Outcome	Actual coef.	Placebo mean	Placebo 95th pct.	RI p_{right}	RI $p_{\text{two-sided}}$
Deal count	1.144^{***}	-0.047	1.361	0.074	0.138

Note: Placebo destination sets drawn from non-SWF countries (1,000 draws). Each draw selects 14 placebo destinations, excluding the 14 realized SWF-status countries as well as the United States and the United Kingdom, and re-estimates the full-controls log specification with three-way FEs. The actual coefficient is then compared to the placebo distribution. The table reports both the one-sided RI p-value (p_{right}: fraction of placebo draws ≥ actual) and the two-sided RI p-value ($p_{\text{two-sided}}$: fraction with |placebo| ≥ |actual|). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E FIRRMA Spillover Robustness

This appendix reports robustness checks for the FIRRMA crowd-in results in Section 5.1. The checks address the main limits of the design. Appendix E.1 expands the destination universe beyond the PitchBook SWF-home destinations used in the main specification. Appendix E.2 reports pre-treatment balance on destination characteristics, Appendix E.3 documents non-U.S. high-dilution SWF deal volume around FIRRMA, and Appendix E.4 displays the cross-country exposure distribution. Appendix E.5 tests placebo shock dates and drops oil-exporting SWF countries, and Appendix E.6 reports leave-one-exposed-SWF-country-out estimates.

E.1 Full-Destination Main Specification

Table 28 repeats the FIRRMA specification on the broader non-U.S. destination universe rather than restricting destinations to the SWF home countries used in the main text. This is a demanding comparison because the additional destinations have zero SWF exposure by construction and need not be on the same investment margin as SWF home countries. The estimates remain positive and statistically significant across the high-dilution, count-based, and value-weighted exposure measures. The full-control high-dilution coefficient is smaller than in the SWF-home destination design, 0.318 rather than 0.663, which is consistent with dilution from adding many zero-exposure destinations outside the main support. I therefore read this table as a robustness check on sign and statistical stability, not as the preferred magnitude.

E.2 Pre-Treatment Balance: Exposure

Figure 10 asks whether the high-dilution exposure measure is proxying for destination-level fundamentals within the SWF-home destination sample. It regresses exposure on six pre-2018 destination covariates: GDP, GDP growth, U.S. trade share, the level and slope of pre-2018 FDI inflows, and

Table 28: FIRRMA Exposure and Third-Party FDI Inflows: Full Destination Sample

	High-Dilution Exposure				Count	Value
				Target Sources		
	(1)	(2)	(3)	(4)	(5)	(6)
SWF US Exposure × Post	0.310***	0.333***	0.318***	0.795***		
	(0.098)	(0.098)	(0.098)	(0.306)		
SWF US Exposure (count) × Post					0.224***	
					(0.086)	
SWF US Exposure (value) × Post						0.238***
						(0.083)
US Trade Share × Post			0.178	0.102	0.202*	0.202*
			(0.116)	(0.244)	(0.116)	(0.116)
Pre-FIRRMA FDI Level × Post			-0.088***	-0.154***	-0.092***	-0.091***
			(0.019)	(0.039)	(0.019)	(0.019)
asinh(Bilateral Trade)		0.105	0.127	0.157	0.129	0.133
		(0.155)	(0.155)	(0.256)	(0.155)	(0.155)
Diplomatic Distance		0.007	0.015	0.082	0.015	0.016
		(0.059)	(0.059)	(0.117)	(0.059)	(0.059)
Other gravity controls	–	✓	✓	✓	✓	✓
Source × Year FE	✓	✓	✓	✓	✓	✓
Pair FE	✓	✓	✓	✓	✓	✓
SE clustered by pair	✓	✓	✓	✓	✓	✓
Source countries	224	224	224	47	224	224
Destination countries	172	172	172	172	172	172
Observations	112,541	112,541	112,541	38,495	112,541	112,541
R^2	0.296	0.296	0.296	0.291	0.296	0.296

Note: Same specifications as Table 8, estimated on the broader non-U.S. destination universe (172 destinations) as a robustness check. Dependent variable: asinh(bilateral FDI inflow), 2014–2020. Columns (3)–(6) include U.S. trade share × Post and standardized pre-FIRRMA FDI level × Post. All columns include source×year and pair FE. Standard errors are clustered by country pair. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the number of source partners. This balance check is necessarily at the destination level because the treatment varies at that level; bilateral covariates are absorbed by source×year and pair fixed effects in the main regression. The figure shows two imbalances. Exposed SWF destinations have greater pre-2018 U.S. trade exposure, which is expected because the treatment is built from U.S.-targeted SWF deals. They also have lower pre-2018 bilateral FDI inflow levels. The other covariates, including destination GDP, GDP growth, pre-2018 FDI growth, and the number of source partners, are not statistically distinguishable across exposure levels. I therefore read the balance test as a reason to include U.S.-trade-share and pre-FDI-level post interactions in the main table, and to read the estimates together with the dynamic and placebo checks.

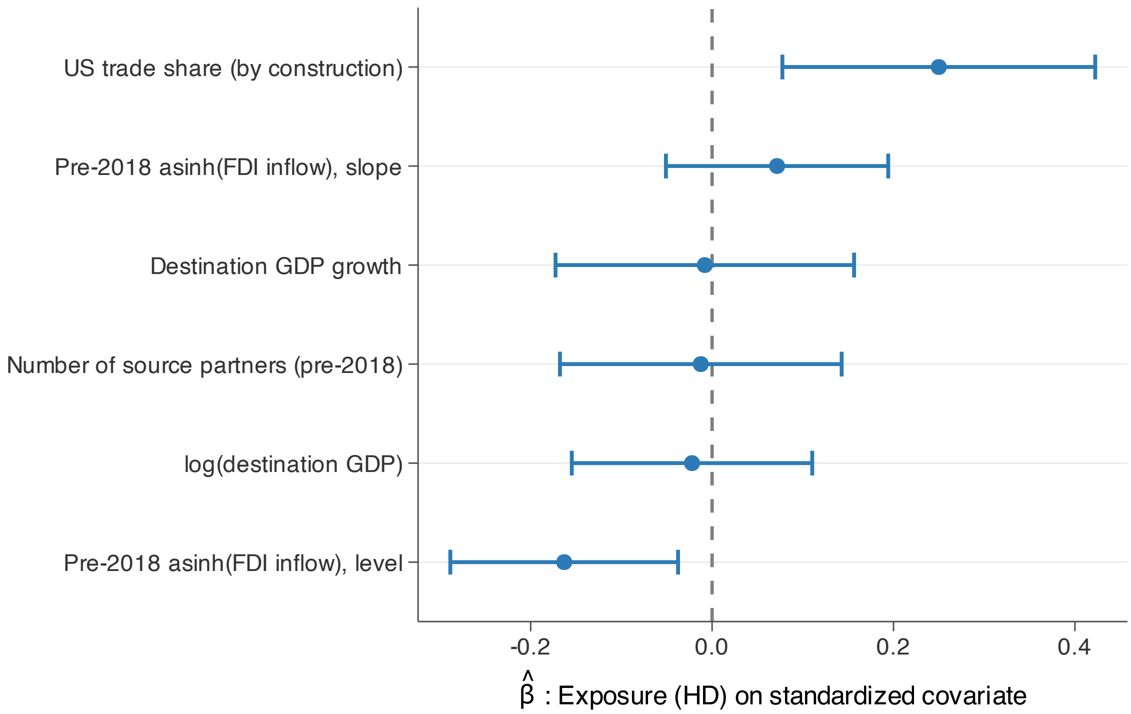

Figure 10: FIRRMA Exposure Balance: Pre-2018 Destination Covariates

Notes: Cross-sectional OLS of destination-level high-dilution exposure on six standardized pre-2018 destination covariates. Sample: 22 effective non-U.S. SWF home destinations in the main FIRRMA panel (Table 8); pre-2018 means computed over 2014–2017. Each covariate is standardized to mean zero and unit standard deviation. Error bars: 95% confidence intervals. The U.S. trade-share coefficient is positive, consistent with the fact that exposure is built from pre-2018 SWF deals targeting the United States. The negative coefficient on pre-2018 FDI inflow levels means that more-exposed destinations start from lower bilateral inflows on average; pair fixed effects absorb fixed bilateral levels in the main regression, while the event-study and placebo tests address differential timing.

E.3 Non-U.S. high-dilution deal volume

Figure 11 reports annual non-U.S. high-dilution SWF deal counts around FIRRMA. Panel A scales deal counts by the number of SWF home countries in each exposure group; Panel B reports raw annual counts. Non-U.S. high-dilution deal volume rises after 2018 in both groups, including among SWF countries with positive pre-FIRRMA U.S. high-dilution exposure. I read the figure as descriptive evidence that the relevant non-U.S. deal margin expanded after FIRRMA.

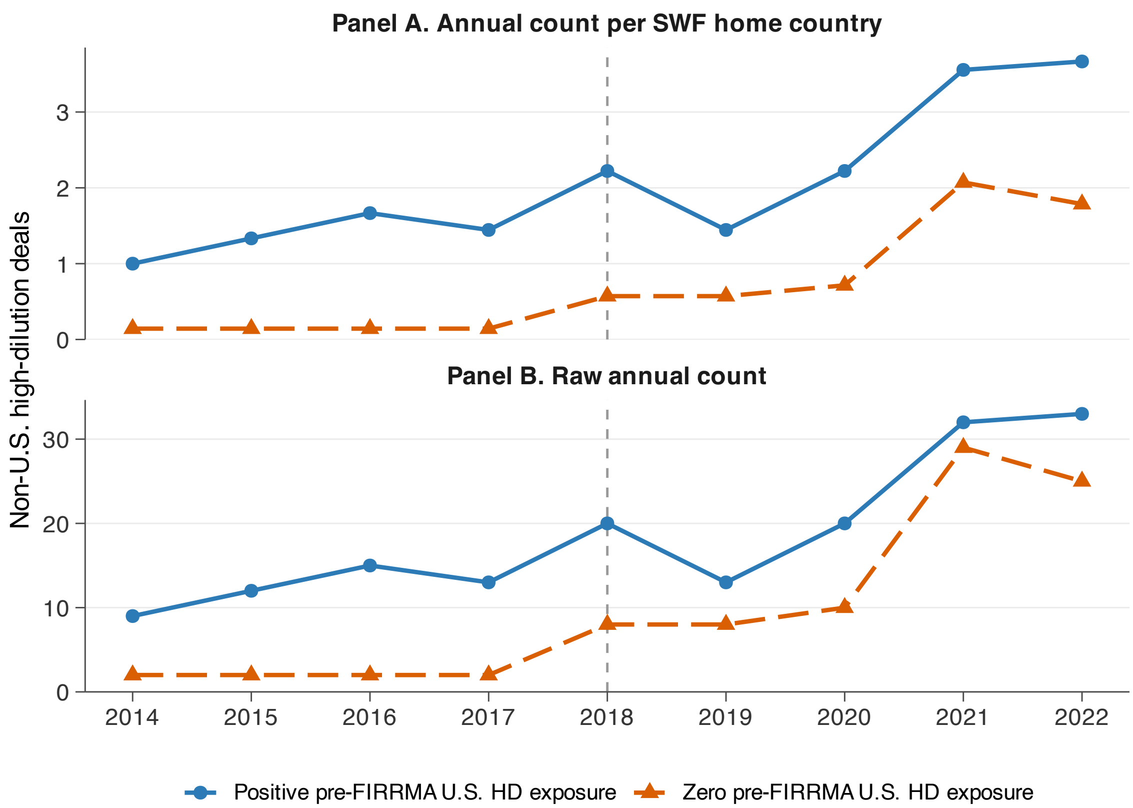

Figure 11: Non-U.S. High-Dilution SWF Deal Volume around FIRRMA

Notes: Annual PitchBook counts of SWF-backed high-dilution deals targeting non-U.S. companies, 2014–2022. High-dilution deals are those with dilution above the median among SWF co-investment deals. Positive-exposure countries are SWF home countries with at least one U.S.-targeted high-dilution deal in 2014–2017; zero-exposure countries have none. Panel A divides annual counts by the number of SWF home countries in the exposure group. Panel B reports raw annual counts. The dashed vertical line marks 2018, the FIRRMA enactment year. The figure is descriptive and does not condition on controls or fixed effects.

E.4 Exposure by SWF Country

Figure 12 plots the cross-country distribution of high-dilution FIRRMA exposure for the 23 non-U.S. PitchBook SWF home countries. Nine countries have positive exposure and fourteen have zero. Within the positive group, three countries reach 67–100% mechanically: Ireland and New Zealand sit at 100% on a single pre-2018 SWF deal each, and South Korea at 67% on four deals. The substantive variation comes from France (40% on 18 deals), Australia (17% on 9), China (14% on 11), Qatar (13% on 29), Malaysia (9% on 17), and Singapore (10.5% on 176). The zero-exposure comparison group includes UAE (n=24) and Kuwait (n=11) alongside smaller zero-exposure SWF home countries. This dispersion across positive-exposure countries motivates the leave-one-out exercise in Appendix Figure 13.

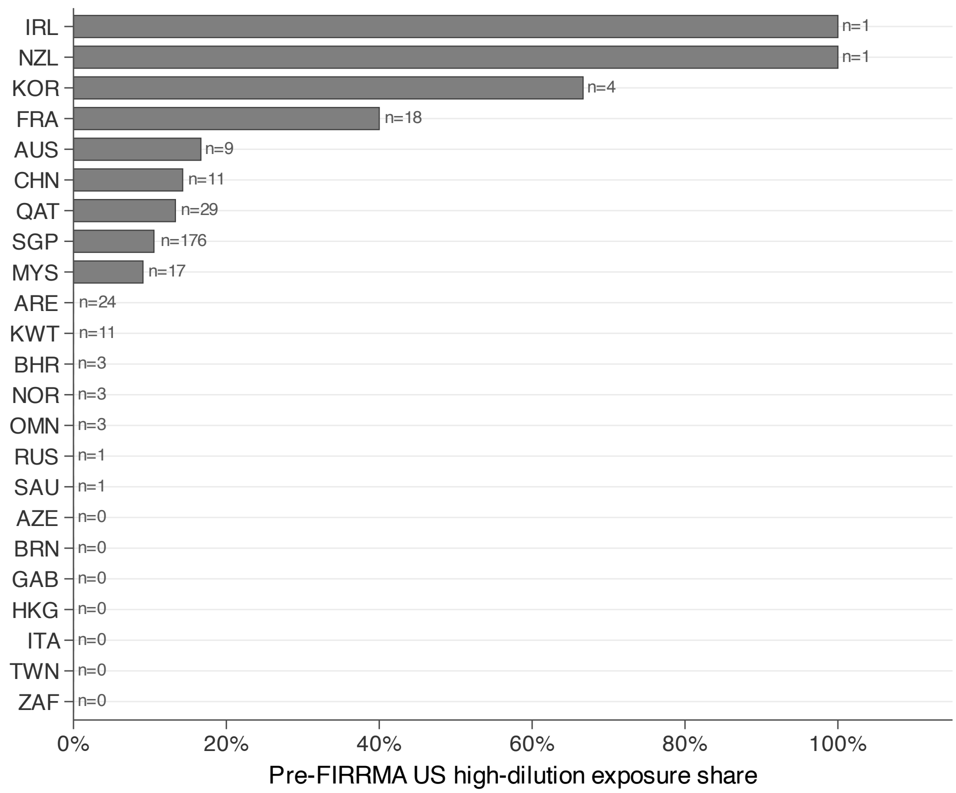

Figure 12: FIRRMA High-Dilution Exposure by SWF Country

Notes: Bars report each SWF home country's pre-FIRRMA high-dilution exposure, defined as the share of 2014–2017 SWF co-investment deals with observed dilution that both targeted the United States and involved above-median dilution. Countries are sorted by exposure. Labels report the total number of pre-2018 SWF home-country deal observations. High-dilution exposure shares are computed over observations with non-missing dilution.

E.5 Placebo Shocks and Oil-Exporter Drop

Table 29 tests two confounds in the FIRRMA design. The placebo columns move the shock date earlier while keeping the panel entirely pre-FIRRMA and holding fixed the realized 2014–2017 exposure measure, asking whether that realized exposure predicts similar pre-2018 FDI shifts when the shock date is moved earlier. The placebo estimates are much smaller than the baseline and statistically insignificant: -0.008 for a 2015 placebo shock and -0.130 for a 2016 placebo shock, compared with the baseline estimate of 0.663. The oil-exporter exclusion asks whether the result is driven by concurrent oil-related shocks in Gulf and other commodity-linked SWF countries, including Norway. Dropping oil-exporter SWF countries leaves a positive and statistically significant coefficient of 0.561. The table therefore separates the FIRRMA interpretation from two obvious alternatives: earlier timing patterns in the realized exposure and oil-linked SWF shocks.

Table 29: FIRRMA Spillover: Robustness

	Baseline (1)	Placebo 2015 (2)	Placebo 2016 (3)	No Oil Exporters (4)
SWF US Exposure (high-dilution) × Post	0.663***	-0.008	-0.130	0.561**
	(0.215)	(0.238)	(0.262)	(0.235)
asinh(Bilateral Trade)	-0.707**	-0.372	-0.637**	-0.945***
	(0.302)	(0.255)	(0.290)	(0.322)
Diplomatic Distance	-0.249	-0.134	-0.270	-0.324
	(0.211)	(0.169)	(0.196)	(0.217)
Gravity controls (GDP, FTA)	✓	✓	✓	✓
Balance controls	✓	✓	✓	✓
Source × Year FE	✓	✓	✓	✓
Pair FE	✓	✓	✓	✓
SE clustered by pair	✓	✓	✓	✓
Observations	17,591	17,147	14,787	14,310
R^2	0.363	0.377	0.386	0.393

Note: Column (1): baseline with full gravity and balance controls (high-dilution exposure). Columns (2)–(3): placebo shocks at 2015 and 2016, respectively, using the same realized 2014–2017 exposure measure as the baseline but restricting the estimation window to pre-FIRRMA years ending in 2017. Column (4): oil-exporter SWF countries dropped (UAE, Saudi Arabia, Qatar, Kuwait, Norway, Oman, Bahrain, Brunei). Dependent variable: asinh(bilateral FDI inflow). Balance controls are U.S. trade share × Post and standardized pre-FIRRMA FDI level × Post. All columns include source×year and pair FEs. SE clustered by country pair, as in the main FIRRMA specification; because treatment varies at the destination-exposure level, inference should be read together with the placebo-timing and leave-one-country-out checks. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.6 Leave-One-SWF-Country-Out

Figure 13 drops each positively exposed high-dilution SWF destination country in turn. This check addresses the finite-country concern in the macro FDI design while focusing on the countries that identify the high-dilution exposure coefficient. The coefficient remains positive across all omissions, so the result is not driven by a single exposed destination. The movement across omissions is still informative: countries whose omission shifts the estimate most are the destinations carrying more of the identifying variation, so the figure should be read as a sensitivity diagnostic rather than as a claim that every SWF country contributes equally.

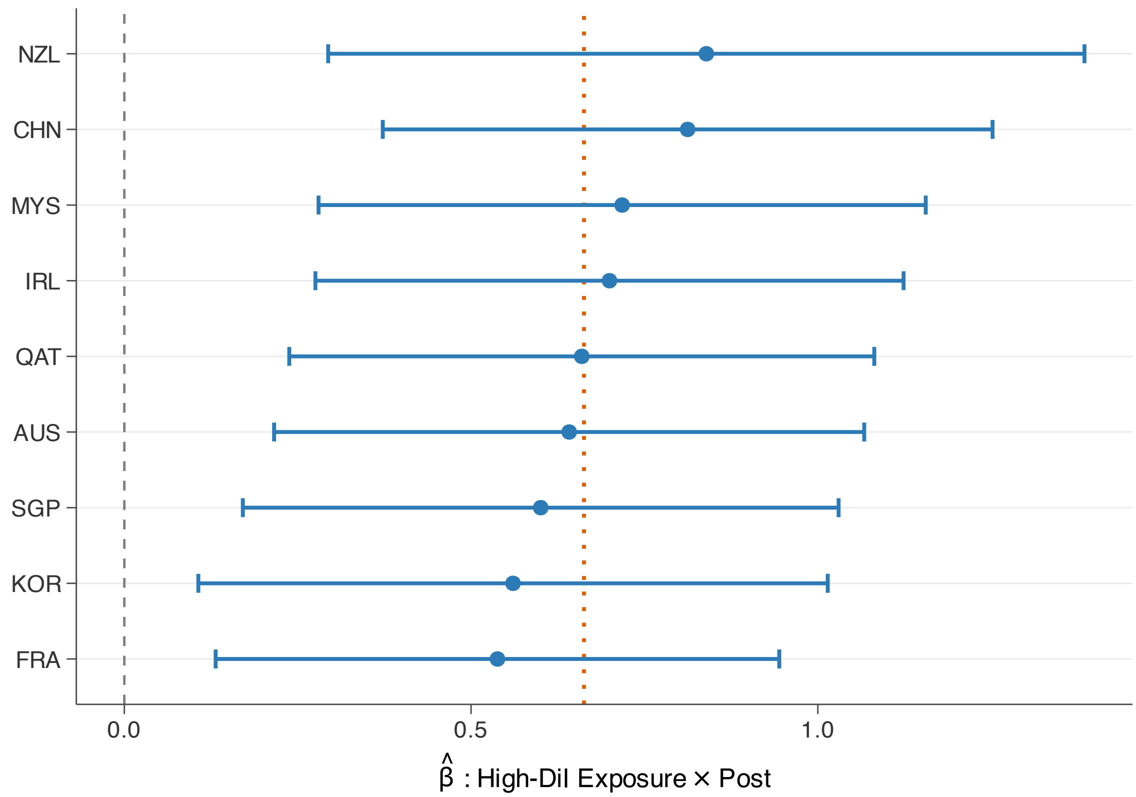

Figure 13: Leave-One-SWF-Country-Out: FIRRMA Spillover
Notes: Each row drops one positively exposed high-dilution SWF destination country and re-estimates the balance-control high-dilution specification from Table 8, column (3). Orange dashed line: full-sample column (3) baseline. Error bars: 95% CIs; SE clustered by country pair.

F BIS Spillover Robustness

This appendix reports robustness checks for the BIS cross-border banking-claims spillover results in Section 5.2. Appendix F.1 reports matching balance, Appendix F.2 tests whether one treated lender–borrower cohort drives the result, and Appendix F.3 checks whether the result survives quarterly timing. Appendix F.4 reports a liability-side test and compares the LBS residence-based estimates with CBS nationality-based claims.

F.1 Matching Balance

Table 30 reports the balance achieved by the cohort-by-cohort Mahalanobis matching used in the main BIS specification. The matching uses pre-treatment levels and slopes of log loans and deposits, the outcome closest to the main banking-claims channel. Standardized differences fall sharply after matching, including for auxiliary outcomes that were not used in the match. The stacked-cohort estimates therefore compare treated pairs with controls that had similar pre-treatment banking relationships, rather than with the full pool of unmatched country pairs.

Table 30: BIS Mahalanobis Matching Balance

	Treated	Pre-Match Pool		Matched Controls (k=20)	
	Mean	Mean	|Std. Diff.|	Mean	|Std. Diff.|
Pre-mean log(loans & deposits)*	8.228	5.013	1.136	7.975	0.114
Pre-slope log(loans & deposits)*	0.268	0.150	0.265	0.221	0.187
Pre-mean log(total claims)	8.433	5.228	1.111	8.218	0.095
Pre-mean log(non-bank loans)	6.881	4.407	1.008	6.964	0.045
Number of pairs (across cohorts)	12	765		172	

Notes: Cohort-aggregated balance for the matched stacked-cohort design used in Table 10. Pre-treatment statistics are computed within the three pre-treatment years before each cohort's first-treatment date. "Treated" pools the 12 treated lender–borrower pairs; "Pre-Match Pool" pools the full set of eligible non-treated pairs in the same lender country (765 unique pairs across cohorts); "Matched Controls (k=20)" pools the cohort-specific Mahalanobis-matched controls (172 unique pairs across cohorts). Because controls can be reused across cohorts, the number of unique matched-control pairs is smaller than the mechanical upper bound of 240 cohort-control slots (12 cohorts times 20 controls), implying 68 control-pair reuses across cohorts. The matching routine uses up to 20 controls per cohort, and in the realized matched sample all 12 cohorts use 20 controls. Absolute standardized differences are computed as $|\bar{x}_{\text{treated}} - \bar{x}_{\text{control}}|/\sqrt{(s^2_{\text{treated}} + s^2_{\text{control}})/2}$. The two starred variables are used in the Mahalanobis distance; total claims and non-bank loans are auxiliary checks not used in the match. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.2 Leave-One-Cohort-Out

Figure 14 drops each treated lender–borrower cohort in turn, including that cohort's matched controls. This is the direct concentration check for the BIS design: it asks whether the pooled coefficient is coming from a single treated pair. The loans coefficient remains positive in every omission. The estimates range from 0.0475 to 0.0768, compared with the full-sample estimate of 0.0566.

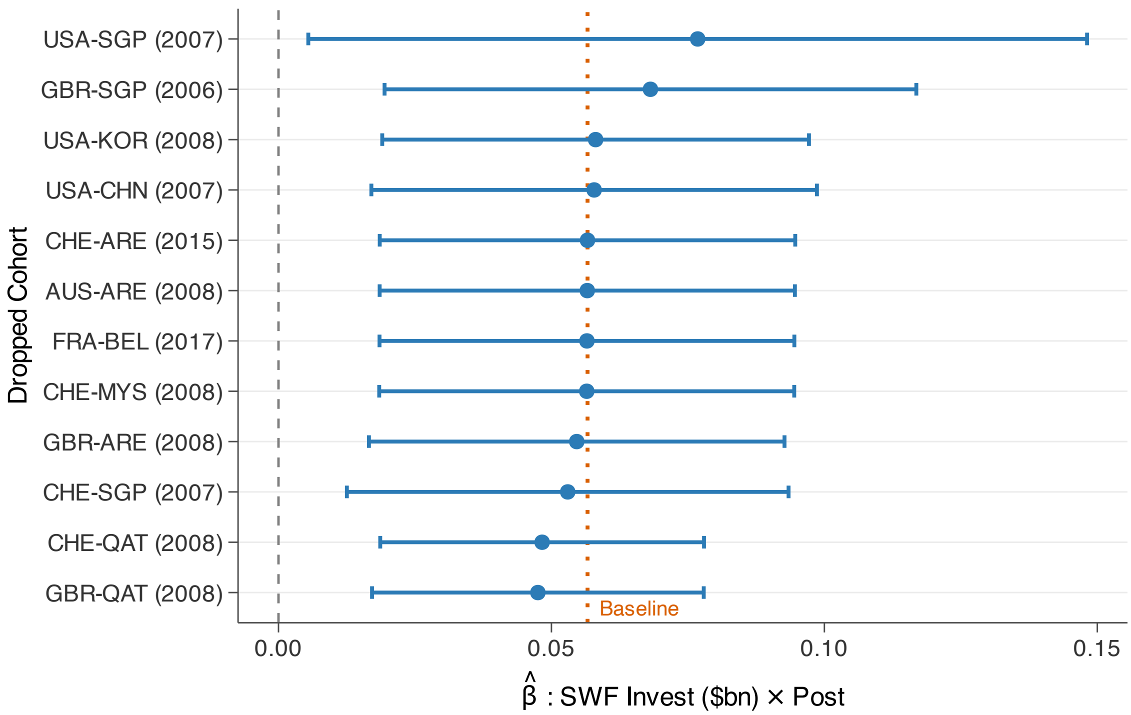

Figure 14: Leave-One-Cohort-Out: LBS Cross-Border Loans Spillover

Notes: Each row drops one treated lender-country × SWF-country cohort, including its matched controls, and re-estimates the intensity-weighted DiD from Table 10 (column 2: log loans & deposits). Orange dashed line: full-sample baseline ($\hat{\beta}$ = 0.0566). Error bars: 95% CIs; SE clustered by cohort. All estimates remain positive; the LOO range is [0.0475, 0.0768]. Eleven of twelve estimates are significant at the 5% level, and the remaining estimate is significant at the 10% level.

F.3 Quarterly Specification

Table 31 re-estimates the BIS design at quarterly frequency. The annual specification is the baseline because BIS stocks are cleaner at year end and match the main stacked-cohort window. The quarterly version is a timing check: similar estimates would mean the crowd-in pattern is not created by annual aggregation. Treatment timing uses the exact deal quarter when it is observed; in the raw post-filter treatment universe, 33 of 48 lender–borrower pairs have dated quarters and 15 are imputed to Q1 of the treatment year. The quarterly analog using all 12 matched cohorts gives a log-loans coefficient of 0.0515, close to the annual baseline of 0.0566. One matched cohort uses a Q1-imputed treatment quarter; dropping that cohort gives the same estimate at displayed precision, so treatment-quarter imputation is not driving the result.

Table 31: Cross-Border Banking Claims Spillover: Quarterly Specification

	(1) Total Claims	(2) Loans & Placed Dep.	(3) Non-Bank Loans
SWF Invest ($bn) × Post	0.0531***	0.0515**	0.0557***
	(0.0165)	(0.0167)	(0.0127)
Outcome transform	log	log	log
Observations	12,276	12,270	12,217
Matched cohorts	12	12	12
Cohort × Quarter FE	✓	✓	✓
Cohort × Pair FE	✓	✓	✓
Cluster	Cohort	Cohort	Cohort

Note: Quarterly analog of the annual specification in Table 10. Same matched stacked-cohort design using BIS Locational Banking Statistics (residence-based). Treatment intensity: total SWF equity injection in $bn. The treatment quarter equals the observed deal quarter when available and Q1 of the treatment year otherwise. Pre-matching window: 12 quarters. Control pairs are Mahalanobis-matched on pre-treatment log-loan levels and slopes. SE clustered by cohort. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.4 Liability Test and Nationality Robustness

Table 32 compares the main LBS residence-based estimates with CBS nationality-based claims. This check matters because LBS assigns claims to the location of the reporting banking office, not to the nationality of the banking group. For example, a loan booked by the London branch of a U.S. bank is a U.K. claim in LBS, not a U.S. claim. CBS instead follows the nationality of the parent banking group, so it provides a complementary check that the result is not created by the residence-based booking convention or by activity routed through international banking centers. Column (5) first reports a liability-side test based on BIS liabilities; the estimate is small and statistically insignificant, which is inconsistent with a symmetric expansion of both sides of the bilateral balance sheet. The CBS estimates are the same sign and of similar order: the total-claims estimate is smaller than the LBS total-claims baseline (0.0426 versus 0.0581), while the CBS international-claims estimate is slightly larger (0.060). The comparison therefore suggests that the macro spillover result is not an artifact of the LBS reporting convention, while still leaving some scope for differences between residence and nationality measurement.

Table 32: Cross-Border Banking Claims Spillover: Liability Placebo and Nationality Robustness

	LBS (Residence)					CBS (Nationality)	
	(1) Total Claims	(2) Loans & Placed Dep.	(3) Non-Bank Loans	(4) Wins. Std. Flow Loans	(5) Liabilities	(6) Total Claims	(7) International Claims
SWF Invest (\$bn) × Post	0.0581**	0.0566**	0.0530***	0.0252**	0.0225	0.0426*	0.0599***
	(0.0195)	(0.0193)	(0.0151)	(0.0113)	(0.0197)	(0.0210)	(0.0181)
Outcome transform	log	log	log	wins. flow/pre-stock	log	log	log
Observations	3,259	3,258	3,250	3,259	3,255	4,325	4,055
Matched cohorts	12	12	12	12	12	17	17
Cohort × Year FE	✓	✓	✓	✓	✓	✓	✓
Cohort × Pair FE	✓	✓	✓	✓	✓	✓	✓
Cluster	Cohort	Cohort	Cohort	Cohort	Cohort	Cohort	Cohort

Note: Same matched stacked-cohort design as Table 10 but at annual frequency (±6 years), using Q4 end-of-year annual stocks. Control pairs are Mahalanobis-matched on pre-treatment log-loan levels and slopes. Columns (1)–(3) are BIS Locational Banking Statistics (residence-based) log stock outcomes: total claims, loans and placed deposits, and non-bank loans. Column (4) is the reported annual LBS loan flow, summed from quarterly flows, divided by pre-treatment mean loan stock and winsorized at the 1st and 99th percentiles. Column (5) is the log BIS liability-side position, used as a placebo outcome. Columns (6)–(7) are BIS Consolidated Banking Statistics (nationality-based, immediate counterparty) log stock outcomes: total claims and international claims. CBS total claims include local claims in local currency; CBS international claims exclude these, making column (7) the closest CBS analog to the LBS cross-border measure. The CBS estimates are the same sign and of similar order as the LBS baseline: CBS total claims are smaller, while CBS international claims are slightly larger. The CBS sample includes 17 cohorts (vs. 12 for LBS) because CBS covers additional reporting countries. SE clustered by cohort. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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